Does Corporate AI Talk Predict Stock Returns? Evidence from LLM-Based Disclosure Analysis and the AI Washing Phenomenon

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Abstract

This paper examines whether the language public firms use to discuss Artificial Intelligence (AI) in 10-K [1]reports, viewed as a signal of their underlying AI strategy, influences subsequent market performance. Using a panel of 799 Russell 3000 constituents from 2020 to 2024, the study applies three Large Language Models (LLMs), GPT 4o mini and two versions of Gemini Flash, to score eight disclosure dimensions, which include strategic depth, sentiment, three risk categories, forward-looking statements, AI washing, and talent & investment. Model scores are averaged to create an AI factor suite and an aggregate index. Cross-sectional regressions relate these measures to three-, six-, nine- and twelve-month buy-and-hold excess returns while controlling for firm characteristics, Fama-French-Carhart six-factor betas, and fixed effects. Higher scores on most AI dimensions predict significantly lower twelve-month excess returns, suggesting investor scepticism toward strong AI rhetoric and a correction of hype during the sample period.

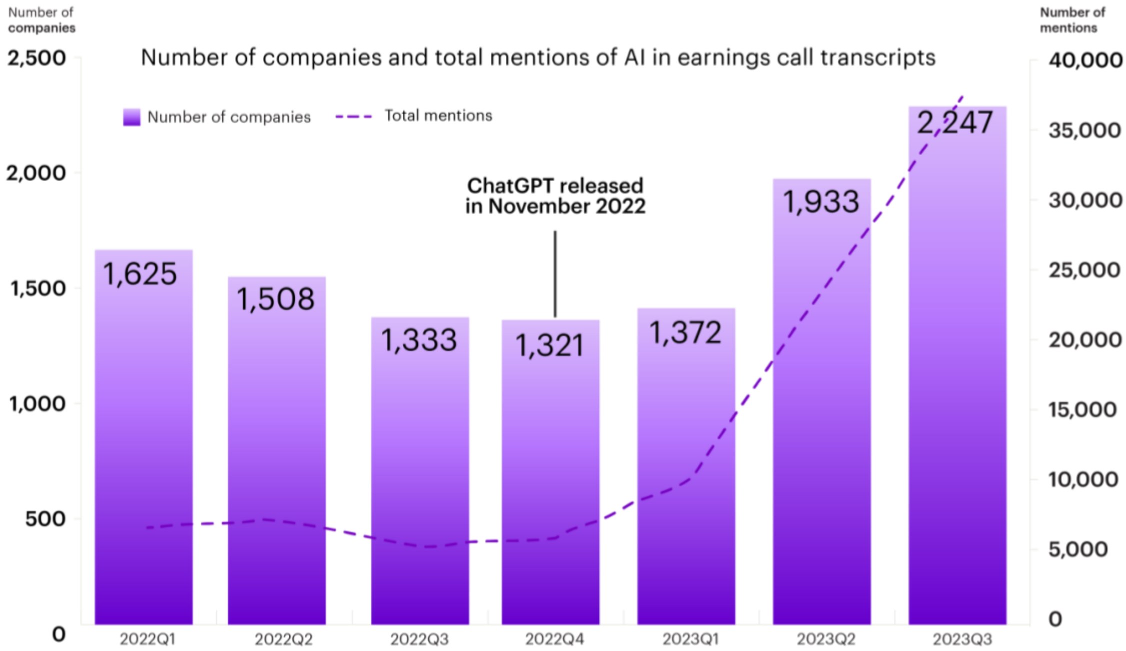
[1] Form 10-K is the official annual report required by the U.S. Securities and Exchange Commission (SEC) for publicly traded companies. This study's analysis focuses on two key narrative sections: Item 1A (Risk Factors) and Item 7 (Management's Discussion & Analysis).

1. Introduction

The Rise of AI in Company Talk and Investor Attention

Artificial Intelligence (AI) is rapidly becoming central to corporate strategy and operations (PwC, 2024; Mishra et al., 2022). It increasingly affects how companies make long-term plans and interact with investors and regulators (Campbell, 2025; Cockburn et al., 2018). This trend is clear in corporate reporting, especially in U.S. Securities and Exchange Commission (SEC) 10-K filings, where terms like “machine learning,” “Large Language Model (LLM),” and “Generative AI (GenAI)” appear more often (Mishra et al., 2024; Alston & Bird, 2024; Heyliger et al., 2023). Figure 1 shows the rising use of AI-related language in these documents (Accenture, 2024).

Figure 1:
Growth in AI Mentions in Corporate Earnings Call Transcripts (2022-2023)



Note. This figure shows the number of companies mentioning AI-related terms (left axis, purple bars) and total number of AI mentions (right axis, dashed line) in quarterly earnings call transcripts. Data from 2,500 companies shows a 6x increase in AI mentions following ChatGPT's release in November 2022. Source: Accenture Technology Vision 2024.

The increasing interest in AI matches its description as a general-purpose technology that is powerful, constantly improving, and able to drive innovation across many sectors (Bresnahan & Trajtenberg, 1995; Cockburn et al., 2018).

However, this trend also presents challenges in interpretation. Investors must now distinguish between real AI strategies and vague statements (Simonian, 2025). The term "AI washing" describes the use of misleading or shallow language about AI to attract investors (Seele & Schultz, 2022; Al Haddi, 2024; Barrios et al., 2025; Uslaner & Coquin, 2025).

As a result, regulatory control has increased. SEC Chair Gary Gensler (2024) stated "Investment advisers should not mislead the public by saying they are using an AI model when they are not. Such AI washing hurts investors" (U.S. Securities and Exchange Commission, 2024). This growing concern has led to specific guidance, as legal reviews have warned companies about disclosure risks and advised them to validate their AI claims to avoid enforcement action. (Heyliger et al., 2023).

Assessing the credibility of AI disclosures is challenging because genuine AI integration depends on intangible assets like proprietary code, training data, or specialised staff, which are difficult for outsiders to verify (Greene, 2025; Campbell et al., 2025). Because these elements are not easily verifiable by external parties, a condition of information asymmetry arises (Akerlof, 1970). This problem is intensified by the "black box" nature of many AI systems, where even its own developers may not fully understand model outputs (Burrell, 2016; Fleisher, 2022).

The rapid increase in AI references may reflect not only real adoption but also impression management. Firms might use AI-heavy narratives to influence stakeholder perceptions (Merkl-Davies & Brennan, 2007). This is consistent with attention-driven investment theories, which argue that investors are more likely to allocate capital to firms that attract media or trading attention (Barber & Odean, 2008). When companies see that AI-related stories get rewarded in the market, others might do the same. This pattern can create a feedback loop typical of the technology hype cycle— where initial excitement leads to very high expectations, and eventually, more realistic applications (Fenn, 1995; Manning & Napier, 2024).

If markets fail to distinguish real AI adoption from empty, promotional language, capital may be misallocated to firms skilled in storytelling rather than operational substance (Cooper et al., 2001). This behaviour reflects earlier periods of mispricing, such as during the dot-com era, when firms received valuation premiums for their use of fashionable terminology (Cooper et al., 2001). As a result, the current AI enthusiasm could risk distorting the efficient allocation of capital, driven by investor overreaction (Barberis et al., 1998).

These dynamics raise the following important question: Do AI-related corporate disclosures signal real technological progress, or are they just attention-seeking statements with no real substance?

The Measurement Challenge: From Basic Word Counts to Language Models That Think

Measuring the actual content and impact of corporate AI language presents significant challenges. Annual reports, especially 10-K filings, have become longer and more complex (Muchnik et al., 2019), with a rising number of AI-related references (Accenture, 2024). This makes manual analysis impossible at scale (Loughran & McDonald, 2016; Bojić et al., 2025). Early Natural Language Processing (NLP) approaches, such as keyword counting and bag-of-words methods, lacked the ability to detect meaning or nuance (Juluru et al., 2021; Loughran & McDonald, 2011). Even more advanced tools like FinBERT had difficulty with irony, sarcasm, and shifting terminology (Araci, 2019; Weitzel et al., 2016; Alexandris, 2024). LLMs, such as GPT-4, provide a more nuanced understanding of the context in financial texts therefore offering improved analysis. (Campbell et al., 2025; Li et al., 2023). However, concerns remain about their transparency and the potential for fabricated or biased outputs (Burrell, 2016; Prabhakar, 2025; Bommasani et al., 2021). Evaluating the truthfulness of AI claims is therefore complex and resembles efforts to detect greenwashing in Environmental, Social, and Governance (ESG) reports (Seele & Schultz, 2022; Simonian, 2025; Moodaley & Telukdarie, 2023). If markets are to reflect available information accurately, as Fama (1970) argued, tools that help distinguish substance from exaggeration are crucial. LLMs could support this aim—provided their limitations are understood and carefully managed.

Research Agenda and Questions

This study investigates the relationship between corporate communication about AI and subsequent stock market performance. The main research question is:

Do companies that communicate more clearly or positively about AI in their 10-K reports receive superior stock market returns, or do investors discount such statements if not backed by clear evidence of action?

This core question is supported by two secondary inquiries.

First, *which aspects of AI-related disclosure matter most to investors?* This includes elements like tone, strategic depth, risk language, future focus, and specific investment information. The analysis not only looks at whether AI is mentioned, but also at how it is discussed. Previous research has shown that narrative features such as tone and clarity affect investor reactions (Feldman et al., 2010; Loughran & McDonald, 2011), and that changes in disclosure sentiment over time provide insights into future company performance (Brown & Tucker, 2011). Specifically, disclosures that indicate credible investment or future strategy may be more significant than vague or overly positive mentions. On the other hand, disclosures that point out risk can either lower perceived value or boost credibility, depending on the context (Kravet & Muslu, 2013; Hope et al., 2016).

Second, *does the industry context affect how AI disclosures are understood?* This study proposes that how informative AI communication is depends on industry norms and expectations. In technology sectors, using AI might be seen as standard and may not draw much market attention. However, when non-technology firms announce AI adoption with a clear purpose, it might indicate real innovation or change. Previous research shows that the economic value of AI investments relies on industry-specific skills and supporting assets (Cockburn et al., 2018; Brynjolfsson et al., 2021). Additionally, asset pricing studies suggest comparing firms within their industry to better estimate disclosure effects (Asness et al., 2000).

These questions address whether financial markets effectively include complex, qualitative information. The study examines the semi-strong form of the Efficient Market Hypothesis (Fama, 1970; Malkiel, 2003), which suggests that all publicly available information should quickly be reflected in prices. If AI-related language in regulatory filings—scored using LLMs—can consistently predict future returns, it implies this 'soft' information is not instantly priced by market.

To test these hypotheses, the study uses LLMs to measure six aspects of AI-related disclosure: strategic depth, sentiment, risk framing, forward-looking statements, investment focus, and signs of AI washing (Seele & Schultz, 2022; Al Haddi, 2024). These scores are then connected to future stock returns using panel regressions with standard controls, including time and industry fixed effects. This method is based on well-known research showing that text features in regulatory filings predict both investor perception and market outcomes (Loughran & McDonald, 2011; Feldman et al., 2010; Andreou et al., 2023).

The aim is to understand if markets reward firms for meaningful AI disclosures or if they are influenced by persuasive but unsupported stories. If the quality of disclosure is accurately reflected in prices, stock returns should not consistently change with language features. However, if the tone, structure, or focus predicts market performance, it would raise concerns about the impact of corporate storytelling on capital distribution.

Data and Methodology Overview

This study investigates the relationship between AI-related disclosures and future stock returns. It uses a sample of 799 firms from the Russell 3000 index, covering the years 2020 to 2024. This broad index covers about 98% of the investable U.S. equity market, offering large representation while focusing on actively traded companies (FTSE Russell, 2025). This timeframe is selected to align with a period of rapid growth in corporate AI engagement (Mollman, 2022), as seen by survey data showing that the regular use of generative AI in organisations nearly doubled in less than a year (McKinsey, 2024). This period is especially useful for studying how markets react to different types of AI stories (Mahat et al., 2024).

The sample includes only those firms with complete data, including financials, 10-K filings, and return history. As market shocks and macroeconomic uncertainty mark these years (Grigoryev, 2023), the models also incorporate control variables to help distinguish AI signals from broader market noise.

The core of the analysis is the textual content of 10-K filings. Two parts of each company's 10-K filing are used:

Item 1A: Risk Factors. This section requires firms to disclose important risks, including those associated with AI. Previous studies have shown that risk disclosures in Item 1A give useful information about specific threats to firms and relate to how investors view risk (Kravet & Muslu, 2013; Beatty et al., 2018).

Item 7: Management's Discussion & Analysis (MD&A). In this section management reflects on its past performance and future goals. It frequently contains references to AI-related plans, Research and Development (R&D) priorities, and operational transformations. MD&A language has been shown to affect investor expectations and capture narrative shifts not reflected in quantitative data (Feldman et al., 2010; Brown & Tucker, 2011; Muslu et al., 2015).

To analyse these sections, three different LLMs are used: OpenAI's GPT-4o-mini, Google's Gemini Flash 1.5, and Gemini Flash 2.5. An ensemble approach, which averages the outputs of the three models, is used to reduce model-specific variance and improve predictive accuracy (Breiman, 1996; Seni & Elder, 2010). This multi-model approach lowers the risk of single-model bias, similar to how inter-coder reliability is established in manual content analysis to ensure coding consistency across human raters (Artstein & Poesio, 2008). Also, analysis uses advanced prompt engineering techniques, including role assignment, Chain-of-Thought (CoT) reasoning, and structured output validation, to ensure reliable extraction of structured information from corporate narratives (OpenAI, 2025; Wei et al., 2022; Dunivin, 2025).

This study creates six key dimensions of AI-related disclosure in 10-K filings, based on prior research in financial text analysis, technological innovation, and impression management. Each factor is measured using ranked scores that allow for comparison across firms and time. These categories serve as the basis for analysing how qualitative AI communication relates to stock returns.

1. **AI Strategic Depth**:
   Examines how much AI is seen as essential to the firm's business model or strategy. Research indicates that firms focusing on technological integration and skill development often show greater innovation and long-term success (Cockburn et al., 2018; Mishra et al., 2022; Babina et al., 2024).
2. **AI Disclosure Sentiment**
   This dimension measures the overall tone of AI discussion, from optimistic to cautious. The tone of corporate language has been shown to influence investor behaviour and market reactions (Loughran & McDonald, 2011; Feldman et al., 2010). Positive language may boost short-term sentiment, while negative tone can signal risk (Kearney & Liu, 2014).
3. **AI Risk Disclosure** (*includes three sub-metrics*)
   Measures firm references to risks related to:
   (a) Own Adoption (implementation risk),
   - Firms that reveal internal risks, like integration issues, legal risks, or technical limits, indicate possible operational problems but also show some transparency in handling AI adoption (Alston & Bird, 2024; Amershi et al., 2019).

   (b) External Threats (AI used by competitors or third parties)
   - Disclosures that recognise competitive threats from external AI developments show an understanding of industry changes, in line with market shifts driven by innovation (Christensen, 1997; Cockburn et al., 2018).

   (c) Non Adoption (the risk of falling behind due to non-adoption)
   - When companies mention the risk of falling behind due to slow or not adopting AI, they stress the need to invest in technology to stay competitive in changing markets (Ang & Goh, 2024; Brynjolfsson & McAfee, 2014).

   Such risk disclosures have been shown to influence investor interpretation and reduce information asymmetry (Beatty et al., 2018; Hope et al., 2016)
4. **Forward-Looking AI Statements**

This score asseses language about future AI investments, product plans, or strategic developments. Forward-looking MD&A content is known to shape investor expectations (Muslu et al., 2015), but vague use of future-tense language may be viewed with scepticism (Karapandza, 2016). Clear and specific plans are more likely to be seen as valuable signals (Campbell et al., 2025).

5. **AI Washing Index**

   Assesses whether AI is presented using vague, promotional language without evidence of action. This measure draws on research in greenwashing and dot-com-era hype cycles (Cooper et al., 2001; Seele & Schultz, 2022; Al Haddi, 2024).

6. **AI Talent & Investment Focus**

   Evaluates references to AI-specific hiring, funding, or partnerships. Prior work shows that disclosures about R&D and human capital investment are associated with innovation and signalling firm commitment (Babina et al., 2024; Spence, 1973).

These scores are then used as independent variables in regression models. The aim is to test whether they help predict stock returns after the filing date. The primary dependent variable is excess returns, calculated at 3, 6, 9, and 12 months after the 10-K date. Returns are adjusted for the risk-free rate to capture true outperformance.

To ensure the effect of AI disclosure is not confounded with other known drivers of stock returns, the models include a complete set of control variables. These cover firm size, valuation (like price-to-book), profitability, and standard risk factor loadings from the Fama-French-Carhart models (Fama & French, 1993; Carhart, 1997).

Put simply, the goal is to see if these AI narrative signals say something new—something not already captured by classic financial metrics (Loughran & McDonald, 2016).

To explain pattern seen, the study combines statistical analysis with detailed case studies (Creswell & Plano Clark, 2018). Case studies examine companies with the highest AI disclosure scores and worst stock returns. These cases show how management pressures, misleading signals, and company self-selection create the observed patterns (Jensen & Meckling, 1976; Spence, 1973; Roy, 1951).

Key Findings Preview

The results reveal a counterintuitive pattern: firms with stronger or more optimistic AI disclosures tend to experience lower subsequent stock returns. Initially, it might seem that companies with detailed or confident AI talk in their 10-K filings would do better. However, the data do not support this assumption (Campbell et al., 2025). In fact, between 2020 and 2024, most aspects of AI disclosure, such as “Strategic Depth”, “Disclosure Sentiment”, and “Forward-Looking” statements, are negatively linked to future performance.

This negative relationship suggests that markets may have responded with caution, or even scepticism, to highly visible AI disclosures. Companies that show strong AI intentions or used a positive tone often did worse than those with more moderate language. These findings indicate that, during the sample period, intensive AI communication did not lead to better returns. Instead, it seems to have been penalised (Greene, 2025).

Another explanation is based on signalling theory: some firms may use assertive AI language to project strength and innovation under asymmetric information (Spence, 1973). However, if firms self-select into such disclosure because they lack strong fundamentals, this introduces selection bias in observed outcomes (Jacobs et al., 2009). In this case, investors may discount the narrative, seeing it as a signal for underlying weakness rather than a reliable sign of value.

Also, companies could have changed their reporting strategies based on market reactions. If investors initially favoured companies highlighting AI in their reports, other firms might have used similar wording to gain attention, regardless of their actual progress. This type of imitation, often driven by a desire for attention, can lead to a gap between what is said and what is real, causing valuation adjustments when expectations are not met (Barber & Odean, 2008; Cooper et al., 2001).

So, what seems like a contradiction may simply be the market learning. Initial excitement fades. Investors ask more challenging questions. Furthermore, the link between words and value becomes more complex.

Contributions and Implications

This study contributes to the fields of finance, accounting, and information systems in several clear ways. First, it shows that LLMs—when carefully designed—can be used to turn complex, soft language in 10-Ks into structured data. Instead of simply counting keywords or examining sentiment, the method scores six detailed aspects of how firms discuss AI. These scores are built using prompts designed to capture nuance and tested across different LLMs for reliability.

Second, the study presents a framework that others can reuse or adapt for topics like ESG, cybersecurity, or sustainability, where complex narratives and signals are important. This helps in gaining valuable insights from large amounts of unstructured information.

Third, the empirical results show an unexpected pattern: firms with stronger narratives often have lower future returns. This challenges the belief that markets reward AI-focused statements. Instead, it suggests that investors may devalue overly enthusiastic or vague language, especially when lacking proof of real capability. The results imply that markets can see excessive disclosure as a sign of weak fundamentals, which aligns with signalling theory and self-selection bias (Spence, 1973; Jacobs et al., 2009).

For investors, this finding suggests that not all AI communication provides useful information for investors. Frequent or overly positive statements might aim to create an impression rather than show real progress. The scoring method developed in this study may help analysts identify AI disclosures that appear inflated or lack supporting evidence, providing an early warning signal for potential overstatement.

For companies writing disclosures or managing investor relations, the message is clear: more talk about AI is not always better. General optimism without specific details can be unhelpful. As SEC Director Gurbir Grewal (2024) warned, exaggerated AI claims, known as "AI washing," (Seele & Schultz, 2022; Al Haddi, 2024) can draw attention and reduce investor trust. Honest, detailed disclosures linked to real investments and capabilities are more likely to support long-term value.

These results provide new evidence on the limits of semi-strong market efficiency (Fama, 1970), suggesting that markets may take time to distinguish between meaningful and shallow AI stories. For regulators, the findings highlight the need to watch for misleading narratives in a time when intangible assets and strategic storytelling have a growing impact on capital allocation.

2. Literature Review and Hypotheses Development

This section reviews prior research on narrative disclosures in annual reports and their relationship with stock performance. It focuses on how companies' AI discussions in 10-K filings might predict future returns. Special focus is on the qualitative parts, such as MD&A

and Risk Factors, which have been found to provide signals beyond quantitative financial data (Brown & Tucker, 2011; Kravet & Muslu, 2013).

Recent advances in LLMs allow for more nuanced interpretations of corporate narratives compared to traditional keyword-based or sentiment-scoring methods (Loughran & McDonald, 2016; Bojić et al., 2025). Unlike dictionary-based tools, LLMs can better understand subtle differences in language, such as tone and irony. This capability allows analysts to distinguish between meaningful and surface level disclosure (Gupta et al., 2024), thereby addressing a critical gap in financial text analysis (Barrios et al., 2025).

This ability allows for the analysis of six specific AI disclosure dimensions developed for this study: strategic depth, sentiment, three risk categories, forward-looking statements, AI washing, and talent/investment focus. Each dimension may signal either real technological engagement or empty talk (Seele & Schultz, 2022; Al Haddi, 2024; Campbell et al., 2025). This section develops hypotheses that connect AI disclosure quality to future stock returns. This research also outlines how LLM-based scoring methods offer novel contributions to understanding AI's capital market implications.

Narrative Disclosures and Market Implications

As noted in Chapter 1, the MD&A and Risk Factors sections of 10-K filings are rich sources of qualitative, non-financial information. While early studies find only modest market reactions to 10-K filings once earnings information is known (Easton & Zmijewski, 1993), this view has since evolved. More recent evidence shows that the textual content of 10-K's provides valuable information about firm prospects that is reflected in stock prices (You & Zhang, 2009).

Key textual properties that are considered informative include narrative tone, readability, and the extent of year-over-year modifications (Feldman et al., 2010; Li, 2008; Brown & Tucker, 2011). For example, MD&A sentiment affects investor responses. Changes in MD&A tone from year to year provide information beyond standard financial measures like earnings surprises (Feldman et al., 2010). Similarly, other work has shown that a higher number of negative words in 10-K filings is associated with lower stock returns, confirming that textual sentiment is priced by the market (Loughran & McDonald, 2011).

Language clarity is another factor that affects market responses. Annual reports with simpler language tend to signal greater transparency, while complex language may indicate that firms are trying to hide poor performance (Li, 2008; Loughran & McDonald, 2014; Garel et al., 2019). You and Zhang (2009) show that investors underreact to information in more complex 10-K filings, suggesting that complex language affects how the market processes disclosure content.

Year-to-year changes in narrative disclosures also carry useful information (Brown & Tucker, 2011). Brown and Tucker (2011) show that firms with larger MD&A modifications experience stronger stock price reactions to 10-K filings, suggesting that investors use this information. However, these modifications do not predict analyst earnings forecast revisions, indicating that professional analysts may not find the same value in year-over-year textual changes. In contrast, companies with low levels of textual modification ("lazy" disclosures) exhibit fewer market reactions, suggesting investors are inattentive to chatter but do price new information when it is provided (Cohen et al., 2020).

The Risk Factors section (Item 1A of 10-K) has drawn attention since the SEC made such disclosures mandatory in 2005. These qualitative disclosures are intended to summarise management's view of significant factors that could negatively affect the company (Nelson & Pritchard, 2016).

Research explores whether these risk narratives inform investors (Kravet & Muslu, 2013; Hope et al., 2016). Risk factor disclosures help shareholders understand a firm's operational risks. Kravet and Muslu (2013) find that increases in risk-related sentences are associated with higher stock return volatility and trading volume, indicating that these disclosures heighten investors' risk perceptions. Hope et al. (2016) show that companies providing more specific risk factors experience stronger market reactions and help analysts assess firm risk more accurately.

This evidence supports examining the MD&A and Risk Factors sections in this study. The documented market reactions to sentiment, readability, textual changes, and risk disclosures suggest that these narrative sections contain useful information that is not immediately reflected in stock prices. This delayed price incorporation challenges the strong form of market efficiency and shows that analysis of qualitative disclosures can reveal information advantages (Fama, 1970; Malkiel, 2003).

Advanced NLP by LLMs

The use of LLMs for textual analysis is motivated by their advantages over traditional NLP models, such as FinBERT (Bojić et al., 2025; Campbell et al., 2025). Advanced LLMs can process and analyse extensive volumes of textual data, such as the "Risk Factors" and "Management's Discussion and Analysis" sections from numerous 10-K filings, with considerable speed. This scalability allows for more effective analysis of a larger market segment compared to manual coding or previous computational methods (Bojić et al., 2025).

Furthermore, LLMs are more suited for nuanced text understanding compared to traditional NLP methods (Zhang, 2023; Gupta et al, 2024). While older approaches might focus on keyword counting or basic sentiment scoring, modern LLMs can identify more nuanced, context-rich expressions related to corporate AI engagement (Gupta et al, 2024; Thapa et al., 2025; Campbell et al., 2025). This deeper understanding is vital for distinguishing between companies with genuine, significant AI integration and those that make only superficial or generic mentions (Campbell et al., 2025; Dunivin, 2025). For instance, models like GPT-4 have shown strong performance on financial text analysis tasks, in some cases outperforming domain-specific models (Li et al., 2023). The capacity of LLMs to perform tasks like sentiment analysis effectively (Bojić et al., 2025) is also pertinent to metrics such as AI "Disclosure Sentiment" in this research.

Finally, the application of LLMs offers the potential to uncover unexplored linguistic patterns or possible themes within corporate disclosures, identifying subtle, context-rich expressions that traditional methods might miss (Thapa et al., 2025). These insights could be new indicators of a company's AI adoption and strategic direction, which could stay undetected by traditional models. While acknowledging LLM capabilities, potential limitations in interpreting highly nuanced or sarcastic statements (Bojić et al., 2025) are also considered, particularly for complex metrics.

The quality of LLM outputs is very dependent on the design of the input prompts. To ensure reliability, the study's prompt design follows best practices for extracting structured and reasoned outputs from LLMs (Wei et al., 2022; Dunivin, 2025).

Based on these analytical abilities, this study identifies six AI disclosure factors using proven theories that connect corporate narratives to economic results. These factors cover various ways companies discuss AI, from strategic use to unsupported claims.

AI/Textual Factors and Financial Performance

Within the 10-K narratives, several dimensions of AI-related disclosure are created that could plausibly affect investor expectations and, therefore, stock returns. Each dimension is grounded in prior literature on how textual or thematic elements relate to firm performance (Babina et al., 2024):

1. AI Strategic Depth:

AI Strategic Depth refers to how much a company's 10-K report highlights AI as a central part of its business strategy rather than just a surface-level mention.

A strong strategic focus on AI may signal a firm's commitment to developing technological capabilities for a sustainable competitive advantage. (Cockburn et al., 2018). This strategic focus matches evidence showing that companies making large investments in new technologies, like AI-driven digital change, often see improved long-term performance (Mishra et al., 2022; Cui, 2025).

Research shows that companies making long-term AI investments in hiring and projects tend to achieve higher growth and innovation (Babina et al., 2024). However, this positive relationship follows the "Productivity J-Curve" effect described by Brynjolfsson et al. (2021).

This theory explains why companies adopting transformative technologies like AI often see their productivity and performance decline initially, before the long-term benefits appear. The decline happens because companies must make significant investments in intangible assets—such as training, process redesign, and organisational change—that are difficult to measure but essential for successful AI adoption.

Furthermore, Mishra et al. (2022) show that a strong AI focus in 10-K reports, which indicates management's emphasis on AI initiatives, is associated with improved operating efficiency and profitability. Thus, by integrating AI into their core business, companies could hint at future success. The market might reward these organisations, seeing them as leaders in AI.

2. Disclosure Sentiment

Disclosure Sentiment is defined by the tone or sentiment surrounding AI discussions that may influence investors' interpretations. If management speaks about AI in optimistic terms (opportunity, growth, competitive strength) versus in cautious or pessimistic terms (concerns, costs, uncertainties), it could impact stock expectations. Indeed, extensive evidence suggests that the sentiment expressed in corporate disclosures correlates with market reactions and future returns (Kearney & Liu, 2014). Positive language in narratives often accompanies higher immediate returns or improved investor sentiment, whereas negative language can predict stock price declines or signal risk (Loughran & McDonald, 2011; Kearney & Liu, 2014).

In the context of AI, a positive tone may signal management's confidence in AI-related projects, whereas a negative tone could reflect operational or strategic challenges (Kearney &

Liu, 2014). Consistent with general findings that textual tone in MD&A and earnings calls affect investor behaviour (Henry, 2008; Feldman et al., 2010), it is anticipated that the sentiment of AI disclosures (measured, for example, via a financial sentiment model) will relate to stock performance. However, sentiment alone may be less informative than substantive content; therefore, it is tested alongside other aspects of AI disclosure.

3. AI-Related Risk Disclosures

In discussions about AI-related risks, several dimensions are considered:

(3a) Implementation Risk (Own AI Adoption)

Adopting new technologies like AI involves significant implementation risks, such as integration problems, cybersecurity threats, and regulatory issues (Alston & Bird, 2024; Amershi et al., 2019). These obstacles can reduce productivity and lead to project failure if not effectively considered (Brynjolfsson et al., 2021)

A company's open disclosure of AI implementation challenges might warn investors of potential setbacks, possibly lowering short-term expectations. However, it can also signal transparency and proactive management of AI-related threats, which means that firms are managing these risks responsibly.

(3b) External AI Threats

Next, companies may describe risks from AI developments in the external environment. External risks include the risk of industry disruption by AI, new AI-enabled competitors, or shifts in customer behaviour due to AI. Such disclosures reflect the concept of technological disruption risk, where companies risk becoming outdated if they fail to keep pace with technological advancements (Christensen, 1997). As discussed in Chapter 1, general-purpose technologies like AI are known to reshape industries (Cockburn et al., 2018). Firms that do not adopt AI could lose market share, as they risk being outcompeted by more agile competitors, a principle of technological disruption theory (Christensen, 1997). When a company warns about external AI threats, it indicates that management is aware of potential competitive disruption. Management awareness could mean two things: it might make investors cautious about the firm's future, but it also shows the firm is conscious of its situation.

(3c) Competitive Risk of Non-Adoption

Additionally, there is a risk of falling behind if AI is not adopted quickly. Companies increasingly recognise that not using AI could mean lagging behind competitors who do, making AI adoption essential rather than optional (Campbell et al., 2025; Ang & Goh, 2024).

This type of competitive risk acts as the opposite of external threats. Economic theory and evidence suggest that firms investing in emerging technologies tend to achieve superior long-term results, while those slow to adopt technology are more likely to underperform (Brynjolfsson & McAfee, 2014; Christensen, 1997). Furthermore, recent findings indicate the existence of an "AI premium," whereby firms with higher levels of AI investment experience better stock performance, suggesting that capital markets reward AI capability (Yang et al., 2024). Therefore, when firms mention the risk of falling behind in AI developments, it shows they see technology adoption as important for staying competitive in the future.

In summary, AI-related risk disclosures, whether about implementation problems or competitive threats, provide useful information. While markets may initially respond negatively due to the uncertainty these risks suggest (Beatty et al., 2018), detailed and specific disclosures over time show that firms are actively dealing with potential challenges. Studies indicate that such disclosures are linked to lower information asymmetry and a reduced cost of capital (Campbell et al., 2019; Hope et al., 2016).

4. Forward-Looking AI Statements

Forward-looking AI statements reflect how firms discuss expected investments, product launches, and long-term strategic plans involving AI. Research indicates that companies with stock prices that poorly reflect future earnings information are more likely to issue forward-looking disclosures in MD&A sections (Muslu et al., 2015). Also, excessive use of future-tense terms such as "will" or "shall" is associated with lower future stock returns, suggesting that markets may interpret such language as overly optimistic or lacking in substance (Karapandza, 2016). This suggests that companies prioritize talk over taking action.

Confirming that firms use forward-looking AI statements to describe their planned investments, product developments, and long-term AI strategies. If these statements are seen as credible and informative, they are linked to higher stock prices (Muslu et al., 2015; Campbell et al., 2025). However, if they lack substance, they can be viewed negatively (Karapandza, 2016).

5. "AI-Washing" Index (Hype vs. Substance)

As discussed in Chapter 1, AI washing (Seele & Schultz, 2022; Al Haddi, 2024)—overstating the use or impact of AI in ways that resemble greenwashing—has attracted significant regulatory attention (Delmas & Burbano, 2011; U.S. Securities and Exchange Commission, 2024; Uslaner & Coquin, 2025). The SEC recently charged two investment advisers for making false claims about their AI use and has warned that inflated AI promises can mislead investors and invite regulatory scrutiny (U.S. Securities and Exchange Commission, 2024).

The use of buzzwords can lead to market mispricings, as seen during the dot-com bubble when firms adding ".com" to their names experienced unjustified stock price jumps not explained by fundamentals (Cooper et al., 2001). While academic research on AI hype in corporate filings is developing, earlier ESG and greenwashing studies have used textual analysis to distinguish genuine reporting from exaggerated claims (Campbell et al., 2019; Seele & Schultz, 2022; Al Haddi, 2024).

This AI-washing index identifies firms that frequently mention AI related terms without providing evidence of meaningful activity. Companies with high-hype, low-substance AI claims may experience temporary positive market reactions that fade when expectations aren't met, leading to underperformance. Conversely, firms providing clear, low-hype disclosures backed by actual AI actions may achieve more lasting stock performance gains.

6. AI Talent & Investment Focus:

Taken together, these AI disclosure dimensions—strategic depth, tone, risk categories, forward-looking content, hype versus substance, and talent & investment—draw on well-established theories that link corporate narratives to economic outcomes. This study evaluates these dimensions using 10-K text and examines their relationship to future stock returns, both individually and in combination.

The final metric captures how firms describe investments in AI talent, infrastructure, and capabilities. Actions such as hiring data scientists, acquiring AI startups, or funding AI-related projects often indicate serious efforts toward implementation (Babina et al., 2024). Research shows that human capital and research and development (R&D) investment are essential to realise value from new technologies. Babina et al. (2024) show that AI-related hiring is associated with increased innovation and firm growth. These disclosures may show more than simple interest. According to signalling theory, visible and costly actions, like hiring specialised talent, act as valid signs of a firm's quality and long-term potential (Spence, 1973).

Together, these AI disclosure dimensions (strategic depth, sentiment, risk types, forward-looking content, hype versus substance, and talent & investment) are based on established theories linking corporate communication to financial performance. This study assesses them using 10-K text and explores how each relates to future stock returns, both separately and combined.

Case Study Applications in Corporate Disclosure Research

Case studies help explain complex disclosure patterns that statistics alone cannot capture (Yin, 2018). Combining statistical analysis with detailed company examples have proven effective in finance research for understanding market reactions (Eisenhardt, 1989).

Recent studies show the value in this approach. Beattie and Smith (2013) use statistics followed by case studies to explain how graphics in reports affect firm value. Linsley and Shrives (2006) combine numbers with specific company examples to understand disclosure strategies. Similarly, Linsley and Shrives (2006) combine quantitative risk disclosure analysis with specific company examples to understand different disclosure strategies.

This approach works particularly well when markets react in unexpected ways to company disclosures, helping researchers identify the behavioral reasons behind statistical patterns (Patton, 2015).

Hypotheses Development

Building on this literature, three key hypotheses are created about AI disclosures and stock returns:

H1 (AI Disclosure and Returns): *Firms with higher quality and extent of AI-related disclosures in their 10-K will experience better future stock returns on average than those with minimal or poor-quality AI disclosures.*

This hypothesis matches studies on how markets react slowly to complex information, like future statements or detailed reports, which affects return predictions (Ball & Brown, 1968; Sadka, 2003; Muslu et al., 2015). Recent research indicates that higher quality AI disclosures, particularly those linked to strategic moves, can indicate innovation and company growth, which markets might initially undervalue (Babina et al., 2024; Campbell et al., 2025).

H2 (Strategic vs. Sentiment Dimensions): *Strategic and talent/investment-focused AI disclosure dimensions will be more predictive of returns (and future firm performance) than sentiment or hype-related measures.*

Research supports this hypothesis by showing that disclosures focused on implementation, like AI hiring, infrastructure investment, or detailed strategies, provide more information than vague, sentiment-heavy language (Cao et al., 2024; Babina et al., 2024; Mishra et al., 2022). This suggests that, tangible evidence of implementation is a more credible signal for long-term value than a positive tone or excessive buzzwords. Objective signs of AI adoption may lead to return premium, while hype-driven sentiment might only result in temporary valuation increases (Karapandza, 2016; Seele & Schultz, 2022).

H3 (Sectoral Differences): *The impact of AI disclosures on returns will be more substantial for firms outside the traditional tech sector.*

Prior work shows that firm-level signals, such as valuation or profitability, are more informative when considered relative to industry peers (Asness et al., 2000). This suggests that the informativeness of AI disclosures varies across industries. Bonsón et al. (2021) find that AI reporting is already common in the technology sector, which reduces differences and makes these disclosures less surprising to investors. In contrast, disclosures from non-tech firms are less common and may serve as stronger signals of innovation. Research on general-purpose technology shows that adopting AI affects sectors differently, especially where companies need to make significant organisational changes (Brynjolfsson et al., 2021; Cockburn et al., 2018). This suggests that AI disclosures might be more informative in industries where such practices are uncommon.

Research Gaps and Contributions

This study offers several contributions in the fields of corporate disclosure, AI economics, and natural language processing (Mishra et al., 2022; Babina et al., 2024). First, it provides one of the first large-scale analyses of AI-related disclosures across the whole sample of public companies' 10-K filings. This study extends prior work that used keyword frequency or hiring patterns to measure AI investment by applying full-text analysis to score multiple nuanced dimensions of AI disclosure and link them to stock returns (Mishra et al., 2022; Babina et al., 2024; Campbell et al., 2025). This study extends narrative disclosure research into the AI era, demonstrating how firms discuss AI and whether such discussions carry financial relevance.

The study introduces a relatively novel method by using structured prompts in LLMs to evaluate AI disclosure quality in context. Instead of depending on word counts or sentiment analysis, the models work as intelligent readers to judge if a firm's AI talk is substantial, forward-looking, or hype-driven (Dunivin, 2025). This approach is based on recent findings that the detail and framing of disclosures provide more insight than tone or frequency alone (Cao et al., 2024). The prompts are designed using new practices in LLM finance applications, with examples to improve validity and consistency (Barrios et al., 2025). To ensure reliability, the study combines results from several models—OpenAI's GPT-4o-Mini, Gemini Flash 1.5, and Gemini Flash 2.5—recognising variations in LLM outputs and using ensemble logic to reduce model-specific bias (Bojić et al., 2025). This ensemble LLM methodology advances the state of text-based financial research.

Third, the study aims to distinguish between genuine AI engagement and inflated claims by constructing an AI-Washing Index. This index is checked for visible company actions, such as AI-related patent filings, R&D spending, and hiring, to see if firms' statements are backed by real actions. This method follows earlier ESG research that looks at the gap between what companies say and what they do (Delmas & Burbano, 2011), and it addresses increasing regulatory concern about overstated AI claims in company reports (U.S. Securities and Exchange Commission, 2024; Uslaner & Coquin, 2025). When firms show high AI disclosures but low relevant investment, the index provides a new way to spot possible exaggerations in their reports.

Finally, the study links these AI disclosure characteristics to stock returns, contributing to the literature on asset pricing and narrative-based mispricing. If specific disclosure dimensions predict abnormal returns, this would suggest that investors underreact to qualitative signals, consistent with prior research on market responses to textual complexity and disclosure changes (Cohen et al., 2020; Chen et al., 2025). This underreaction shows how narratives about new technologies can affect company value and lead to pricing errors. The paper contributes both empirically and methodologically by developing improved tools for analysing corporate narratives surrounding AI and other general-purpose technologies.

3. Data

3.1 Sample Selection and Filtering Process

The sample construction follows a sequential filtering process to maintain data quality and methodological consistency (Regulwar et al., 2023). As explained in the Introduction, the study focuses on corporate AI disclosures over 2020-2024 to analyse the recent increase in AI engagement (Mollman, 2022; McKinsey, 2024). The initial sample included all 2,629 constituents of the iShares Russell 3000 index (BlackRock, 2025; FTSE Russell, 2025). After applying a four-stage filtering process, the final sample consists of 799 firms (see Table 1).

Stage 1: Balanced Panel Requirement. To create a balanced panel and ensure consistency over time, only firms with complete 10-K filings available for all five fiscal years (2020-2024) are retained. This reduces the sample from 2,629 to 1,635 firms, eliminating companies that delisted, merged, or had incomplete filing histories during the study period.

Stage 2: Section Extraction Success. A successful extraction of both Item 1A (Risk Factors) and Item 7 (Management's Discussion & Analysis) is required for each filing. These sections serve as the primary textual input for the LLM analysis (Nelson & Pritchard, 2016; Brown & Tucker, 2011; Feldman et al., 2010; Kravet & Muslu, 2013; Andreou et al., 2023; Huang et al., 2016). Firms missing either section in any year are excluded, yielding 1,445 firms.

Stage 3: Technical Processing Constraint. To ensure reliable, untruncated analysis by LLMs, firms whose combined Risk Factors and MD&A text exceeds 200,000 characters in any given year are excluded (OpenAI, 2023). This threshold represents the upper limit for consistent LLM processing given current model context sizes. While this filter affects only a small number of firms with exceptionally lengthy filings (primarily large conglomerates), it is necessary to maintain methodological consistency across all observations. This reduces the sample to 828 firms.

Stage 4: Return Data Availability. Firms with insufficient trading history or missing price during the study period are excluded. This results in a final sample of 799 unique firms corresponding to 3,995 firm-year observations.

The requirement for complete data across all five years introduces survivorship bias, favouring more stable, established firms that remained listed and filed regularly throughout the period (Brown et al., 1992; Carpenter et al., 2002). Although this reduces the ability to apply findings to smaller or less stable firms, it improves the reliability of the panel analysis by keeping a consistent group of firms over time. This eliminates confounding effects that could arise from a changing sample composition (Aptech Education, 2019).

Table 1:
Sample Construction

Stage	Description	Companies	Firm-Years
Russell 3000 Universe	iShares ETF constituents	2,629	13,145
Complete 10-K	All 5 – Years (2020-2024)	1,635	8,175
Section Extraction Success	Valid Risk Factors & MD&A	1,445	7,225
Text Length Sample	<200k characters combined	828	4,140
Final Sample	With returns data	799	3,995

Note. Filters are applied sequentially to Russell 3000 constituents (FY 2020-2024). “Firm-years” = firm × fiscal-year observations.

3.2 Data Sources and Collection

This study collects the data through two different data sources: (i) raw 10-K text directly from the SEC and (ii) numerical series accessed through WRDS

- **Textual data.** Annual 10-K filings are downloaded through the SEC's EDGAR API. For each filing, *Item 1A: Risk Factors* and *Item 7: MD&A* are programmatically extracted. When amended filings (10-K/A) existed, the most recent version is retained to ensure reliance on the final information set available to investors (Nelson & Pritchard, 2016).
- **Market and accounting data.**
  - *Center for Research in Security Prices (CRSP).* Provided daily stock returns used to construct the study's dependent variables.
  - *Compustat North America Fundamentals Quarterly.* Supplied firm-level accounting items (e.g., book value of equity, total assets) for traditional control variables.
  - *Kenneth French's Data Library.* Delivered daily factor returns for the Fama–French–Carhart models (Fama & French, 1993, 2015; Carhart, 1997) and the daily risk-free rate, necessary for calculating excess returns and estimating factor exposures.

Separating textual data (EDGAR) from numerical data (CRSP/Compustat/French Library via WRDS) clarifies the research design by distinguishing raw inputs from the analysis that follows.

3.3 Sample Characteristics

Table 2 shows the industry composition, categorised according to the Global Industry Classification Standard (GICS). The sample represents a broad cross-section of sectors, with Industrials (20.7%), Consumer Discretionary (15.4%), and Information Technology (13.6%) forming the three largest groups.

Comparing the final sample to the full Russell 3000 index reveals some compositional differences. The sample slightly overrepresents Industrials (+5.7 percentage points) and Consumer Discretionary (+4.4 percentage points), while underrepresenting Financials (-5.3 percentage points) and Health Care (-8.4 percentage points). These differences likely reflect the study's data requirements—firms in highly regulated sectors (Financials, Health Care) may have more complex or lengthy filings that were filtered out during the text processing stage. Nonetheless, all 11 GICS sectors remain well-represented, ensuring that results are not driven by a single industry's dynamics (Andreou et al., 2023).

Table 2:
Sector Composition - Final Sample vs Russell 3000 Index

Sector	Companies	Sample (%)	Cumulative %	Russell 3000 (%)
Industrials	165	20.7%	20.7%	15.0%
Consumer Discretionary	123	15.4%	36.0%	11.0%
Information Technology	109	13.6%	49.7%	12.0%
Financials	101	12.6%	62.3%	17.9%
Health Care	70	8.8%	71.1%	17.2%
Real Estate	62	7.8%	78.8%	6.1%
Consumer Staples	47	5.9%	84.7%	4.2%
Materials	46	5.8%	90.5%	4.4%
Energy	33	4.1%	94.6%	4.8%
Communication	26	3.3%	97.9%	4.5%
Utilities	17	2.1%	100.0%	2.5%

Note. Distribution shows coverage across all major Global Industry Classification Standard (GICS) sectors for 2020–2024. Sample percentages refer to final analytical sample (N=799); Russell 3000 percentages refer to index composition as of 2024, excluding cash funds and derivative instruments (0.2% of index constituents). Company counts sum to 100 per cent within each column.

The requirement for complete and manageable filings over five years skews the sample towards larger, more established firms. This is because such firms are more likely to meet these criteria. However, the broad sector coverage offers a reliable basis for generalisable conclusions about how capital markets interpret corporate AI narratives (Kravet & Muslu, 2013).

3.4 Descriptive Statistics

3.4.1 Dependent Variable Properties

Table 3:
Descriptive Statistics for Excess-Return Dependent Variables

Variable	N	Mean	Std	Min	P25	Median	P75	Max
3m Excess Returns	3325	0.0003	0.185	-0.465	-0.116	-0.008	0.106	0.625
6m Excess Returns	3270	0.0235	0.2541	-0.575	-0.129	0.003	0.156	0.975
9m Excess Returns	3220	0.0451	0.3399	-0.667	-0.178	0.016	0.215	1.389
12m Excess Returns	3174	0.0412	0.376	-0.73	-0.191	0.001	0.213	1.672

Note. This table presents summary statistics for the dependent variable, buy-and-hold excess returns, calculated over 3-, 6-, 9-, and 12-month horizons following the 10-K filing date. N represents the number of firm-year observations. Returns are annualised and expressed as decimals (e.g., 0.10 = 10%). The mean return is positive and increases with the horizon, as does the standard deviation, reflecting the higher volatility associated with longer-term equity holdings.

Table 3 shows expected patterns across investment horizons. Mean excess returns increase from essentially zero at 3 months to approximately 4% over 12 months, reflecting equity risk premium (Damodaran, 2025). Return volatility similarly increases from 18.5% to 37.6%, consistent with risk accumulation over time (Fama & French, 1993). The wide range of outcomes (-73% to +167% for 12-month returns) provides necessary variation for cross-sectional analysis.

3.4.2 Control Variables Properties

Table 4:
Descriptive Statistics for Control Variables

Variable	N	Mean	Std	Min	P25	Median	P75	Max
Book Control Variables								
Log Market Cap	3989	22.1377	1.6426	18.908	20.935	22.015	23.235	26.405
Price-to-Book	3985	3.9108	4.0677	0.347	1.338	2.352	4.66	16.546
ROA	3984	0.0122	0.0236	-0.08	0.002	0.011	0.022	0.087
Factor Beta Controls (12-Month)								
Market Beta (MKTRF)	3843	0.9708	0.4706	-0.587	0.75	0.983	1.223	2.124
Size Beta (SMB)	3843	0.6748	0.6995	-1.125	0.227	0.649	1.102	2.448

Variable	N	Mean	Std	Min	P25	Median	P75	Max
Value Beta (HML)	3843	0.1546	0.7864	-2.027	-0.261	0.13	0.554	2.517
Profitability Beta (RMW)	3843	0.1418	0.8669	-2.622	-0.236	0.197	0.599	2.445
Investment Beta (CMA)	3843	0.0365	1.1002	-3.212	-0.522	0.05	0.619	3.191
Momentum Beta (UMD)	3843	-0.0296	0.615	-1.879	-0.322	-0.014	0.307	1.608

Note. Book controls from Compustat; factor returns from Ken French data library (WRDS). All variables winsorized.

Table 4 shows sensible characteristics for both book and factor controls. The dynamic factor loading procedure is very effective, achieving 96% estimation coverage with R-squared ranging from 39.7% to 41.7%. The mean market beta is 0.97, close to the theoretical expectation of 1.0 (Fama & French, 1993), with 85% of estimates statistically significant. The SMB factor shows a positive mean (0.67), indicating a slight small-cap tilt consistent with Russell 3000 coverage.

The Summary Statistics for Factor Betas over the other time horizons *(3-, 6-, 9-Month Horizons)* can be found in table … in the Appendix

4. Methodology

4.1 AI Disclosure Measurement Framework

This study uses LLMs to convert qualitative AI disclosures into quantitative metrics. As discussed in the introduction and literature review, this LLM-based approach overcomes the limitations of traditional textual analysis methods, which struggle with context and nuance (Loughran & McDonald, 2011, 2016; Araci, 2019). Traditional keyword-counting methods may falsely emphasise AI mentions with buzzwords rather than meaningful discussions using different terminology (Campbell et al., 2025; Hansen et al., 2018). LLMs evaluate how firms discuss AI, by looking at context, tone, and content to distinguish useful disclosure from empty AI talk (Gupta et al., 2024; Li et al., 2023).

To improve robustness and mitigate single-model bias, this study uses three distinct LLMs (OpenAI's GPT-4o-mini, Google's Gemini Flash 1.5 & 2.5). By averaging scores across models, the study produces a reliable measure similar to establishing inter-coder reliability in manual content analysis (Artstein & Poesio, 2008). This ensemble method uses the strengths of different models together to reduce variance and increase accuracy, which is a common practice in machine learning (Wei et al., 2022).

4.1.1 AI Factor Design and Categories

The study uses structured, categorical outputs (A–E scales) rather than continuous scores to improve classification reliability. Research on "LLM-as-a-judge" indicates that models perform more consistently on well-defined classification tasks with clear categories compared to generating nuanced continuous scores (Evidently AI, 2024; Anthropic, 2025).

Initial tests with continuous scoring (1-5 scales) created interpretability problems. The difference between scores of 4.3 and 3.8 is not clearly meaningful and subjective. The methodology therefore uses categorical buckets with clear grade definitions (A-E scales with detailed criteria). This categorical approach offers advantages supported by recent research (Bojić et al., 2025). Clearly defined categorical buckets improve inter-rater reliability, leading to higher agreement scores when comparing outputs across different LLMs or against human annotators (Wang et al., 2025).

The eight disclosure metrics are organised into six broader themes. First, “Strategic Depth” measures the degree to which AI is central to the firm's long-term plan (Mishra et al., 2022; Cockburn et al., 2018). Second, “Disclosure Sentiment” assesses the tone firms use when describing AI initiatives (Loughran & McDonald, 2011; Feldman et al., 2010). Third, AI-related risk is split into three subtypes: risks from the firm's own adoption (Amershi et al., 2019), external threats (Christensen, 1997), and the risk of falling behind by not adopting (Brynjolfsson & McAfee, 2014). Fourth, “Forward-Looking” statements capture the extent to which organisations mention future AI projects (Muslu et al., 2015). Fifth, the “AI Washing Index” assesses whether the language is hyped or substantiated (Seele & Schultz, 2022; Al Haddi, 2024). Lastly, “Talent & Investment” indicates if firms talk about hiring, teams, or financial commitment to AI (Babina et al., 2024; Spence, 1973)."

Table 5:
AI Disclosure Grading Framework: Categories and Scores

Factor	Scale	Grades
1. Strategic Depth (Mishra et al., 2022; Cockburn et al., 2018)	A-E (5-1)	A=Core Strategic, B=Significant Integration, C=Emerging, D=Superficial, E=No Mention

Factor	Scale	Grades
2. Disclosure Sentiment (Loughran & McDonald, 2011; Feldman et al., 2010)	A-E (5-1)	A=Clearly Positive, B=Mostly Positive, C=Neutral, D=Cautious, E=Negative/N/A
3.a Risk - Own Adoption (Alston & Bird, 2024; Amershi et al., 2019)	A-C (5-3)	A=Detailed Discussion, B=General Mention, C=No Mention
3.b Risk - External Threats (Christensen, 1997; Cockburn et al., 2018)	A-C (5-3)	A=Detailed Discussion, B=General Mention, C=No Mention
3.c Risk - Non-Adoption (Brynjolfsson & McAfee, 2014)	A-C (5-3)	A=Explicitly Discussed, B=Implicitly Suggested, C=No Mention
4. Forward-Looking (Muslu et al., 2015)	A-D (5-2)	A=Specific Plans, B=General Intent, C=Implicit Only, D=No Mention
5. AI Washing Index (Seele & Schultz, 2022; Al Haddi, 2024)	A-E (5-1)	A=Substantive, B=Mostly Substantive, C=Mixed, D=Mostly Hype, E=Pure Hype/N/A
6. Talent & Investment (Babina et al., 2024; Spence, 1973)	A-D (5-2)	A=Explicit Focus, B=General Mention, C=Implicit Only, D=No Mention

Note. Letter grades are mapped to numeric scores (A = 5, …, E = 1 or 2/3 depending on scale). See text below for detailed criteria.

Each theme has its grading system. “Strategic Depth”, “Disclosure Sentiment”, and the “AI Washing Index” use a five-level scale ranging from A (5) to E (1). Meanwhile, the three risk types adopt a simpler three-point scale, from A (5) to C (3), due to their more focused nature. “Forward-Looking” statements and “Talent & Investment” use four-point scales, ranging from A (5) to D (2), aiming to strike a balance between detail and clarity.

This approach ensures scores reflect disclosure substance rather than mere presence, addressing concerns about the gap between corporate AI claims and actual capabilities (Barrios et al., 2025; Campbell et al., 2025). Table A.2 in the appendix provides illustrative examples of actual corporate disclosure language for each factor, demonstrating the qualitative differences between high-scoring and typical AI communications.

4.1.2 Prompt Engineering and Validation

The quality of LLM-based analysis depends heavily on prompt design quality (OpenAI, 2024; Prompting Guide, 2025). This study follows a prompt engineering approach across four key dimensions.

Categorical Output Design

The categorical structure is implemented using Python's Literal types for strict validation:

```
AIStrategicDepthBucket = Literal[
    "A (Core Strategic Enabler)",
    "B (Significant Operational/Product Integration)",
    "C (Emerging/Exploratory Mentions)",
    "D (Superficial/Generic Mentions OR Risk of Non-Adoption Only)",
    "E (No Mention of Own AI Strategy/Adoption)"]
```

For example, classifying a company's AI "Strategic Depth" as 'A (Core Strategic Enabler)' offers clearer qualitative distinction than a numerical score. Research indicates that LLMs achieve higher consistency on well-defined classification tasks compared to nuanced continuous scoring (Wang & Wang, 2025).

Core Prompt Design Principles

Role Assignment and Context Provision: Each prompt begins by assigning the LLM a specific analytical persona for its evaluation:

```
"You are an expert financial analyst and AI strategist, specializing in evaluating corporate disclosures for insights into a company's engagement with Artificial Intelligence (AI). Your task is to meticulously analyse the provided text from a company's 10-K filing..."
```

This approach helps the model focus on key financial, strategic, and risk elements for each metric, following established practices in financial text analysis (Loughran & McDonald, 2016). Assigning roles is a crucial part of structured prompts for qualitative coding using LLMs (Dunivin, 2025). Sufficient context is provided to ensure understanding of the domain, including specifying the type of document and offering guidance on the analytical perspective.

Clear and Exact Instructions: Prompts are designed for maximum clarity, with explicit definitions provided for each categorical grade. For classification tasks, detailed criteria are included directly in the prompts, consistent with the "LLM-as-a-judge" concept (Evidently AI, 2024). For example, the AI “Strategic Depth” evaluation includes:

> "A = AI is deeply integrated into core business strategy with multiple concrete, specific examples of application and impact. B = [Specific criteria]... E = AI is mentioned superficially or not at all, with no connection to core strategy."

Chain-of-Thought Reasoning: The methodology incorporates Chain-of-Thought (CoT) prompting, where the model is instructed to provide its reasoning before delivering a final classification. Chain-of-thought prompting has shown effectiveness in complex analytical tasks (Wei et al., 2022; Li et al., 2023). CoT extracts more robust reasoning pathways within the model, leading to more accurate and reliable results, particularly for complex judgment tasks (Wei et al., 2022). This approach provides a transparent audit trail for the classification process, allowing for qualitative checks on the model's decision-making logic (Dunivin, 2025).

Structured Output Requirements: Critical output formatting requirements specify the exact JSON structure:

```json
{
  "ai_strategic_depth": {
    "bucket_assessment": "A (Core Strategic Enabler)",
    "supporting_evidence": ["quote1", "quote2"],
    "chain_of_thought_reasoning": "Your reasoning for this factor."}
        }
```

This explicit output structure prevents ambiguous responses and ensures information extraction in financial tasks (Wang & Wang, 2025). The structured format follows guidance from OpenAI (2025) and established prompting guides.

Quality Control and Validation Mechanisms

Schema Enforcement for Structured Output: To ensure consistent formatting and complete information, the study uses Pydantic models and the instructor library to define and enforce expected output structures for each prompt (Liu, 2024; Pydantic, 2025). For each AI metric, these models specify fields, data types, and allowed values using Literal types. This approach transitions from ‘hoping for good outputs’ to a more deterministic method of structured data extraction (Liu, 2024).

Temperature Control: All LLM generation uses a temperature of zero to ensure reproducibility by eliminating randomness and selecting the highest probability tokens (Schmalbach, 2025)

4.1.3 Cross-Model Reliability Testing

The same firm filings are evaluated by all three models using identical prompts to assess reliability. This cross-validation follows established inter-coder reliability methods (Artstein & Poesio, 2008). The reliability assessment uses multiple statistical measures to capture different aspects of agreement between the LLMs' ordinal classifications:

Kendall's Tau Correlation Coefficient

Kendall's tau measures rank correlation between ordinal classifications and is calculated as:

$$\tau = \frac{C - D}{C + D + T}$$

Where, *C* represents concordant pairs (when both models rank firms in the same order), *D* are discordant pairs (when models rank firms in opposite order), *T* represents tied pairs (when one or both models give the same grade). Kendall's tau is appropriate for evaluating agreement between the LLMs' ordinal (A–E) classifications because it assesses the similarity of data ordering and remains robust to non-normally distributed data and tied ranks (Kendall, 1938; Goodwin & Leech, 2006). Both characteristics are common in categorical rating scales.

Spearman's Rank Correlation Coefficient

Spearman's rho provides an alternative rank-based correlation measure, calculated as:

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2 - 1)}$$

Where, d_i represents the difference in ranks between models for each firm, *n* is the number of firms being compared (Spearman, 1904). If GPT-4o ranks a firm 15th out of 100 firms and Gemini ranks the same firm 20th, then d_i = 5 for that firm. Spearman's rho calculates how consistently the two models rank firms across the entire sample. Spearman's rho complements Kendall's tau by providing a different perspective on rank agreement, though it is more sensitive to outliers.

Cohen's Weighted Kappa

Cohen's weighted kappa adjusts for chance agreement and accounts for the ordinal structure by allowing partial credit for near-miss disagreements:

$$\kappa w = \frac{p_o - p_e}{1 - p_e}$$

Where, p_o is the observed weighted agreement, and p_e is the expected agreement by random chance. The weights are defined as $w_{ij} = 1 - \frac{|i - j|^2}{(k - 1)^2}$ where i and j represent grade categories and k is the total number of categories.

This weighted approach is particularly suitable for ordinal AI disclosure grades because disagreements between near categories (e.g., A vs B) receive partial credit, while disagreements between distant categories (e.g., A vs E) are penalised more heavily. This reflects the underlying assumption that near-miss disagreements are less problematic than fundamental disagreements about disclosure quality (Cohen, 1960, 1968).

Exact Match and Within-One-Grade Agreement

Exact match rates calculate the percentage of observations where all models assign identical grades. Within-one-grade agreement measures the percentage of cases where models' ratings differ by at most one category level (e.g., A vs B, or B vs C). These measures provide clear benchmarks for practical agreement levels.

Section 5.1.2 presents the reliability results and cross-model agreement patterns using these statistical measures.

4.1.4 Composite AI Score Construction

A Cumulative AI Score addresses potential multicollinearity concerns and creates a single measure of overall AI disclosure quality. Multicollinearity occurs when explanatory variables are highly correlated, which can make regression results unstable when using multiple AI factors simultaneously (Brooks, 2019). This composite measure includes information across all eight disclosure dimensions while keeping the benefits of the multi-model validation approach described earlier.

The Cumulative AI Score follows a two-step calculation process. First, for each firm-year observation and each AI factor, the numeric scores from the three LLMs are averaged to create a consensus score for that dimension. This averaging reduces model specific biases while keeping the useful information from all three. Second, the eight averaged factor scores are summed to create the Cumulative AI Score.

Calculated, for firm i in year t:

$$\text{Cumulative AI Score}_{i,t} = \sum_{k=1}^{8} \frac{Score_{i,t,k}^{GPT} + Score_{i,t,k}^{Flash1.5} + Score_{i,t,k}^{Flash2.5}}{3}$$

Where k indexes the eight AI disclosure factors. The resulting composite measure ranges from 16 (the lowest possible AI disclosure across all dimensions) to 38 (the highest possible AI disclosure). This approach retains the information from each individual factor. At the same time, it provides a single composite measure that mitigates the multicollinearity that would arise from including the correlated AI factors simultaneously in a regression (Brooks, 2019).

4.2 Control Variable Construction

This section details the construction of all non-AI variables used in the analysis. Data sources and collection procedures are described in Chapter 3; this section focuses on variable construction methodology.

4.2.1 'Book' Control Variables

The regressions control for firm characteristics that affect stock returns to isolate AI disclosure effects. Following Fama and French (1992), the analysis includes controls for firm size, value, and profitability.

The traditional control variables are constructed as follows. Firm size equals the natural logarithm of market capitalisation, calculated at the quarterly filing date using shares outstanding and closing prices. The price-to-book ratio measures value, computed as market capitalisation divided by shareholders' equity from the most recent quarterly filing. Return on assets (ROA) measures profitability, calculated as trailing twelve-month net income divided by total assets. These variables follow definitions used in asset pricing studies (Fama & French, 1992).

4.2.2 Factor Risk Loadings

Time-varying factor loadings account for each firm's changing exposure to systematic risk using the Fama-French-Carhart six-factor model (Fama & French, 1993, 2015; Carhart, 1997). This approach measures how risk profiles change over time.

The six-factor model includes: Market Risk Premium (MKT-RF), the market return minus the risk-free rate; Size factor (SMB), excess returns of small-cap over large-cap stocks; Value factor (HML), excess returns of high book-to-market over low book-to-market stocks (Fama & French, 1993); Profitability factor (RMW), excess returns of firms with robust over weak profitability; Investment factor (CMA), excess returns of conservative over aggressive investment firms (Fama & French, 2015); and Momentum factor (UMD), excess returns of past winning over losing stocks (Carhart, 1997).

For each firm-year and investment horizon $h \in \{3,6,9,12\}$ months, time-varying factor loadings are estimated using the following regression:

$$(R_{i,t} - RF_t) = \alpha_i + (\beta_{MKT} MKT_t - RF_t) + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \beta_{RMW} RMW_t + \beta_{CMA} CMA_t + \beta_{UMD} UMD_t + \varepsilon_{i,t}$$

Where, $R_{i,t}$ is the return on stock *i* in period *t*, RF_t refers to the risk-free rate in period *t*, is the α_i = firm-specific intercept. The β_{MKT}, β_{SMB}, β_{HML} β_{RMW}, β_{CMA}, β_{UMD} are the factor loadings, MKT_t, SMB_t, HML_t, RMW_t, CMA_t, UMD_t refer to the factor returns in period *t*, and $\varepsilon_{i,t}$ = idiosyncratic error term

The analysis employs rolling windows to prevent look-ahead bias whilst measuring time-varying risk characteristics. Look-ahead bias occurs when future information is inadvertently used to make decisions about past periods, creating unrealistically favourable results (Roberts & Whited, 2013). For each firm and horizon, loadings are estimated using six-month windows with approximately 130 daily observations. A minimum of 100 valid daily returns is required within each window. To prevent look-ahead bias, estimation windows always end before the 10-K filing date. For example, when calculating factor loadings for the three-month return period following a 10-K filing, the estimation uses daily data from six months before the filing date up to the filing date itself.

4.2.3 Data Treatment and Quality Controls

Buy-and-hold excess stock returns serve as dependent variables, measured over 3-, 6-, 9-, and 12-month horizons following each 10-K filing date. Excess returns equal stock returns minus the daily risk-free rate, compounded over each horizon.

Winsorization addresses outliers using variable-specific thresholds (Adams et al., 2019). Daily returns below 0.1% are dropped to address thin trading effects (Lesmond et al., 1999). Daily returns are winsorized at ±100% (Ince & Porter, 2006), with final return measures bounded at -95% to +500% (Hou et al., 2020). Price-to-book ratios are winsorized at 5th/95th percentiles due to negative book equity issues, factor betas at 2nd/98th percentiles, and other variables at 1st/99th percentiles (Boudt et al., 2020).

These horizon-specific factor loadings control for time-varying systematic risk exposure, ensuring results reflect relationships rather than outlier effects.

4.3 Portfolio Analysis Methods

This section describes the portfolio analysis used to examine relationships between AI disclosure and stock performance before applying multivariate controls. The analysis uses standard quintile sorts and sector-neutral techniques to separate firm-level effects from industry patterns.

4.3.1 Quintile Portfolio Construction

The study constructs value-weighted portfolios using standard sorting procedures (Fama & French, 1993; Daniel et al., 1997). Each year, following 10-K filing dates, firms are ranked by their AI disclosure scores and sorted into five equal-sized portfolios (quintiles). Q1 contains firms with the lowest AI scores (predominantly E grades), while Q5 contains firms with the highest scores (frequent A grades).

Portfolios are rebalanced annually based on new AI scores from the latest filings. Buy-and-hold excess returns are tracked over 3-, 6-, 9-, and 12-month horizons following each rebalancing date. This approach follows established asset pricing methodology for characteristic-based portfolio studies (Jegadeesh & Titman, 1993).

A high-minus-low (HML) spread portfolio measures return differences between Q5 and Q1, following the zero-investment strategy approach of Fama and French (1993). The HML return tests whether stronger AI disclosure predicts superior stock performance. Statistical significance is assessed using Newey-West standard errors to address potential serial correlation (Newey & West, 1987).

4.3.2 Sector-Neutral Analysis

Industry composition may confuse the AI-return relationship since both AI adoption and return patterns vary across sectors. Technology firms typically show higher AI engagement and different risk profiles than utilities or materials companies. Following Asness et al. (2000), the analysis uses sector-neutral sorting to isolate firm-level effects.

The procedure ranks firms within each GICS sector separately. Due to varying sector sizes, terciles replace quintiles to ensure adequate sample sizes across all industries. The top third of each sector forms the "High AI" portfolio, while the bottom third creates the "Low AI" portfolio. This approach ensures both portfolios have similar industry composition, removing sector bias from performance comparisons.

4.3.3 Statistical Testing

Return differences are tested using appropriate statistical methods for the data structure. For HML portfolios with time-series returns, standard t-tests assess whether mean returns differ significantly from zero

$$t = \frac{\bar{R}}{\frac{\sigma_{HML}}{\sqrt{T}}}$$

Where $\bar{R}$ is the average HML return, σ_{HML} is its standard deviation, and T is the number of periods.

For sector-neutral portfolios comparing cross-sectional groups, Welch's t-test accounts for potentially unequal variances and sample sizes (Welch, 1947):

$$t = \frac{\bar{x}_{Low} - \bar{x}_{High}}{\sqrt{\frac{s_{High}^2}{n_{High}} + \frac{s_{low}^2}{n_{Low}}}}$$

Where $\bar{x}_{High}$, s_{High}^2, n_{High} are the mean return, variance of returns, and number of firms in the 'High AI' portfolio, and $\bar{x}_{Low}$, s_{Low}^2, n_{Low} are the corresponding values for the 'Low AI' portfolio.

This test is more robust than Student's t-test for comparisons between groups with different characteristics (Ruxton, 2006; Delacre et al., 2017).

4.3.4 Sector-Specific Analysis

The analysis examines each GICS sector individually to identify whether results reflect broad patterns or sector-specific effects. This approach follows Roberts and Whited (2013) guidance for cross-sectional studies and helps identify uneven effects across industries. The same portfolio construction methodology applies within each sector, enabling comparison of AI disclosure effects across different economic environments.

4.4 Regression Framework

This section describes the regression approach used to test the research hypotheses and addresses key econometric challenges.

4.4.1 Baseline Specification and Fixed Effects Strategy

The study uses panel regressions to test the hypotheses, controlling for sector and time fixed effects. The baseline specification is:

$$ExRet_{i,t+h} = \alpha + \beta AIScore_{i,t} + \sum_{s} \delta_s \times \mathrm{Sector_{i,s}} + \sum_{y} \gamma_y \times \mathrm{Year_y} + \varepsilon_{i,t+h}$$

where $ExRet_{i,t+h}$ is the buy-and-hold excess return of firm i over the risk-free rate, measured over a horizon $h \in \{3,6,9,12\}$ months following the 10-K filing date. $AIScore_{i,t}$ represents one of the eight LLM-derived AI scores for that firm-year (converted to numeric 1-5 or 1-3 scale as appropriate). The model includes sector dummies (δ_s) that absorb any constant return differences across industries (GICS sectors) and time dummies (γ_y) that absorb year-specific shocks. All models use HC3 heteroskedasticity-robust standard errors (MacKinnon & White, 1985).

The coefficient of interest β measures the excess return associated with a one-grade increase in the AI disclosure score after controlling for broad industry and year effects. This baseline specification tests whether there is a predictive relationship between AI disclosure dimensions and future returns.

The identification strategy relies on fixed effects inclusion. Omitted variable bias occurs when unobserved factors correlate with both the explanatory variable (AI disclosure) and the dependent variable (stock returns), leading to biased coefficient estimates (Roberts & Whited, 2013). Sector and year fixed effects absorb all time-invariant sector-level and year-level unobserved heterogeneity. This ensures that the estimated AI effect is identified from within-sector, within-year variation, strengthening the model's internal validity (Roberts & Whited, 2013). This approach mitigates concerns about omitted variable bias from stable industry characteristics or broad economic trends.

4.4.2 Progressive Control Inclusion Strategy

The study uses a stepwise control inclusion strategy to ensure omitted variables do not drive observed AI disclosure effects. Controls are progressively added to the baseline model while monitoring the stability of the AI factor coefficient. A significant shift in the coefficient following the inclusion of a particular control suggests that the AI factor may proxy for that omitted variable. Stable coefficients provide confidence that the AI factor reflects an independent effect (Roberts & Whited, 2013).

Two control specifications are used. The "Book Controls" model adds standard firm attributes: firm size (logarithm of market capitalisation), valuation (price-to-book ratio), and profitability (return on assets). These are standard predictors in cross-sectional return studies

(Fama & French, 1992) and account for size, value, and quality effects that may correlate with AI disclosure tendencies.

The "Factor Controls" model includes the six Fama-French-Carhart factor exposures (Fama & French, 1993; Carhart, 1997). These betas are estimated using daily returns over the six months preceding the 10-K filing date, as detailed in Section 4.2. Factors are added incrementally: first the three-factor model (MKT-RF, SMB, HML), then the five-factor version (adding RMW and CMA), and finally the sixth factor (UMD). This addresses the possibility that firms with high AI scores are simply growth stocks with low HML exposure or strong momentum characteristics.

4.4.3 Individual Factors vs. Composite Score Approach

Regressions are run separately for each of the eight AI disclosure dimensions rather than simultaneously. Including all eight factors together would raise multicollinearity concerns as individual scores tend to correlate (Brooks, 2019; see Section 4.1.5). Eight individual models are estimated, one for each AI factor under each control specification. This produces a coefficient for each AI metric, allowing examination of significance and stability across models.

The Cumulative AI Score addresses multicollinearity whilst capturing overall AI disclosure (Brooks, 2019). Two main composite models are estimated: one regressing excess return on the Composite AI Score with book controls and fixed effects, and another including six-factor betas alongside fixed effects. The Composite AI Score ranges between 16 and 38, with the coefficient interpreted as the return impact per one-point increase in AI disclosure quality.

4.4.4 Sector-Specific Analysis

To examine whether AI disclosure effects vary systematically across sectors, separate regressions are estimated for each of the 11 GICS sectors. This approach allows for sector-specific AI effect estimation without the complications of interaction terms, providing clearer coefficient interpretation (Grace-Martin, 2009). Both book controls and factor control specifications are estimated separately for each sector *s*:

(1) Book Controls with the AI Scores:

$$ExRet_{i,t+h} = \alpha_s + \beta_{1s} AIScore_{i,t} + \sum_{y} \gamma_y \times \mathrm{Year}_y + \beta_{2s} Size_{i,t} + \beta_{3s} Value_{i,t} + \beta_{4s} Profitability_{i,t} + \varepsilon_{i,t+h}$$

(2) Factor Controls with the AI Scores:

$$ExRet_{i,t+h} = \alpha_s + \beta_{1s} AIScore_{i,t} + \sum_{y} \gamma_y \times \mathrm{Year_y} + \beta_{2s} MktRF_{i,t} + \beta_{3s} SMB_{i,t} + \beta_{4s} HML_{i,t} + \beta_{5s} RMW_{i,t} + \beta_{6s} CMA_{i,t} + \beta_{7s} UMD_{i,t} + \varepsilon_{i,t+h}$$

Where, the subscript s denotes sector-specific coefficients, i indexes firms, t indexes time, and γ_t represents year fixed effects. All models use heteroscedasticity-robust standard errors (HC3).

4.5 Explanatory Case Study Design

This study uses a mixed-methods approach where qualitative case analysis follows the quantitative findings. The quantitative analysis identifies the statistical relationship between AI disclosure intensity and stock returns. The case study component examines what drives this negative relationship.

4.5.1 Methodological Justification

Case studies allow investigation of contemporary market behaviour in its real-world context (Yin, 2018). The AI disclosure paradox represents complex market behaviour that cannot be fully understood through statistical analysis alone. Case studies examine the specific language, timing, and contextual factors that quantitative measures cannot capture, providing the depth needed to understand causal mechanisms (Eisenhardt, 1989).

This approach follows established practice in accounting and finance research. Beattie and Smith (2013) use a similar design to understand how graphics in corporate reports affect firm value, combining large-scale quantitative analysis with detailed examination of specific company disclosures. Similarly, Linsley and Shrives (2006) follow quantitative risk disclosure analysis with case examples to show the nature and quality of different disclosure strategies.

4.5.2 Case Selection Strategy

The study uses extreme case sampling. This sampling strategy selects cases that represent the most significant patterns of the phenomenon under investigation (Patton, 2015).

From the three sectors showing statistically significant negative relationships (Health Care, Industrials, Consumer Discretionary), the study identifies companies with the highest AI disclosure scores that subsequently experienced the most severe negative returns. These are termed 'AI Hyper' firms.

This extreme case selection strategy serves two critical purposes. First, it maximises the potential for identifying clear causal mechanisms by examining instances where the phenomenon is most pronounced (Patton, 2015). Second, it follows Eisenhardt's (1989) recommendation that theory-building case research should select cases that are likely to replicate or extend the theoretical framework being developed.

The selection criteria were:

- Sectoral Focus: Cases drawn only from sectors showing statistically significant negative AI disclosure effects
- Disclosure Intensity: Companies ranking in the top quartile of AI disclosure scores within their sector
- Performance Impact: Firms experiencing returns in the bottom decile of their sector following high AI disclosure
- Temporal Alignment: Clear temporal sequence where high AI disclosure precedes negative returns

4.5.3 Data Collection and Analysis

Case study data consists of direct quotes from 10-K filings. This follows the approach used by Linsley and Shrives (2006) in their analysis of risk reporting practices. Each case examines the specific language and framing around AI disclosure. The analysis also considers other external factors such as earnings results, market conditions, and corporate events. These factors may affect how markets interpret disclosure strategies.

The analysis applies established theoretical frameworks to explain the patterns found in disclosure language. These frameworks include agency theory (Jensen & Meckling, 1976), signalling theory (Spence, 1973), and legitimacy theory (Suchman, 1995). This theory testing approach follows Eisenhardt's (1989) framework for using case studies to refine and extend existing theoretical models in new contexts.

4.5.4 Validity and Limitations

The extreme case sampling strategy improves validity by selecting instances where theoretical relationship should be most clearly observable (Yin, 2018). However, this approach limits how broadly findings apply to firms with moderate AI disclosure levels. The cases aim to show and explain what drives the statistical relationship. They do not confirm how common these patterns are across all firms.

The use of multiple cases across different sectors supports external validity by showing that similar patterns occur across varied industry contexts (Yin, 2018). The application of established theoretical frameworks enhances reliability by providing consistent analytical criteria for interpreting disclosure patterns (Eisenhardt, 1989).

4.6 Identification Challenges and Robustness Considerations

Several identification challenges require consideration. Omitted variable bias poses the primary threat to causal inference, occurring when an unobserved variable correlates with both AI disclosure and stock returns. For example, managerial overconfidence could drive both aggressive AI language and poor investment decisions (Malmendier & Tate, 2015). The inclusion of firm characteristics, factor betas, and sector and year fixed effects is the primary strategy to mitigate this bias, though establishing full causality remains challenging (Cohen et al., 2020).

Measurement error in AI scores is addressed through the multi-model ensemble approach described in Section 4.1. Three different LLMs independently rate each firm's disclosures. The strong inter-model agreement supports agreement validity, and averaging reduces individual model bias (Artstein & Poesio, 2008; Seni & Elder, 2010).

Statistical robustness is ensured through several diagnostic procedures. Tests for equal error variance (Breusch-Pagan and White tests) reject homoskedasticity at the 1% level (Breusch & Pagan, 1979; White, 1980), justifying HC3 robust standard errors throughout (Brooks, 2019). The Durbin-Watson statistic (~1.91) suggests no significant autocorrelation in residuals (Durbin & Watson, 1950, 1951; Brooks, 2019). Outliers are handled through layered winsoriation with different thresholds applied based on distributional properties (Tukey, 1962). Variance Inflation Factors (VIFs) remain well below conventional thresholds, see table … in the Appendix (Brooks, 2019).

Appendix Table [X] shows the Pearson correlation coefficients among the key variables. The correlations between the control variables are generally low to moderate. For example, the highest correlations are between Log Market Cap and Price-to-Book (0.43) and between Log Market Cap and the SMB beta (-0.42). These levels are well below thresholds typically associated with problematic multicollinearity (Brooks, 2019), which confirms that the selected control variables capture distinct dimensions of firm characteristics and risk. This provides confidence that their inclusion in the regression models will not unduly inflate standard errors. Variance Inflation Factors (VIFs) remain well below conventional thresholds, see table … in the Appendix.

The assumption of normally distributed residuals is formally required for t-tests in small samples but becomes unnecessary in large samples due to the Central Limit Theorem (Fischer, 2011). With 3,995 firm-year observations, the CLT provides a strong theoretical foundation for the validity of reported p-values and confidence intervals (Lumley et al., 2002).

This framework follows widely accepted standards in empirical asset pricing (Fama & French, 1993; Jegadeesh & Titman, 1993) and supports the argument that observed patterns reflect genuine relationships rather than methodological artefacts.

5. Intermediate Results

This section provides the foundation for the regression analysis by validating the LLM-derived AI measures, showing their characteristics, and examining the basic relationships between AI disclosure and stock returns.

5.1 AI Score Characteristics and Patterns

This subsection examines the AI disclosure scores produced by the LLM methodology described in Chapter 4. The analysis first characterises the patterns that emerge across firms and industries, then validates the reliability of the scoring process.

5.1.1 AI Score Characteristics and Patterns

Table 6:
Descriptive Statistics for LLM-Derived AI Disclosure Factors

Variable	Obs	Mean	Std	Min	Q25	Median	Q75	Max	Skew	Kurt	CV
AI Washing Index	3995	1.43	0.73	1.00	1.00	1.00	1.67	5.00	1.75	2.10	0.51
Disclosure Sentiment	3995	2.16	0.80	1.00	1.67	2.33	2.33	4.67	0.71	0.25	0.37
Forward-Looking	3995	2.31	0.59	2.00	2.00	2.00	2.33	5.00	1.99	3.29	0.26
Strategic Depth	3995	1.72	1.01	1.00	1.00	1.00	2.00	5.00	1.33	0.59	0.59
Talent & Investment	3995	2.33	0.54	2.00	2.00	2.00	2.33	5.00	1.95	3.66	0.23
Risk - External	3995	3.32	0.41	3.00	3.00	3.33	3.33	5.00	1.47	1.97	0.12
Risk - Non Adoption	3995	3.49	0.52	3.00	3.00	3.33	3.67	5.00	1.10	0.56	0.15
Risk - Own Adoption	3995	3.19	0.45	3.00	3.00	3.00	3.00	5.00	2.58	5.74	0.14

Variable	Obs	Mean	Std	Min	Q25	Median	Q75	Max	Skew	Kurt	CV
AI Cumulative Score	3995	19.95	4.01	16.00	17.00	18.33	21.67	38.00	1.41	1.34	0.20

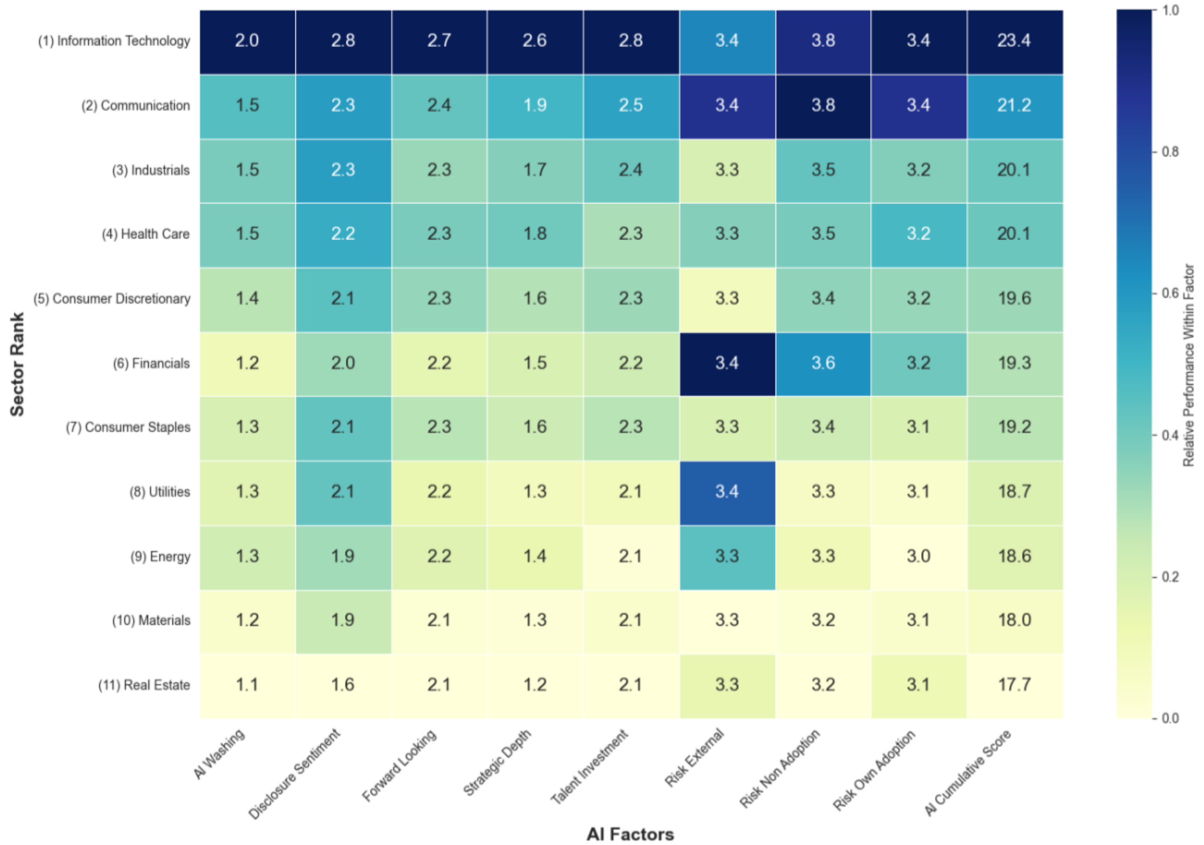
Note. Scores are the average of GPT-4o-Mini, Gemini 1.5 and 2.5 outputs. CV = coefficient of variation.

Table 6 shows that substantive AI discussion in 10-K filings was relatively uncommon during 2020-2024. “Strategic Depth” and “AI Washing Index” exhibit low means (1.72 and 1.43 respectively) with median scores at minimum levels, indicating most firms made only superficial AI mentions. The risk factors require careful interpretation since they use a 3-5 scale where 3 represents "No Mention" rather than low performance. The apparently high averages (3.19-3.49) therefore indicate minimal risk disclosure rather than comprehensive assessment.

These patterns provide initial evidence relevant to Hypothesis H3's prediction of sectoral variation, with technology sectors showing markedly different AI disclosure behaviour than traditional industries. Despite generally limited substantive AI engagement, meaningful variation exists for empirical analysis. The “AI Cumulative Score” ranges from 16 to 38 with a standard deviation of 4.01, while individual factors show coefficients of variation from 0.12 to 0.59.

Figure 2:

Heatmap of AI Disclosure Factors by Sector



Note. This figure presents mean AI factor scores by GICS sector, ranked by composite AI performance. Rows represent GICS sectors ordered by “AI Cumulative Score” from highest (1) to lowest (11): Information Technology, Communication, Industrials, Health Care, Consumer Discretionary, Financials, Consumer Staples, Utilities, Energy, Materials, and Real Estate. Columns represent individual AI factors converted to numeric. Color coding indicates relative performance within each factor, with darker blue representing higher sector performance and lighter yellow representing lower sector performance within that specific AI dimension. All values represent sector means across the sample period.

Figure 2 reveals clear industry patterns. Technology-intensive sectors show highest AI engagement, with Information Technology leading (“AI Cumulative Score of 23.4) and showing strength across multiple dimensions. Communication Services follows (21.2) with particularly high awareness of competitive AI risks. Traditional sectors like Real Estate (17.7) and Materials (18.0) show consistently low engagement across most dimensions, though all sectors converge on standardised risk reporting practices.

Table 7:

Sector-Level Distribution of Composite AI Disclosure Scores

Sector	Obs	N	Mean	Std	Min	Q25	Median	Q75	Max	Skew	Kurt
Information Technology	545	109	23.44	4.86	16.00	19.33	23.00	27.00	38.00	0.58	-0.50

Sector	Obs	N	Mean	Std	Min	Q25	Median	Q75	Max	Skew	Kurt
Communication	130	26	21.16	4.12	16.33	18.33	19.67	22.58	32.00	1.16	0.31
Industrials	825	165	20.08	3.79	16.00	17.33	18.67	21.67	33.67	1.27	0.67
Health Care	350	70	20.08	4.04	16.00	17.00	18.33	22.33	32.33	1.20	0.33
Consumer Discretionary	615	123	19.57	3.89	16.00	17.00	18.00	20.67	36.33	1.60	1.92
Financials	505	101	19.32	2.99	16.00	17.33	18.33	20.33	31.00	1.49	2.02
Consumer Staples	235	47	19.24	3.44	16.00	16.83	17.67	20.67	32.33	1.55	1.99
Utilities	85	17	18.74	2.50	16.00	17.33	18.00	19.00	27.00	1.70	2.21
Energy	165	33	18.64	3.05	16.00	17.00	17.33	19.00	30.00	1.80	2.40
Materials	230	46	18.03	2.22	16.00	16.67	17.33	18.33	29.00	2.43	6.26
Real Estate	310	62	17.68	2.47	16.00	16.33	16.67	18.00	27.67	2.32	4.83
Overall	3995	799	19.95	4.01	16.00	17.00	18.33	21.67	38.00	1.41	1.34

Note. Composite = sum of eight factor scores. Obs = firm-years.

Table 7 confirms that Information Technology not only leads in average performance but also shows the greatest within-sector variation (standard deviation = 4.86), suggesting differences between AI-focused and traditional technology firms. Traditional sectors exhibit both lower means and reduced variation, with most firms clustering near minimum scores while technology sectors drive the overall sample distribution.

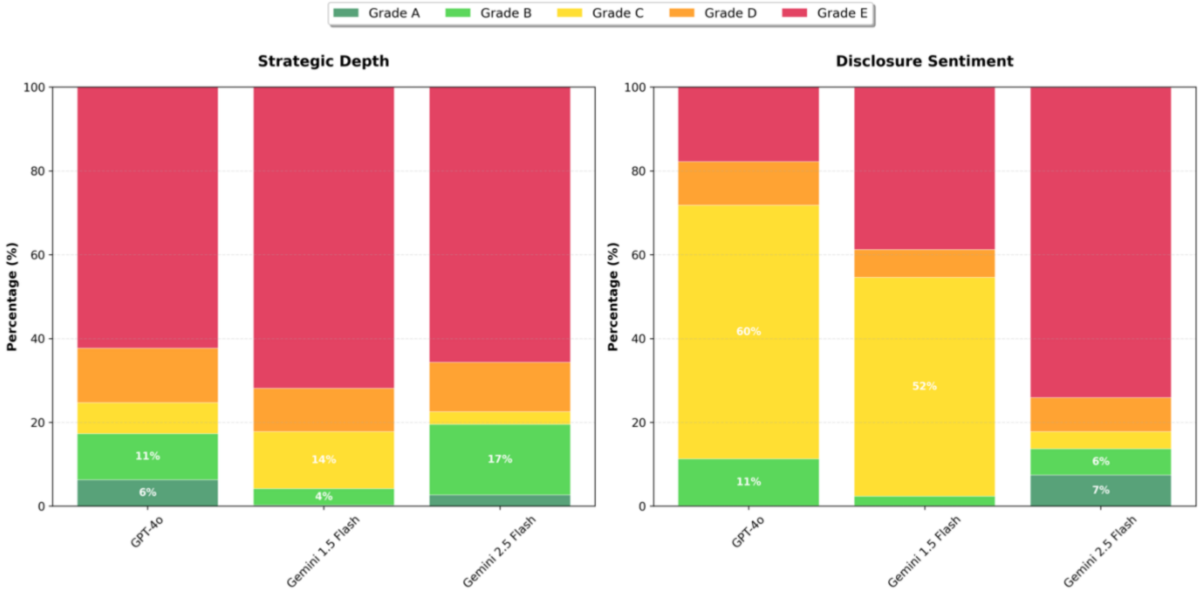
The substantial variation in AI disclosure both within and across sectors provides the empirical foundation for testing whether markets systematically price this qualitative information, as examined in the portfolio analysis that follows.

5.1.2 Cross-Model Reliability Analysis

The credibility of the empirical analysis depends on the quality of the LLM-derived AI disclosure scores. This subsection examines agreement patterns across the three models to validate the ensemble approach.

Figure 3:

Distribution of AI Factor Grades by LLM Model (Strategic Dept and Disclosure Sentiment)



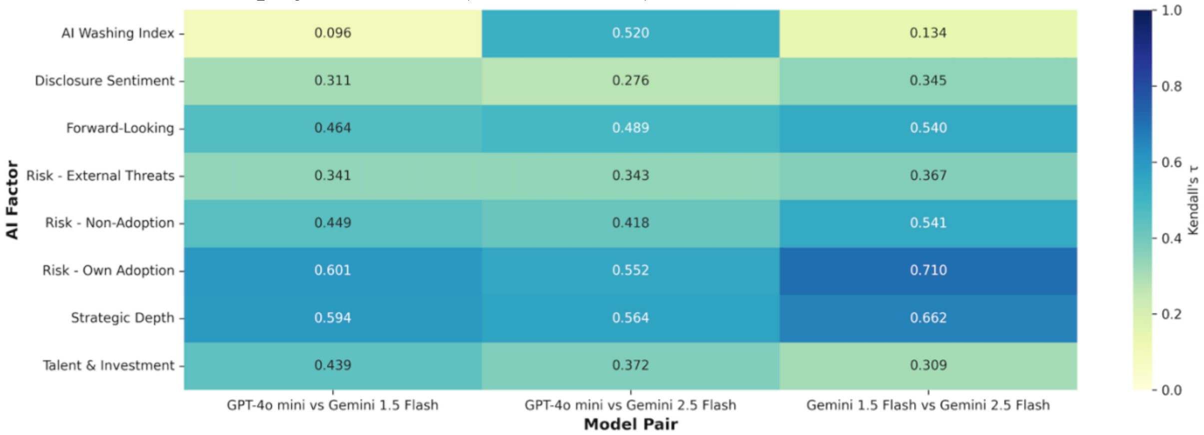
Note. Each panel shows frequency of letter grades (A–E) issued by GPT-4o-Mini, Gemini 1.5, and Gemini 2.5. Additional factors shown in Appendix.

Figure 3 reveals both agreement and systematic differences across models. “Strategic Depth” shows broadly similar patterns, though GPT-4o assigns more top grades (A-B) while Gemini 2.5 is more conservative. “Disclosure Sentiment” shows more pronounced differences, with Gemini 1.5 assigning far more D grades than other models. These patterns underscore the value of the ensemble approach, which balances individual model biases to produce more reliable scores.

The remaining factors show additional patterns of interest (see Appendix Figures …). Risk factors exhibit high agreement across models, with most firms receiving C grades ("No Mention") across all three LLMs, reflecting the limited AI risk disclosure noted in Table 6. “Forward-Looking” statements and “Talent & Investment” show moderate variation but generally consistent patterns across models. However, the “AI Washing Index” reveals a clear anomaly: Gemini 1.5 assigns nearly universal E grades, essentially classifying almost all corporate AI content as "Pure Hype" or "Not Applicable," while GPT-4o and Gemini 2.5 show much more varied distributions with substantial portions of firms receiving higher substantive scores. This extreme divergence suggests fundamental differences in how models interpret the distinction between genuine AI implementation and promotional language, making the ensemble approach particularly valuable for this subjective dimension.

Figure 4:

Correlation Heatmap of AI Factors (Kendall's Tau)



Note. Darker tiles denote stronger rank correlation (τ). Sample = 3,995 firm-years, 2020-24.

Figure 4 shows strong inter-model agreement for objective dimensions. "Strategic Depth" achieves Kendall's tau values of 0.59-0.66 across model pairs, while "Risk - Own Adoption" shows even stronger agreement (τ = 0.55-0.71). These factors have well-defined scoring criteria, making high agreement both expected and reassuring (Artstein & Poesio, 2008). "Forward-Looking" statements and "Risk - Non-Adoption" show moderate agreement (τ = 0.42-0.54), while the "AI Washing Index" exhibits the weakest consistency, particularly between GPT-4o and Gemini 1.5 (τ = 0.096), reflecting the subjective nature of distinguishing "hype" from "substance."

Table 8:

LLM Agreement Metrics (GPT-4o-Mini vs. Gemini Flash 2.5)

Factor	Kendall's tau	Spearman's rho	Cohen's weighted kappa	Exact Match	Within 1	N
Strategic Depth	0.56	0.61	0.65	65.60	87.30	3995
Disclosure Sentiment	0.28	0.31	0.23	26.80	41.50	3995
Risk - Own Adoption	0.55	0.57	0.65	82.70	98.60	3995
Risk - External Threats	0.34	0.35	0.37	70.70	97.30	3995
Risk - Non Adoption	0.42	0.45	0.33	52.10	90.70	3995
Forward-Looking	0.49	0.51	0.50	78.90	86.70	3995

Factor	Kendall's tau	Spearman's rho	Cohen's weighted kappa	Exact Match	Within 1	N
AI Washing Index	0.52	0.54	0.53	66.40	72.70	3995
Talent & Investment	0.37	0.39	0.36	79.50	89.60	3995

Note. Kendall's tau (Kendall, 1938), Spearman's rho (Spearman, 1904), Cohen's weighted kappa (Cohen, 1960, 1968), Exact Match = % identical grades, Within 1 = % agreeing within one grade, N are the number of valid paired assesments.

Table 8 demonstrates acceptable reliability across most factors. Exact match rates exceed 65% for most dimensions, while within-one-grade agreement surpasses 87% across nearly all factors. "Risk - Own Adoption" achieves the strongest agreement (82.7% exact matches, 98.6% within one grade), while "Disclosure Sentiment" shows the weakest (26.8% exact matches, 41.5% within one grade). These agreement levels compare favourably to manual content analysis studies (Goodwin & Leech, 2006), supporting the validity of using averaged scores for empirical analysis.

The reliability assessment confirms that the ensemble approach produces sufficiently consistent measures for examining AI disclosure effects on stock returns, as explored in the portfolio analysis that follows.

5.2 Portfolio Analysis

The portfolio analysis examines whether the AI disclosure patterns identified in Section 5.1 translate into systematic return differences, providing initial tests of the research hypotheses before controlling for firm characteristics in Chapter 6.

5.2.1 Individual AI Factor Quintile Sorts

Table 9:
Portfolio Quintile Sorts by Individual AI Factors – 12 M Excess Return

AI Factor	Q1 (Low)	Q2	Q3	Q4	Q5 (High)	High-Low
AI Washing Index	3.92	8.1	5.79	3.54	-0.78	-4.70**
Disclosure Sentiment	6	5.55	5.46	3.96	-0.39	-6.39***
Forward-Looking	4.19	7.95	5.96	3.42	-0.94	-5.12**
Strategic Depth	5.43	7.81	3.37	4.72	-0.75	-6.17***
Talent & Investment	4.73	7.65	4.36	6.15	-2.32	-7.04***

AI Factor	Q1 (Low)	Q2	Q3	Q4	Q5 (High)	High-Low
Risk - External	4.97	6.08	2.82	4.06	2.66	-2.31
Risk - Non Adoption	5.19	6.65	5.37	4.24	-0.87	-6.06***
Risk - Own Adoption	5.01	6	5.25	4.81	-0.49	-5.50**
AI Cumulative Score	7.75	4.78	4.08	5.8	-1.97	-9.72***

Note. Firms are sorted into quintiles (Q1 = lowest score). Mean excess return and Newey-West t-stat in parentheses (Newey & West, 1987). High-Low is Q5 minus Q1 portfolio return. *p<0.1, **p<0.05, ***p<0.01 for High-Low t-test. Returns are annualised and in %.

Using standard quintile sorting methodology (Fama & French, 1993), Table 9 shows a consistent negative relationship between AI disclosure intensity and subsequent returns. “Strategic Depth” shows a Q5-Q1 spread of -6.17% (significant at 1%), while “Talent & Investment” has -7.04%. The “AI Cumulative Score” produces the largest spread of -9.72%, which is highly significant, suggesting that AI disclosure is associated with substantial underperformance.

This pattern **challenges Hypothesis H1** and traditional signalling theory predictions, where credible disclosure should benefit quality firms (Spence, 1973). The consistent negative relationship across AI dimensions suggests that H1 may be rejected, though formal hypothesis testing requires controlling for firm characteristics and market factors in the regression analysis that follows. These patterns align with technology bubble literature showing negative market reactions to fashionable terminology during hype cycles (Cooper et al., 2001; Ofek & Richardson, 2003).

Regarding Hypothesis H2, the individual factor results provide **mixed preliminary evidence**. Strategic dimensions like “Talent & Investment” (-7.04%) and “Strategic Depth” (-6.17%) show substantial negative effects, but the sentiment-based “Disclosure Sentiment” factor (-6.39%) also exhibits a large negative coefficient that falls within the range of strategic factor effects. While the largest individual effect comes from “Talent & Investment” (a strategic factor), formal testing in the regression analysis will be needed to determine whether strategic dimensions are systematically stronger predictors than sentiment measures, as predicted by H2.

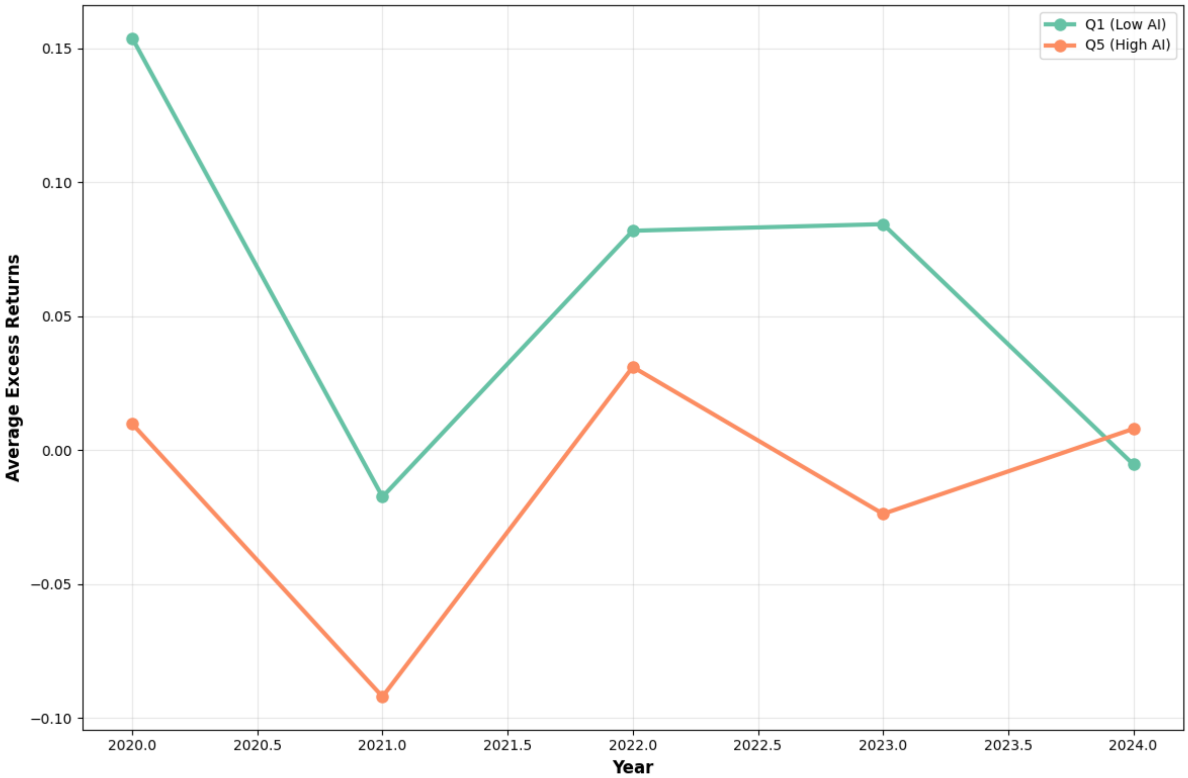
Table 10:
Portfolio Quintile Sorts by Composite AI Score – Excess Return

Time Horizon	Q1 (Low)	Q2	Q3	Q4	Q5 (High)	High-Low
3 Month	1.65	0.68	-0.06	0.24	-2.32	-3.97***
6 Month	3.18	2.85	3.2	3.7	-0.93	-4.11***
9 Month	6.91	4.51	5.79	6.1	-0.28	-7.19***
12 Month	7.75	4.78	4.08	5.8	-1.97	-9.72***

Note. Firms are sorted into quintiles (Q1 = lowest score). Mean excess return and Newey-West t-stat in parentheses (Newey & West, 1987). High-Low is Q5 minus Q1 portfolio return. *p<0.1, **p<0.05, ***p<0.01 for High-Low t-test. Returns are in annualised and in %. As in Table 9 but using the aggregate score.

Table 10 shows this underperformance is not temporary but grows steadily over time, from -3.97% at 3 months to -9.72% at 12 months.

Figure 5:
Cumulative Returns of Highest vs. Lowest AI-Disclosure Quintiles



Note. Q5 = top-quintile composite AI score; Q1 = bottom quintile. Returns are value-weighted and net of risk-free rate (Daniel et al., 1997).

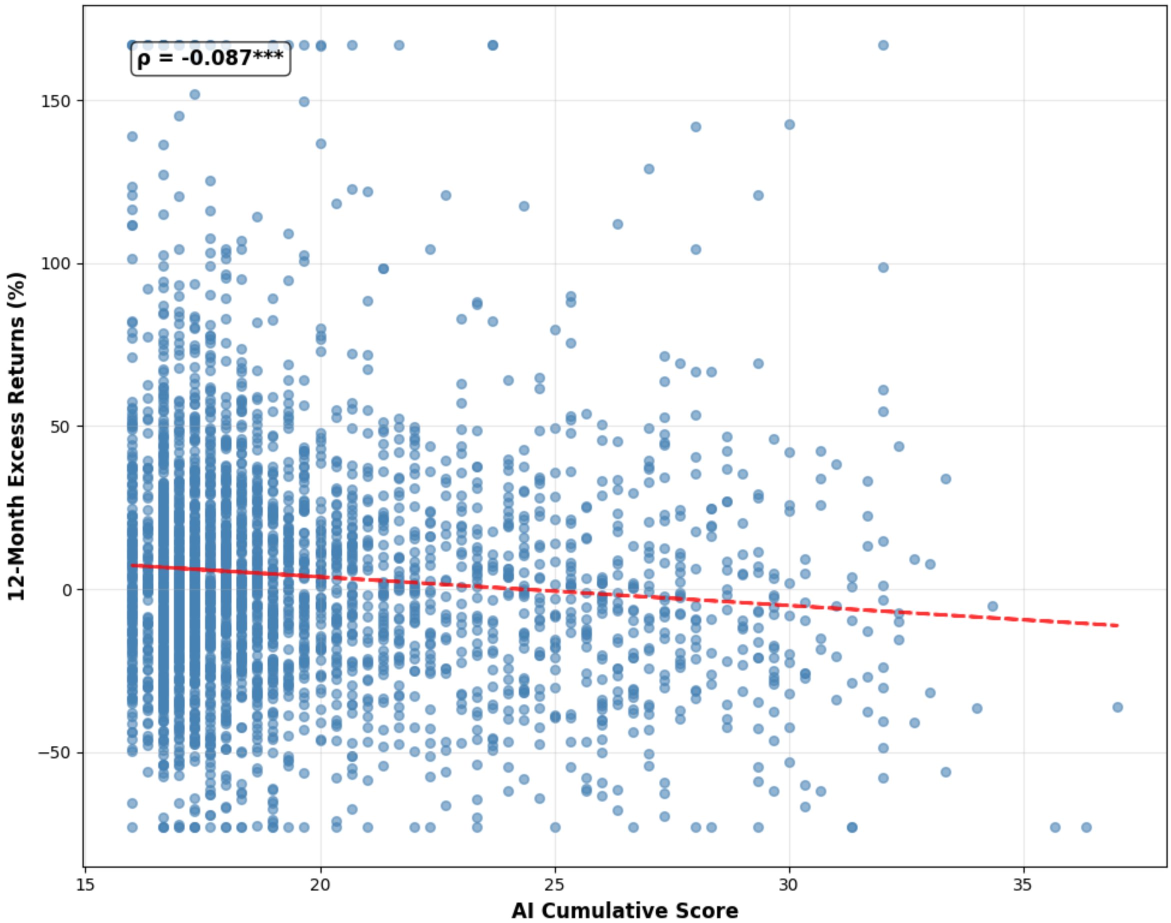
Figure 5 visualises the year-by-year average excess returns for the lowest (Q1) and highest (Q5) “AI Cumulative Score” portfolios. In every year except 2024, Q5 underperforms

Q1, with especially large gaps in 2020 and 2022. This time-series pattern reinforces the notion of consistent underperformance among high-AI firms.

Figure 6 plots 12-month excess returns against the "AI Cumulative Score" at the firm level. The negative slope and significant Pearson correlation (ρ = -0.087, $p < 0.01$) show that higher AI disclosure scores associate with lower future returns on average. Despite considerable noise, the downward trend line supports the portfolio sorting results and confirms the negative relationship exists across the full distribution of firms rather than being driven by extreme quintiles alone.

Figure 6:

Scatterplot of Composite AI Score vs. 12-Month Excess Return



Note. Each point = firm-year. Black line is OLS fit.

5.2.2 Sector-Specific Quintile Sorts

Table 11:

Sector-Specific Quintile Sorts: Excess Return by Composite AI Score

Sector	High-Low Spread (%)	T-Stat	P-Value	N Firms
Communication	13.04	1.19	0.237	66

Sector	High-Low Spread (%)	T-Stat	P-Value	N Firms
Consumer Discretionary	-12.24***	-2.71	0.007	331
Consumer Staples	-10.95*	-1.76	0.08	136
Energy	8.64	0.81	0.418	93
Financials	1.32	0.33	0.743	274
Health Care	-9.71*	-1.76	0.08	201
Industrials	-12.57***	-3.39	0.001	471
Information Technology	-3.72	-0.8	0.424	307
Materials	-9.09	-1.44	0.152	138
Real Estate	-1.38	-0.33	0.745	178
Utilities	-2.25	-0.35	0.725	44

Note. Returns reported relative to sector means to control for industry drift.

The sector-neutral analysis in Table 11 shows that the negative effect persists within several sectors (Asness et al., 2000), indicating that industry composition does not fully explain the patterns. Consumer Discretionary (-12.24%) and Industrials (-12.57%) show the strongest within-sector effects, both significant at 1%. Information Technology shows a smaller, insignificant spread (-3.72%), while Communication Services shows a positive (though insignificant) spread (13.04%).

These results provide **preliminary support for Hypothesis H3**, which predicted systematic variation across industry sectors. The dramatic differences between traditional sectors and technology sectors suggest that industry context matters critically for interpreting AI disclosures, consistent with competency-based disclosure theory where market reactions depend on perceived alignment between firm capabilities and strategic commitments (Campbell et al., 2025).

The portfolio analysis reveals a clear puzzle: firms with extensive AI communication experience poor subsequent stock performance. This challenges expectations about technology disclosure based on prior research showing positive associations between AI communication and firm performance (Mishra et al., 2022; Babina et al., 2024). The relationship appears robust across multiple AI dimensions, persists over various horizons, and exists within industry peer groups, motivating the multivariate regression analysis that follows.

6. Main Results

This chapter tests whether the negative relationship between AI disclosure and future returns shown in Chapter 5 remains after controlling for known stock performance drivers. The analysis progresses from testing individual AI factor regressions to analysing sector-specific data. It demonstrates that the intermediate results reveal meaningful economic relationships rather than false patterns caused by missing control variables.

The regression framework tests the three core hypotheses from Chapter 2. Hypothesis H1 predicted a positive relationship between AI disclosure quality and future returns, consistent with signalling theory (Spence, 1973). Hypothesis H2 suggested that strategic dimensions would be stronger predictors than sentiment-based measures, following disclosure theory literature (Muslu et al., 2015). Hypothesis H3 anticipated variation across industry sectors, based on competency-based signalling research (Asness et al., 2000). The results challenge conventional assumptions about corporate technology disclosure whilst highlighting the importance of industry context.

6.1 Individual AI Factor Analysis

6.1.1 Build-up Model Demonstration

Table 12:
Regressions of 12-Month Excess Returns on the AI Cumulative Score with Book Controls

Variable	Base	Model 1	Model 2	Model 3
Cum AI Score	-0.0114*** (0.0020)	-0.0108*** (0.0020)	-0.0102*** (0.0020)	-0.0094*** (0.0021)
Log Market Cap		-0.0060 (0.0046)	-0.0018 (0.0050)	-0.0064 (0.0051)
Price-to-Book			-0.0044** (0.0019)	-0.0054*** (0.0019)
ROA				1.1734*** (0.4014)
R^2	0.0459	0.0465	0.0483	0.0526
Adj. R^2	0.0414	0.0417	0.0431	0.0472
Observations	3,174	3,174	3,174	3,171

Note. *p<0.1, **p<0.05, **p<0.01. HC3 standard errors in parentheses. Table shows stepwise inclusion of book-based controls (Base model: AI score + fixed effects; Model 1: +firm size; Model 2:

+price-to-book; Model 3: +ROA) with “AI Cumulative Score” coefficient matching Table 13. R^2 increases from 4.59% to 5.26% across specifications. Dependent variable is 12-month buy-and-hold excess returns over risk-free rate. Sample period: 2020-2024.

Table 12 shows four regression specifications with progressive control inclusion. The “AI Cumulative Score” coefficient declines modestly from -0.0114 in the Base model to -0.0094 in the full Model 3 specification, a 17% decline that indicates stability across control specifications.

A one-unit increase in the “AI Cumulative Score” implies a 0.94 percentage point decrease in 12-month excess returns in the full specification. Given the standard deviation of approximately 4.01, a one-standard-deviation increase implies a 3.77 percentage point decrease in annual returns.

This stability contradicts traditional signalling theory predictions (Spence, 1973; Ross, 1977), which suggest that quality firms should benefit from credible disclosure. Instead, the consistently negative coefficients align with dot-com era evidence of technology buzzword penalties (Cooper et al., 2001) and contemporary AI washing concerns highlighted by regulators (U.S. Securities and Exchange Commission, 2024).

6.1.2 Factor Results

Table 13:

AI Factors Comparison Book Control Variables vs Fama French 6 Factor (t+12)

AI Factor	Book Control Variables Model	Fama-French 6-Factor Model
AI Washing Index	-0.0456*** (0.0104)	-0.0516*** (0.0103)
Disclosure Sentiment	-0.0351*** (0.0097)	-0.0395*** (0.0095)
Forward-Looking	-0.0570*** (0.0128)	-0.0653*** (0.0125)
Strategic Depth	-0.0302*** (0.0077)	-0.0336*** (0.0076)
Talent & Investment	-0.0568*** (0.0136)	-0.0672*** (0.0135)
Risk - External	-0.0093 (0.0207)	-0.0118 (0.0205)
Risk - Non Adoption	-0.0516*** (0.0162)	-0.0579*** (0.0165)

AI Factor	Book Control Variables Model	Fama-French 6-Factor Model
Risk - Own Adoption	-0.0435** (0.0216)	-0.0552*** (0.0214)
AI Cumulative Score	-0.0094*** (0.0021)	-0.0106*** (0.0020)
Average R^2	0.0502	0.036
Average Observations	3,171	3,031

Note. *p<0.1, **p<0.05, **p<0.01. HC3 standard errors in parentheses. Table shows regression coefficients for eight individual AI factors plus composite score under book controls (Column 1) and Fama-French 6-factor controls (Column 2). Separate regressions for each AI factor (18 total regressions: 9 factors × 2 specifications) to avoid multicollinearity. Seven of eight factors show significant negative effects. Dependent variable is 12-month buy-and-hold excess returns over risk-free rate. Sample period: 2020-2024.

Table 13 shows seven of eight individual factors exhibit negative coefficients significant at the 5% level or better in both specifications, with only Risk External showing insignificant effects.

These results **reject Hypothesis H1**. With seven of eight AI disclosure dimensions showing significant negative effects after controlling for firm characteristics and market factors, the multivariate analysis confirms that higher quality AI disclosure systematically associates with lower future returns, contradicting signalling theory predictions (Spence, 1973; Ross, 1977).

"Forward-Looking" statements show the largest negative effects, with coefficients of -0.0570 and -0.0653. A one-unit increase in "Forward-Looking" AI language associates with a 5.70 to 6.53 percentage point decrease in 12-month excess returns. "Talent & Investment" shows similar large negative effects (-0.0568 and -0.0672), while "Strategic Depth" shows coefficients of -0.0302 and -0.0336.

"Disclosure Sentiment" shows coefficients of -0.0351 and -0.0395, indicating that more positive AI discussion tone associates with lower returns. The "AI Washing Index" shows effects of -0.0456 and -0.0516.

"Risk Non-Adoption" shows strong negative effects (-0.0516 and -0.0579), while "Risk Own Adoption" shows coefficients of -0.0435 and -0.0552. "Risk External Threats" remains the sole exception, with small, insignificant coefficients.

These patterns directly contradict theoretical predictions. The severe penalties for "Forward-Looking" statements contradict predictions that credible forward-looking disclosures should enhance firm value (Muslu et al., 2015). The negative "Talent &

Investment" contradict signalling theory expectations that visible investments should signal firm quality (Spence, 1973). Instead, suggesting investor concerns about agency costs and overinvestment in fashionable technologies (Jensen, 1986; Malmendier & Tate, 2015). "Strategic Depth" penalties contradict research suggesting AI strategic integration should improve performance (Mishra et al., 2022).

These findings provide **partial support for Hypothesis H2,** though not in the expected direction. Strategic factors generally emerge as stronger predictors than sentiment-based measures, with Forward-Looking statements and Talent & Investment showing the largest negative effects, while Disclosure Sentiment exhibits relatively smaller coefficients. However, formal statistical tests comparing coefficient magnitudes would be needed to confirm whether these differences are statistically significant. The pattern suggests that markets distinguish between different types of AI communication and react most negatively to substantive strategic commitments rather than superficial AI mentions, indicating that investors view concrete AI strategies as particularly risky signals in the current market environment.

6.2 Sector Regression Analysis

The analysis examines whether AI disclosure effects vary across industry sectors, testing Hypothesis H3's prediction of industry-specific variation. This approach follows research showing that firm-level signals are more informative when considered relative to industry peers (Asness et al., 2000), and that AI adoption patterns differ significantly across sectors based on technological infrastructure and competency requirements (Cockburn et al., 2018; Brynjolfsson et al., 2021).

6.2.1 Cross-Sector Regression Results

Table 14:
Sector-Level Regressions: Composite AI Score (Controls vs. FF6 Models)

	(Book Controls Model)			(Fama French 6 Factor Model)			
Sector	Coefficient	P-Value	R^2	Coefficient	P-Value	R^2	N
Communication	-0.08% (1.29)	0.951	0.125	-1.11% (1.18)	0.346	0.261	102
Consumer Discretionary	-1.68%*** (0.49)	5.55E-04	0.077	-1.85%*** (0.48)	1.17E-04	0.064	489

	(Book Controls Model)			(Fama French 6 Factor Model)			
Consumer Staples	-0.34% (0.76)	0.657	0.095	-0.64% (0.69)	0.358	0.045	187
Energy	1.18% (1.80)	0.510	0.092	1.63% (1.60)	0.308	0.168	131
Financials	0.30% (0.78)	0.698	0.121	0.43% (0.69)	0.538	0.124	403
Health Care	-1.66%*** (0.58)	0.004	0.039	-0.91% (0.56)	0.106	0.045	276
Industrials	-1.76%*** (0.39)	6.79E-06	0.060	-1.72%*** (0.40)	1.53E-05	0.057	658
Information Technology	-0.13% (0.53)	0.805	0.035	-0.46% (0.53)	0.383	0.032	432
Materials	-0.74% (1.13)	0.511	0.120	-0.018	0.063	0.089	184
Real Estate	-0.66% (0.79)	0.405	0.262	-0.14% (0.75)	0.849	0.304	244
Utilities	-0.45% (0.82)	0.581	0.265	-0.008	0.097	0.242	68

Note. *p<0.1, **p<0.05, **p<0.01. HC3 standard errors in parentheses. Table shows “AI Cumulative Score” coefficients estimated separately for each GICS sector under book controls (Column 1) and Fama-French 6-factor controls (Column 4), with corresponding p-values, R^2, and sample sizes. Sector-specific regressions for “AI Cumulative Score” only (22 total regressions: 11 sectors × 2 specifications). Individual AI factor sector regressions in Appendix Tables 24-31. Sample sizes range from 68 (Utilities) to 658 (Industrials). Dependent variable is 12-month buy-and-hold excess returns over risk-free rate. Sample period: 2020-2024.

Table 14 shows three sectors with significant negative coefficients: Industrials (-1.76%, p<0.001), Consumer Discretionary (-1.68%, p<0.001), and Health Care (-1.66%, p=0.004), whilst Information Technology shows no significant relationship (-0.13%, p=0.805).

Traditional Sector Penalties: Industrials shows the strongest negative relationship, with coefficients of -0.0176 and -0.0172. A one-unit increase in the “AI Cumulative Score” implies a 1.76 to 1.72 percentage point decrease in annual returns within this sector. Given the within-sector standard deviation of AI scores (approximately 3.8 for Industrials), a one-standard-deviation increase implies roughly 6.7 percentage point lower returns.

Consumer Discretionary shows similar strong patterns with coefficients of -0.0168 and -0.0185. Combined with within-sector AI score variation (standard deviation ≈ 3.9), this translates to approximately 6.5-7.2 percentage point return differences between high and low AI disclosure firms within the sector.

These negative reactions in traditional sectors may reflect self-selection bias, where firms under operational pressure use AI-heavy language to signal strength while potentially lacking fundamental AI capabilities (Roy, 1951; Ang & Goh, 2024).

Technology Sector Neutrality: Information Technology shows virtually no relationship between AI disclosure and returns, with small coefficients (-0.0013 and -0.0046) that are statistically insignificant. This neutrality indicates that within the technology sector, AI disclosure carries minimal informational content for investors. This aligns with market efficiency theory, where AI disclosures provide minimal new information in sectors with rich information environments and sophisticated investor bases (Fama, 1970).

These sector regressions provide **support for Hypothesis H3**, showing systematic variation across industry contexts. The large differences in both coefficient magnitudes and statistical significance across sectors confirm that industry context critically matters for interpreting AI disclosures. Traditional sectors show highly significant negative effects, while Information Technology shows no significant relationship (p=0.805). This pattern strongly supports competency-based disclosure theory, where market reactions depend on perceived alignment between firm capabilities and strategic commitments (Campbell et al., 2025).

6.2.2 Individual Factor Analysis by Sector

APPENDIX - References to "Appendix Tables … verified which tables in the end"

Sector-specific analysis of individual AI factors reveals the factors driving aggregate patterns. This analysis tests each of the eight AI factors separately within each sector (8 factors × 11 sectors × 2 specifications = 176 total regressions).

"Strategic Depth" emerges as a particularly strong negative predictor in traditional sectors. Consumer Discretionary shows coefficients of -6.61% and -7.27% for "Strategic Depth", indicating that a one-unit increase in strategic AI positioning implies a 6.61 to 7.27 percentage point return penalty. Industrials shows similar patterns with coefficients of -6.57% and -6.13%. These magnitudes are much larger than the effects shown in Table 14, suggesting strategic AI positioning is viewed as especially risky when undertaken by firms outside their core technological competencies, consistent with competency-based signalling theory (Campbell et al., 2025).

Market reactions become even more severe for future-oriented AI commitments in traditional sectors. Industrials shows the largest negative coefficients for "Forward-Looking" statements (-10.85% and -10.38%), while Consumer Discretionary follows with coefficients of -8.76% and -10.06%. These severe penalties contradict theoretical predictions that credible "Forward-Looking" disclosures should enhance firm value (Muslu et al., 2015), instead suggesting markets view such commitments as evidence of overoptimism in traditional sectors.

"Talent & Investment" discussions trigger particularly strong negative reactions in sectors lacking technical infrastructure. Consumer Discretionary shows coefficients of -9.46% and -10.25%, Health Care shows -11.48% and -6.73%, and Industrials shows -9.19% and -9.57%. These patterns suggest investor concerns about the costs and execution risks of building AI capabilities in sectors lacking established technical infrastructures, contradicting signalling theory predictions that talent hiring and investments should signal firm commitment and quality (Spence, 1973; Babina et al., 2024).

Acknowledging AI-related risks amplifies rather than reduces negative market reactions in traditional sectors, directly contradicting traditional disclosure theory (Healy & Palepu, 2001). Consumer Discretionary firms discussing AI adoption risks face severe return penalties across different risk dimensions. The "Risk Non-Adoption" factor shows particularly pronounced effects in Consumer Discretionary (-13.70% and -13.91%) and Industrials (-9.35% and -9.11%). "Risk Own Adoption" shows even more severe effects, with Consumer Discretionary coefficients of -15.98% and -18.31%.

Information Technology consistently shows small, insignificant coefficients across all individual factors, reinforcing that AI discourse carries minimal informational content within technology-native sectors. This neutrality extends across all AI disclosure dimensions, supporting theoretical predictions that AI capabilities represent baseline expectations rather than differentiating factors within technology-native industries (Bonsón et al., 2021).

These sector-specific patterns provide strong support for competency-based disclosure theory (Campbell et al., 2025), where market reactions depend critically on perceived alignment between firm capabilities and strategic commitments.

6.3 Hypothesis Testing and Theoretical Framework

6.3.1 Hypothesis Testing Results and Economic Significance

The clearest and most surprising finding is that firms with higher scores on AI-related disclosure dimensions tend to show significantly lower excess stock returns in the following 12 months. This negative relationship remains strong even after accounting for firm-specific characteristics, factor exposures, and fixed effects for both sector and year (Roberts & Whited, 2013).

The economic significance of these relationships is substantial and exceeds many traditional asset pricing anomalies. At the composite level, a one-standard-deviation increase in AI disclosure associates with 3.77 percentage point lower annual returns. The classic size effect confirmed by Fama and French (1992) typically produces annual return differences of 2-4 percentage points, indicating that the observed AI disclosure effects represent economically meaningful relationships. Within traditional sectors, return spreads reach 6.7 percentage points between high and low AI disclosure firms, creating potential opportunities for systematic investment strategies (Barber & Odean, 2008).

Hypothesis H1 (AI Disclosure and Returns) - Rejected: Rather than signalling future strength or innovation consistent with signalling theory (Spence, 1973; Ross, 1977), more detailed or frequent AI communication aligns with lower future returns. This negative pattern appears across nearly all disclosure categories, contradicting the fundamental premise that firms with superior private information should benefit from credible disclosure.

Hypothesis H2 (Strategic vs. Sentiment Dimensions) - Partial Support (Unexpected Direction): Factors tied to deliberate planning and investment, such as "Strategic Depth", "Talent & Investment", and "Forward-Looking" language, are indeed the strongest predictors of returns, but their effect is negative rather than positive as predicted by strategic disclosure theory (Muslu et al., 2015) and talent investment research (Babina et al., 2024). The finding that substantive strategic discourse is penalised more heavily than superficial mentions suggests investors distinguish between different types of AI communication and view strategic commitments as particularly risky signals, aligning with research on managerial overconfidence where concrete commitments in uncertain domains may signal overoptimism rather than strategic clarity (Malmendier & Tate, 2015).

Hypothesis H3 (Sectoral Differences) - Supported: The strength and direction of the AI disclosure effect vary sharply by industry, with effect magnitudes varying dramatically from significant negative effects in traditional sectors to neutral effects in technology sectors. In traditional sectors like Industrials, Consumer Discretionary, and Health Care, the return penalty for high AI disclosure is especially pronounced, whilst in Information Technology, where AI is already integrated into business models, the effect is close to zero. This pattern strongly supports competency-based theories of corporate disclosure, where market reactions depend on perceived alignment between firm capabilities and strategic commitments (Campbell et al., 2025).

6.3.2 Theoretical Framework for AI Disclosure Paradox

The observed negative relationship between AI disclosure and returns is best explained by three related theoretical frameworks. These theories work together to explain both the overall negative effects and why results differ across sectors.

Information Asymmetry and Market Scepticism

Self-Selection and Signalling Distortions: A primary explanation for the negative relationship concerns self-selection bias, where firms under operational or financial pressure may be more likely to use AI-heavy language to signal strength or distract from underlying problems—a classic dynamic in markets with information asymmetry (Roy, 1951; Spence, 1973). This interpretation aligns with recent evidence suggesting that companies may use disclosure tone to hide weak company fundamentals (Ang & Goh, 2024). The self-selection mechanism creates adverse selection problems where investors cannot distinguish genuine AI capabilities from "cheap talk," leading to market-wide discounting of all AI disclosures (Akerlof, 1970).

Evidence from earlier technology bubbles supports this view. During the blockchain boom, underperforming firms rebranded or made vague claims about blockchain use to ride investor excitement, only to later underdeliver (Jain & Jain, 2019). The dot-com bubble showed a similar pattern. During this bubble companies with weak fundamentals used internet-related terminology to inflate valuations (Ofek & Richardson, 2003). The AI disclosures may represent a similar pattern, where struggling firms use technological buzzwords to distract and signal strength.

Proprietary Cost Considerations: The strong negative reactions to “Strategic Depth” disclosures could reflect proprietary cost concerns, where detailed AI disclosures reveal competitive information to rivals (Verrecchia, 1983). In highly competitive sectors, investors may expect that strategic AI disclosures will decrease future competitive advantages, leading to negative valuation effects despite potentially positive underlying projects (Admati & Pfleiderer, 2000).

Agency Problems and Managerial Behaviour

Agency Costs and Overinvestment: The negative relationship may reflect investor concerns about managerial overinvestment in fashionable but unproven technologies. Agency theory suggests that managers with substantial free cash flow may invest in negative-NPV projects to build empires rather than return cash to shareholders (Jensen, 1986). The particularly strong negative reactions to “Talent & Investment” disclosures support this interpretation, suggesting markets view AI hiring and other investment announcements as potential evidence of value-destroying overinvestment rather than strategic capability building (Malmendier & Tate, 2015).

Hype Cycle Effects: The sample period (2020-2024) covered a significant AI hype cycle (PwC, 2024; Manning & Napier, 2024). Following the emergence of generative models, firms raced to mention AI in filings during increasing investor excitement (Manning & Napier, 2024). This period may reflect a pattern similar to earlier technology bubbles, where expectations outpaced results (Fenn, 1995). Such dynamics potentially expose firms with optimistic AI disclosures to negative revaluation as implementation challenges emerge. The progressive strengthening of negative effects over investment horizons suggests markets are learning to increasingly discount AI narratives (Fama, 1970; Barberis et al., 1998).

Competency-Based Signalling and Sectoral Context

Linking Signalling and Core Competency Theory: The identified sectoral differences suggest linking signalling theory (Spence, 1973) with core competency theory (Prahalad & Hamel, 1990). Traditional signalling models assume that expensive, hard-to-fake disclosures should be credible signals of firm quality. However, these findings suggest that signalling effectiveness also depends on whether the disclosed strategy aligns with the companies' core business.

When traditional manufacturing or retail firms make detailed AI disclosures, markets could interpret these as negative signals that management is moving away from proven business areas rather than positive signals of innovation (Barney, 1991). This may explain why Consumer Discretionary and Industrials firms face negative reactions to AI announcements. On the contrary, when technology firms make similar AI disclosures, markets view these as routine updates about expected capabilities rather than meaningful new information, leading to neutral reactions (Teece et al., 1997). This could explain why Information Technology firms show no significant market response to AI disclosures.

Legitimacy and Stakeholder Concerns: The sectoral variation also reflects differential stakeholder reactions to AI adoption. Legitimacy theory suggests that firms disclose activities to align with stakeholder expectations and social norms (Suchman, 1995). In sectors where AI raises ethical, employment, or safety concerns, disclosures may signal potential legitimacy threats rather than strategic opportunities (Freeman, 1984). The negative reactions in traditional sectors may partly reflect stakeholder concerns about job displacement, safety risks, or departure from established industry practices.

Market Efficiency and Information Processing: The null effects in Information Technology suggest that AI disclosures in this sector provide little new information to already well-informed investors and analysts. This matches market efficiency theory. In sectors with many informed investors and analysts, new information gets processed quickly and accurately (Fama, 1970). The contrast is true with traditional sectors, where information asymmetries are greater and information processing ability weaker. This supports the view that disclosure effectiveness varies with market structure and investor knowledge (Lowry & Shu, 2002).

Summary: The overall negative relationship likely comes from a mix of self-selection problems, agency costs, and hype cycle effects. The differences between sectors arise from how well AI fits with a firm's core business, stakeholder reactions, and how much investors know about each industry. This thesis suggests the market's negative reaction to AI disclosures during 2020-2024 has a clear explanation: Rather than viewing AI announcements as positive innovation signals, investors treated them as warning signs.

6.4 Explanatory Case Studies

To illustrate how the theoretical framework explains the AI disclosure paradox, this section examines cases representing the most extreme case of the phenomenon—companies with the highest AI disclosure scores that experienced severe negative returns. For analysis of additional cases across all significant negative sectors, see Appendix A.

6.4.1 Case Study Analysis

Case Study 1: Chegg, Inc. (CHGG) - Consumer Discretionary Sector

- *2023 Form 10-K, Filed: 20 February 2024*
- *AI Score: 36.33 (Highest in their industry)*
- *12-Month Return: -72.99% (Worst performance)*

Chegg provides the most dramatic example of the AI disclosure paradox. The company achieved the highest AI disclosure score while experiencing the worst stock performance across the Consumer Discretionary industry sample. The company's traditional homework assistance model was disrupted by free generative AI tools like ChatGPT, with Q4 2023 earnings showing a 21% year-over-year decline in subscribers and decreasing revenues. Management announced the company was undergoing strategic review to consider all options, including a potential sale.

This context possibly created extreme agency pressures where management needed to maintain stakeholder confidence while acknowledging fundamental competitive threats (Jensen & Meckling, 1976). The extensive AI disclosure represents an attempt to reframe existential disruption as strategic opportunity through impression management (Merkl-Davies & Brennan, 2007).

The company's 2023 10-K filing reveals the underlying vulnerability:

"Our competition may develop products and technologies that are similar or superior to our technologies or are more cost-effective to develop and deploy... Given the long history of development in the AI sector, other parties may have (or in the future may obtain) patents or other proprietary rights that would prevent, limit, or interfere with our ability to make, use, or sell our own AI products." (Chegg, Inc., 2024)

This admission shows information so damaging that firms would only disclose it if hiding it posed greater regulatory risks (Spence, 1973; Verrecchia, 1983). The self-selection pattern becomes clear: Chegg felt forced to discuss AI extensively precisely because its business model had become outdated, creating a situation where extensive disclosure signals weakness rather than strength (Akerlof, 1970; Li & Prabhala, 2007).

6.4.2 Systematic Behavioral Patterns

The Chegg case shows specific patterns that extend across all firms with high AI disclosure scores and negative returns. Analysis of additional companies across the three significant sectors reveals consistent patterns that explain the AI disclosure paradox through five interconnected theoretical frameworks.

The case study analysis of AI-intensive firms across Health Care, Industrials, and Consumer Discretionary sectors reveals that extensive AI disclosure consistently occurs under conditions of business pressure rather than strategic strength. Every case shows the same sequence: underlying business challenges create agency pressures for management, leading to impression management strategies that create negative signalling effects through self-selection bias.

The analysis validates the explanatory power of the integrated theoretical framework across different industry contexts, showing that the AI disclosure paradox results from systematic responses to business pressures rather than random market behaviour:

Agency Theory and Impression Management (Jensen & Meckling, 1976; Merkl-Davies & Brennan, 2007): Cases reveal management teams under pressure to maintain stakeholder confidence while facing performance challenges, competitive threats, or strategic failures. The extensive AI discourse represents attempts to reframe negative developments within aspirational technological narratives.

Signalling Theory Distortions (Spence, 1973; Ross, 1977): Cases reveal systematic "over-signalling without substance" where extensive AI language attempts to signal strength while concrete evidence reveals weakness, creating contradictory signals that markets correctly interpret as negative.

Self-Selection and Market for Lemons (Roy, 1951; Li & Prabhala, 2007; Akerlof, 1970): Companies systematically self-select into extensive AI disclosure when facing business model threats, failed investments, or competitive disadvantage, creating adverse selection where disclosure intensity correlates with underlying vulnerability rather than strategic capability.

Legitimacy Theory (Suchman, 1995; Freeman, 1984): In traditional sectors, extensive AI disclosure often signals potential stakeholder concerns about employment displacement or operational risks rather than competitive advantages.

Proprietary Cost Theory (Verrecchia, 1983): Cases reveal companies inadvertently disclosing competitive vulnerabilities while attempting to demonstrate AI capabilities.

The consistent negative market reactions across all significant sectors suggest sophisticated investor interpretation of patterns rather than simple punishment of transparency. Markets appear capable of distinguishing genuine strategic communication from impression management driven by agency costs or business model threats. The subsequent performance of these companies generally validates the market's initial negative assessments, confirming that extensive AI communication often accompanies rather than contradicts fundamental business weaknesses (Fama, 1970).

6.5 Practical Implications

For Investors: The findings provide clear guidance for portfolio construction. High AI disclosure creates significant negative return effects in Consumer Discretionary, Health Care, and Industrials while showing neutral effects in Information Technology and Financials. These patterns create potential for sector-specific investment strategies that avoid high AI disclosure firms in Consumer Discretionary, Health Care, and Industrials sectors.

For Management: The results suggest firms may benefit from exercising restraint when discussing AI initiatives in formal disclosures. The significant negative effects in Consumer Discretionary, Health Care, and Industrials indicate that markets interpret extensive AI communication in these sectors as potential overinvestment or "AI washing" rather than strategic capability (Jensen, 1986; Campbell et al., 2025). Technology and Financial sector firms face less severe penalties, suggesting that AI commitments within core competencies trigger weaker negative reactions.

For Regulators: The negative market reactions in specific sectors validate current regulatory concerns about AI disclosure practices. The SEC's recent enforcement actions align with market evidence showing sector-dependent penalties for AI disclosure intensity (U.S. Securities and Exchange Commission, 2024). The sector-specific results suggest regulatory frameworks could account for industry-specific factors when evaluating AI disclosure adequacy.

The findings indicate that markets distinguish between sectors when evaluating AI disclosure of any kind. Even firms making substantive disclosures about AI strategies, talent hiring, AI-related investments, and forward-looking plans face substantial penalties in Consumer Discretionary, Health Care, and Industrials, suggesting that markets view detailed AI commitments in these sectors as potentially value-destroying rather than credible signals of future performance.

7. Conclusion

Research Focus and Empirical Design

This thesis addresses a critical question at the intersection of finance and technology: does the narrative content of corporate disclosures on AI predict future stock market performance? The research analyses a sample of Russell 3000 constituents from 2020 to 2024 to investigate whether financial markets reward substantive AI strategies or penalise unsupported claims.

The study's core methodological contribution employs LLMs to analyse AI-related narratives in 10-K filings, moving beyond traditional keyword approaches to capture nuanced disclosure dimensions. These LLMs are prompted with structured instructions to score AI communication across eight dimensions, such as strategic depth, sentiment, and an 'AI washing' index. The resulting scores are then tested using two empirical strategies: quintile portfolio sorts and fixed-effects panel regressions that control for a number of firm characteristics and market risk factors.

Broad Take-aways of Main Findings

The empirical results present a counterintuitive conclusion that fundamentally challenges signalling theory (Spence, 1973; Ross, 1977). Rather than rewarding credible disclosure, markets systematically penalise firms with more extensive AI communication, with higher disclosure scores associated with lower annual returns. This 'AI disclosure penalty' persists across multiple dimensions and remains robust after controlling for known return drivers.

The market reaction varies significantly by industry context. Traditional sectors—Industrials, Consumer Discretionary, and Health Care—experience penalties for AI discourse, suggesting markets interpret such communication as signals of overreach beyond core competencies. By contrast, Information Technology shows neutral effects, indicating AI capability represents baseline expectations rather than differentiating factors in technology-native industries (Bonsón et al., 2021).

Contributions and Practical Relevance

Methodologically, this research shows that ensemble LLM frameworks can convert nuanced corporate narratives into reliable, multi-dimensional data for large-scale financial analysis, providing a replicable template for analysing complex disclosure topics.

Theoretically, the findings challenge classic signalling theory by documenting how credible disclosures can signal weakness rather than strength when firms venture beyond their core competencies (Spence, 1973). The results are consistent with market-wide scepticism about 'AI washing' (Seele & Schultz, 2022), support agency cost theories of managerial overinvestment (Jensen, 1986), and align with competency-based disclosure frameworks where market reactions depend on strategic alignment (Campbell et al., 2025).

For practitioners, the evidence offers clear guidance. Investors should review AI claims from non-tech sectors where negative effects are strongest. Management should prioritise substance over ambitious disclosure, as extensive AI discourse without verifiable action appears to signal weakness rather than strength. Regulators gain quantitative support for continued oversight of potentially misleading technological claims.

Limitations and Future Research

This study has several limitations which point to areas for future research. While validated, LLM assessments can have inherent biases and their reasoning can be unclear. Future work could use human review to refine these assessments. Secondly, the analysis focuses on corporate language, not action. Linking disclosures to tangible data, such as AI spending or patents, would provide a more direct test of the 'talk versus action' hypothesis. Thirdly, the research focuses on US 10-K reports; applying the framework to other documents or countries could reveal different patterns. Finally, the 2020–2024 sample period covers a distinct AI hype cycle during which markets may have been particularly sceptical of technological claims (Manning & Napier, 2024; Accenture, 2024). Ongoing research should track if the observed AI disclosure penalty persists as the technology matures and its productivity gains are realised.

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Appendix

Appendix A:

Table 15:
Illustrative Examples of AI Disclosure Factor Scoring

Factor	Quote – High (A)	Quote – Generic
1. Strategic Depth (Mishra et al., 2022; Cockburn et al., 2018)	"We believe that AI capabilities are central to products and solutions across our markets and we have a broad technology roadmap and products targeting AI training and inference spanning cloud, edge and intelligent endpoints. We offer products that include capabilities to support AI deployment and we expect this part of our business to grow." (AMD 2024-01-31)	"We may face increased competition due to the rapid development and rising use of artificial intelligence (AI) and machine learning technologies. Failure to adopt and incorporate such technologies to improve productivity, manufacturing technology or support functional teams may put us at a long-term competitive disadvantage." (HUBB 2025-02-13)
2. Disclosure Sentiment (Loughran & McDonald, 2011; Feldman et al., 2010)	"We have established a leadership position in the emerging market segment for powering high-performance processors used for acceleration of applications associated with artificial intelligence ('AI'). Our customers in the AI market segment include the leading innovators in processor and accelerator design, as well as early adopters in cloud computing and high performance computing." (VICR 2023-02-28)	"There are risks associated with artificial intelligence ('AI'), any or all of which could adversely affect our business." (SPG 2025-02-21)
3.a Risk - Own Adoption (Alston & Bird, 2024; Amershi et al., 2019)	"We believe generative AI will have a significant effect on how we provide services to our clients and how we enhance the productivity of our people. As with any new technology, we are working closely with our clients and technology partners to take advantage of the benefits of AI while being mindful of its limitations, risks, and privacy concerns." (OMC 2024-02-07)	"We are evaluating the use of artificial intelligence technologies ('AI') within our business and we recognize that third parties that provide services to us may independently use AI. The use of AI, which involves reliance on substantial data volumes, introduces risks that may have a material adverse effect on our business." (PJT 2024-02-28)
3.b Risk - External	"Our systems and third-party systems with which we interact, as well as	"Any future cybersecurity incidents or other similar disruptions impacting

Factor	Quote – High (A)	Quote – Generic
Threats (Christensen, 1997; Cockburn et al., 2018)	systems those third-parties utilize, are subject to increasingly sophisticated methods of attack, including the deployment of artificial intelligence to find and exploit vulnerabilities, such as "deep fakes," long-term persistent attacks referred to as advanced persistent threats and the use of the IT supply chain to introduce malware." (MMC 2021-02-17)	Core Molding Technologies or significant customers and/or suppliers could adversely affect us." (CMT 2025-03-11)
3.c Risk - Non-Adoption (Brynjolfsson & McAfee, 2014)	"Competitors have substantially greater resources to invest in technological improvements and have invested more heavily than us, and will continue to be able to do so, in developing and adopting new technologies, which may put us at a competitive disadvantage." (SNV 2024-02-23)	"The risk of cybersecurity attacks may increase as artificial intelligence capabilities improve and are increasingly used to identify vulnerabilities and construct increasingly sophisticated attacks, with the possibility of additional vulnerabilities being introduced through our own use of artificial intelligence and its use by our stakeholders, including vendors and customers." (NVT 2025-02-18)
4. Forward-Looking (Muslu et al., 2015)	"For retail, we will continue leveraging data science and AI to offer even better digital experiences for associates and consumers, driving conversion and efficiency. We plan to improve our online vehicle transfer experience and expand Skye's functionality with additional data and new architecture." (KMX 2024-04-15)	"We are also continuing to explore the use of machine learning, AI and deep learning in our data and analytics strategies." (TRU 2021-02-16)
5. AI Washing Index (Seele & Schultz, 2022; Al Haddi, 2024)	"With the acquisitions of Mipsology SAS and Nod, Inc. in 2023, we expanded our AI software capabilities to accelerate our AI growth strategy centered on an open software ecosystem to help lower the barriers of entry for customers through developer tools, libraries and models." (AMD, 2025-02-05)	"Some of our computer systems currently, and may in the future, incorporate AI solutions, including machine learning and generative AI tools that collect, aggregate, and analyze data to assist in the development of our services and products and in the use of internal tools that support our business." (ABR, 2025-02-21)
6. Talent & Investment	"We expect spending in technology and infrastructure to increase over time	"Our ability to attract and retain executives and other qualified

Factor	Quote – High (A)	Quote – Generic
(Babina et al., 2024; Spence, 1973)	as we continue to add employees and infrastructure. These costs are allocated to segments based on usage. The increase in technology and infrastructure costs... is primarily due to an increase in spending on infrastructure and increased payroll and related costs associated with technical teams..." (AMZN, 2024-02-02)	personnel... is important to our success. We must also attract and retain a large number of skilled employees... The competition for these employees is intense and we may not be able to hire and retain the number of employees we need in each of our locations." (CNDT, 2024-02-21)

Note. This table provides real examples from corporate 10-K filings demonstrating high-scoring (A-grade) versus typical/generic disclosure language for each AI factor. Company tickers and filing dates are shown in parentheses. These examples illustrate the qualitative differences captured by the grading framework.

Appendix A: Complete LLM Prompt

The prompt template below is used to instruct the LLM for analyzing the 10-K filing excerpts. The {document_text} placeholder is filled with the combined "Risk Factors" and "Management's Discussion and Analysis" sections for each company.

```
You are an expert financial analyst and AI strategist, specializing in evaluating corporate disclosures for insights into a company's engagement with Artificial Intelligence (AI). Your task is to meticulously analyse the provided text from a company's 10-K filing (specifically excerpts from "Item 1. Risk Factors" and "Item 7. Management's Discussion and Analysis") and provide a structured JSON output assessing the company's AI preparedness based on the defined factors and criteria.

**DOCUMENT CONTEXT:**

The provided text below, enclosed in triple backticks (```), contains excerpts from "Item 1. Risk Factors" and "Item 7. Management's Discussion and Analysis (MD&A)" of a single company's 10-K report.

{document_text}

**INSTRUCTIONS:**

1.  Carefully read the entire provided text.

2.  For each of the AI Assessment Factors listed below, evaluate the text based on the defined Buckets & Criteria.
```

3. Your entire response MUST be a single, valid JSON object. Do not include any explanatory text before or after the JSON object. The JSON object MUST conform to the Pydantic schema `CompanyAIAnalysis`.

4. For each factor, you MUST include in the JSON:
   * `bucket_assessment`: The exact string for the category you've assigned.
CRITICAL: This value MUST be one of the precise string values listed in the Buckets & Criteria for that factor (e.g., for AI Strategic Depth, it must be exactly "A (Core Strategic Enabler)" or "B (Significant Operational/Product Integration)", etc. Do NOT use abbreviations or variations).
   * `chain_of_thought_reasoning`: A brief explanation (1-2 sentences) of your thought process for arriving at the bucket assessment, referencing the criteria and evidence.
   * `supporting_evidence`: Key verbatim quotes (or concise summaries of multiple related statements) from the text that justify your assessment. Limit to 2-3 key pieces of evidence. If no direct evidence is found for a factor, you MUST provide an empty list `[]` for this field.

5. **CRITICAL FOR `ai_risk_disclosure` FIELD**: The `ai_risk_disclosure` field in the JSON output MUST be an object containing three specific nested assessment objects as its keys: `risk_from_own_ai_adoption`, `risk_from_external_ai_threats`, and `risk_of_not_adopting_ai_or_competitive_ai_threat`. Each of these three nested objects must follow the `BaseAssessment` structure (i.e., have `bucket_assessment`, `chain_of_thought_reasoning`, and `supporting_evidence`). Refer to the example JSON structure below.

6. If information for any other factor or sub-factor (within `ai_risk_disclosure`) is not present in the text, use the appropriate "No Mention" bucket assessment for that factor (e.g., "E (No Mention of Own AI Strategy/Adoption)" or "C (No Mention of Own AI Adoption Risk)"). For `supporting_evidence` in such cases, provide an empty list `[]`.
   IMPORTANT: Ensure ALL main assessment factor objects (ai_strategic_depth, ai_disclosure_sentiment, ai_risk_disclosure, forward_looking_ai_statements, ai_washing_hype_index, ai_talent_and_investment_focus) are ALWAYS present as keys in your JSON output. If a factor has no relevant information, its `bucket_assessment` should reflect its defined "No Mention" or equivalent lowest grade for that factor's rubric, and `supporting_evidence` should be an empty list. Do not omit these top-level keys from the JSON.

7. For the `company_identifier` field, extract the company name from the "Company Name (if known):" line in the DOCUMENT CONTEXT. If it's "Unknown Company" or not clearly identifiable, set it to null.

8. For the `overall_ai_preparedness_summary_cot` field, you MUST provide a 1-2 sentence summary of the company's overall AI posture based on your analysis of all factors.

9. For the `key_ai_related_terminology_found` field, you MUST list any explicit AI-related terms found in the text (e.g., "artificial intelligence", "machine learning", "NLP", "generative AI"). If no such terms are found, provide an empty list `[]`.

AI ASSESSMENT FACTORS, BUCKETS & CRITERIA:

1. AI Strategic Depth:

* *Concept:* How deeply and explicitly AI is integrated into the company's core business strategy and future plans as evidenced by specific examples, resource allocation, or stated strategic importance. Evaluation should consider the scale and impact of AI initiatives, not just mentions.
* *Buckets & Criteria:*
    * "A (Core Strategic Enabler)": AI is explicitly and repeatedly identified as a core component of the company's overall strategy, with multiple concrete, specific examples of current or planned AI applications that are central to key products, services, or operational efficiency. Evidence of significant investment, dedicated AI leadership, or AI-driven competitive advantages.
    * "B (Significant Operational/Product Integration)": AI is discussed as being integrated into significant aspects of operations or specific product lines with some concrete examples, but may not be portrayed as a universal, core strategic driver for the entire enterprise. Impact is identifiable but perhaps more localised than 'A'.
    * "C (Emerging/Exploratory Mentions)": AI is mentioned in the context of emerging opportunities, ongoing exploration, pilot projects, or general capabilities the company is developing or monitoring. Specific strategic integration or widespread impact is not yet evident.
    * "D (Superficial/Generic Mentions OR Risk of Non-Adoption Only)": AI mentions are high-level, generic, or boilerplate, lacking specific examples or connection to core strategy. Alternatively, AI is only mentioned in the context of the *risk of not adopting it* or competitive pressures, without detailing the company's own strategic AI adoption.
    * "E (No Mention of Own AI Strategy/Adoption)": There is no substantive discussion of the company's own AI strategy, adoption, or integration efforts. (Note: General discussion of AI as an external market trend or risk factor without connection to the company's own plans falls here).

2. AI Disclosure Sentiment:
* *Concept:* The overall tone of the company's *own statements* regarding its AI initiatives, capabilities, and outlook. This excludes sentiment about external AI risks not directly tied to the company's actions.
* *Buckets & Criteria:*
    * "A (Clearly Positive)": Disclosures about the company's AI are predominantly optimistic, highlighting successes, opportunities, competitive advantages, and positive impacts. Language used is confident and forward-looking regarding their AI.
    * "B (Mostly Positive/Balanced)": Disclosures are generally positive but may include some neutral statements or acknowledge minor challenges that are framed as manageable. The overall impression is favourable.
    * "C (Neutral/Factual)": Disclosures about AI are presented in a purely factual, objective manner, without significant positive or negative framing. Focus is on description rather than promotion or concern.
    * "D (Cautious/Mixed)": Disclosures express caution, highlight significant uncertainties, or present a mixed picture of benefits and challenges regarding the company's own AI. May include discussion of ongoing struggles or unproven AI ventures.
    * "E (Negative OR Not Applicable - No Own AI Discussion)": Disclosures about the company's own AI are predominantly negative, focusing on failures, significant unresolved problems, or downsides. Also applies if there is no substantive discussion of the company's own AI initiatives to assess sentiment from.

3. AI Risk Disclosure: (This section requires careful attention to the nested structure)

* *Concept:* How thoroughly and specifically the company acknowledges and discusses potential risks associated with AI. This factor itself is an object in the JSON, containing the following three sub-assessments:
* **3a. `risk_from_own_ai_adoption`:** Risks arising directly from the company's own development, deployment, or use of AI systems (e.g., AI system failure, data bias, model inaccuracies, ethical concerns, implementation challenges, security of own AI systems, intellectual property issues related to own AI).
    * *Buckets & Criteria for 3a:*
        * "A (Detailed & Specific Discussion)": The company provides a detailed and specific discussion of one or more risks related to its own AI adoption, potentially outlining specific scenarios or impacts.
        * "B (General Mention of Own AI Adoption Risk)": The company makes a general mention of risks associated with its own AI adoption, without detailing specific scenarios or impacts extensively.
        * "C (No Mention of Own AI Adoption Risk)": The company does not mention any risks specifically arising from its own AI adoption or systems.
* **3b. `risk_from_external_ai_threats`:** Risks posed by AI used by external actors or general AI advancements in the environment (e.g., AI used in cyberattacks by others against the company, AI-driven misinformation campaigns affecting the company, AI capabilities of competitors if not framed as a 'failure to adopt' risk for the company itself).
    * *Buckets & Criteria for 3b:*
        * "A (Detailed & Specific Discussion)": The company provides a detailed and specific discussion of one or more external AI threats.
        * "B (General Mention of External AI Threat)": The company makes a general mention of external AI threats.
        * "C (No Mention of External AI Threat)": The company does not mention external AI threats.
* **3c. `risk_of_not_adopting_ai_or_competitive_ai_threat`:** Risks associated with the company failing to adopt AI, keep pace with AI developments, or risks from competitors leveraging AI more effectively.
    * *Buckets & Criteria for 3c:*
        * "A (Explicitly Discussed)": The company explicitly discusses the risk of falling behind competitors or negative impacts from not adopting/leveraging AI.
        * "B (Implicitly Suggested)": The risk of not adopting AI or competitive AI threats is implicitly suggested (e.g., by discussing competitors' AI advancements without stating it as a direct risk, or general statements about needing to innovate).
        * "C (No Mention of Non-Adoption/Competitive AI Risk)": The company does not mention risks related to its own non-adoption of AI or competitive AI threats.

4. Forward-Looking AI Statements:
* *Concept:* The extent and specificity of discussion about future plans, intentions, or expectations regarding AI.
* *Buckets & Criteria:*
    * "A (Specific & Detailed Future Plans)": The company provides specific and detailed statements about future AI plans, projects, investments, or expected milestones (e.g., "launching product X with AI feature Y in Q3," "investing Z amount in AI R&D over next year").
    * "B (General Future Intent)": The company expresses general intentions or expectations regarding future AI use or development, without providing specific details, timelines, or quantifiable targets (e.g., "we plan to explore AI," "AI will be important for our future").

        * "C (Implicit Future Focus Only)": Future AI focus is only implied through discussion of ongoing AI initiatives that naturally have a future dimension, but no explicit forward-looking statements are made.
        * "D (No Mention of Future AI Plans)": There is no mention of future plans, intentions, or expectations specifically related to AI.

5. "AI Washing" Hype Index: (Factor is only applicable if positive claims about own AI are made. If no positive claims, or only risks are discussed, use "E")
    * *Concept:* Degree to which a company's positive AI claims appear substantive and grounded in verifiable details versus being exaggerated, vague, or aspirational without clear evidence.
    * *Buckets & Criteria:*
        * "A (Substantive & Grounded)": Positive AI claims are consistently backed by specific examples, data, achievements, or clear explanations of functionality and impact. Claims appear realistic and verifiable.
        * "B (Mostly Substantive)": Most positive AI claims are substantive, but there may be occasional instances of slightly vague or aspirational language. Overall, claims seem credible.
        * "C (Mixed - Some Substance, Some Hype)": There's a mix of substantive claims and more hyped, vague, or aspirational statements about AI. Difficult to clearly distinguish hype from reality throughout.
        * "D (Mostly Hype)": Positive AI claims are predominantly vague, aspirational, or use buzzwords without concrete evidence or specific details. Claims seem exaggerated or difficult to substantiate.
        * "E (Pure Hype OR Not Applicable - No Positive AI Claims)": Positive AI claims are entirely generic, buzzword-laden, and lack any substance, or the company makes no positive claims about its own AI to evaluate for hype (e.g., only discusses AI risks or general market trends).

6. AI Talent & Investment Focus:
    * *Concept:* Explicit discussion of focus on acquiring, developing, or retaining AI talent, or specific investments (financial, infrastructural) in AI.
    * *Buckets & Criteria:*
        * "A (Explicit & Significant Focus)": The company explicitly and significantly discusses efforts to acquire/develop AI talent (e.g., hiring initiatives, training programs) AND/OR details specific, material investments in AI (e.g., R&D spending, acquisitions, infrastructure development).
        * "B (General Mention of AI Talent/Investment)": The company makes general mentions of the importance of AI talent or investment in AI, but without providing specific details on initiatives, scale, or amounts.
        * "C (Implicit Focus Only)": Focus on AI talent or investment is only implicitly suggested (e.g., by discussing AI projects that would logically require talent/investment, but without explicitly stating this focus).
        * "D (No Mention of AI Talent/Investment)": There is no mention of a focus on AI talent acquisition, development, or specific AI investments.

EXPECTED JSON OUTPUT STRUCTURE (Example for LLM guidance, actual structure enforced by Pydantic):
```json
{
  "company_identifier": "EXTRACT_COMPANY_NAME_IF_POSSIBLE_ELSE_NULL",
```

```
  "overall_ai_preparedness_summary_cot": "LLM_GENERATED_1_2_SENTENCE_SUMMARY_OF_OVERALL_AI_POSTURE_BASED_ON_ALL_FACTORS",
  "ai_strategic_depth": {
    "bucket_assessment": "A (Core Strategic Enabler)",
    "supporting_evidence": ["quote1 from text about core AI strategy", "quote2 detailing specific AI application"],
    "chain_of_thought_reasoning": "The company clearly states AI is central to its strategy and provides multiple concrete examples of its application in key products, aligning with criteria for 'A'."
  },
  "ai_disclosure_sentiment": {
    "bucket_assessment": "B (Mostly Positive/Balanced)",
    "supporting_evidence": ["quote expressing optimism about AI initiative"],
    "chain_of_thought_reasoning": "The language used when discussing their AI is generally positive, highlighting benefits, though some statements are factual, fitting 'B'."
  },
  "ai_risk_disclosure": {
    "risk_from_own_ai_adoption": {
      "bucket_assessment": "B (General Mention of Own AI Adoption Risk)",
      "supporting_evidence": ["The company mentions risks related to 'implementation of new technologies' including AI."],
      "chain_of_thought_reasoning": "A general risk of adopting new technologies including AI is mentioned, but not highly specific to AI, fitting 'B'."
    },
    "risk_from_external_ai_threats": {
      "bucket_assessment": "C (No Mention of External AI Threat)",
      "supporting_evidence": [],
      "chain_of_thought_reasoning": "No specific external AI threats like AI-cyberattacks by others were identified in the text."
    },
    "risk_of_not_adopting_ai_or_competitive_ai_threat": {
      "bucket_assessment": "A (Explicitly Discussed)",
      "supporting_evidence": ["'Failure to innovate or adopt new technologies such as AI could harm our competitive position.'"],
      "chain_of_thought_reasoning": "The company explicitly states that not adopting AI poses a competitive risk, aligning with 'A'."
    }
  },
  "forward_looking_ai_statements": {
    "bucket_assessment": "A (Specific & Detailed Future Plans)",
    "supporting_evidence": ["'We plan to invest $X million in AI R&D in the next fiscal year.'", "'A new AI-powered feature for product Y is scheduled for Q4.'"],
    "chain_of_thought_reasoning": "Specific investment figures and product launch timelines for AI are provided, meeting criteria for 'A'."
  },
  "ai_washing_hype_index": {
    "bucket_assessment": "A (Substantive & Grounded)",
    "supporting_evidence": ["The claims about AI improving efficiency are supported by specific operational examples mentioned earlier."],
```

"chain_of_thought_reasoning": "Positive AI claims are linked to concrete examples and described functionalities, indicating substance, hence 'A'."
},
"ai_talent_and_investment_focus": {
"bucket_assessment": "B (General Mention of AI Talent/Investment)",
"supporting_evidence": ["'We are focused on attracting skilled personnel in technology areas like AI.'"],
"chain_of_thought_reasoning": "A general statement about attracting AI talent is made, but lacks specifics on initiatives or scale, fitting 'B'."
},
"key_ai_related_terminology_found": ["artificial intelligence", "machine learning", "automation"]
}

Appendix B: Case Study Analysis

B.1 Introduction and Case Study Overview

This appendix analyses five additional AI Hyper firms that support the Chegg case from the main text. These cases show that the same behavioral patterns appear across all three sectors with significant negative relationships between AI disclosure and stock returns.

Table 16:
Complete Case Study Overview

Company	Sector	AI Score	12m Return	Key Issue	Analysis Location
Chegg (CHGG)	Consumer Discretionary	36.33	-72.99%	Core Business Disrupted by AI	Main Text (6.4.1)
3D Systems (DDD)	Industrials	29.67	-62.04%	Failed AI Investment	Appendix B.2.1
Omnicell (OMCL)	Health Care	30.33	-60.14%	AI Creates New Risks	Appendix B.2.2
Corvel (CRVL)	Health Care	24.67	-64.23%	Regulatory Pressure	Appendix B.2.3
TTEC Holdings (TTEC)	Industrials	31.33	-72.99%	AI Threatens Core Business	Appendix B.3.1

Company	Sector	AI Score	12m Return	Key Issue	Analysis Location
iRobot (IRBT)	Consumer Discretionary	31.33	-72.99%	Losing Key AI Talent	Appendix B.3.2

Note. This table summarizes six companies with high AI disclosure scores and severely negative stock returns. Each case demonstrates how extensive AI discussion can signal underlying business problems rather than strategic strength. The case Chegg is analysed in detail in the main text, while the remaining cases provide supporting evidence in the appendix sections.

B.2 Multi-Sector Case Analysis

B.2.1 Case Study: 3D Systems Corp. (DDD) - Industrials Sector

- *2022 Form 10-K, Filed: 16 March 2023*
- *AI Score: 29.67 (15th highest)*
- *12-Month Return: -62.04% (7th worst)*

External Context and Failed Investment Pressures

3D Systems' high AI disclosure came with disappointing Q4 2022 earnings. The company missed analyst estimates, showed declining revenues, and reported significant losses. Rising interest rates put extra pressure on unprofitable technology companies to show clear paths to profitability.

This created strong agency pressures for management. They had made large AI investments that failed to generate expected returns. Managers faced the challenge of explaining poor performance while maintaining credibility about their strategy.

Critical Evidence: Failed Investment Signalling

"On November 1, 2021, we acquired Oqton for $187.8 million. Oqton is a software company that creates an intelligent, cloud-based MOS platform... leveraging artificial intelligence, and machine learning technologies... However, the acquisition's impact on the Company's financial position, results of operations and cash flows has been dilutive."
(3D Systems Corporation, 2023)

Theoretical Analysis

This disclosure shows clear **Failed Investment Signalling** (Spence, 1973; Ross, 1977). Management explicitly admits their $187.8 million AI acquisition "has been dilutive." At the same time, they use positive language about "intelligent, cloud-based platforms" and "artificial intelligence and machine learning technologies."

This creates what signalling theory calls "contradictory signals" (Spence, 1973). The company uses extensive positive AI language while admitting concrete value destruction. This pattern shows management trying to maintain credibility about AI strategy despite clear execution failure. This contradicts signalling theory's requirement that credible signals must be costly and hard to fake (Ross, 1977).

Agency Theory implications are severe (Jensen & Meckling, 1976). Management faced a substantial investment that destroyed shareholder value. They confronted classic pressures to justify strategic decisions while maintaining stakeholder confidence. The extensive AI discourse represents impression management designed to frame failure within technological sophistication narratives (Merkl-Davies & Brennan, 2007).

Markets rationally interpreted this pattern as evidence of poor strategic execution and inadequate accountability. The combination of positive AI language and explicit value destruction signals fundamental problems with management's capital allocation decisions. This aligns with research on managerial overinvestment in fashionable technologies (Malmendier & Tate, 2015; Jensen, 1986).

B.2.2 Case Study: Omnicell, Inc. (OMCL) - Health Care Sector

- *2021 Form 10-K, Filed: 25 February 2022*
- *AI Score: 30.33 (7th highest)*
- *12-Month Return: -60.14% (7th worst)*

External Context and Agency Pressures

Omnicell's extensive AI disclosure came with severely disappointing Q4 2021 earnings. The company missed analyst estimates, faced rising supply chain costs, and gave forward guidance well below market expectations. This created classic agency pressures where management needed to offset negative financial news with strategic narratives (Jensen & Meckling, 1976; Healy & Palepu, 2001).

Critical Evidence: Legitimacy Theory and Proprietary Costs

"An effective attack on our solutions could disrupt the proper functioning of our solutions, allow unauthorized access to sensitive and confidential information of our customers (including protected health information)... These risks are likely to increase as we continue to grow our cloud-based offerings, including in support of the industry vision of the Autonomous Pharmacy, and as we receive, store, and process more of our customers' data." (Omnicell, Inc., 2022)

Theoretical Analysis

This disclosure shows how **Legitimacy Theory** explains negative market reactions to AI announcements (Suchman, 1995). Rather than positioning "Autonomous Pharmacy" as a competitive advantage, management frames their most ambitious AI initiative primarily in terms of escalating cybersecurity and privacy risks. This suggests management views AI expansion as increasing rather than decreasing vulnerability. This contradicts traditional signalling theory expectations that strategic investments should signal strength (Spence, 1973).

Proprietary Cost Theory implications are severe (Verrecchia, 1983). By explicitly linking their strategic "Autonomous Pharmacy" vision to increased data security risks, the company provides competitors and stakeholders with specific information about potential vulnerabilities. Markets rationally interpret such admissions as evidence of poor strategic positioning that will harm future competitive advantages (Admati & Pfleiderer, 2000).

The timing amplifies these effects through what **Impression Management Theory** predicts (Merkl-Davies & Brennan, 2007). When extensive AI disclosure accompanies earnings disappointments, sophisticated investors correctly identify the pattern as defensive impression management rather than genuine strategic communication.

B.2.3 Case Study: Corvel Corp. (CRVL) - Health Care Sector

- *2023 Form 10-K, Filed: 24 May 2024*
- *AI Score: 24.67 (36th highest)*
- *12-Month Return: -64.23% (3rd worst)*

External Context and Self-Selection

Corvel's extensive AI disclosure occurred alongside significant insider selling by executives. This created contradictory signals about management confidence. This combination shows self-selection bias where defensive disclosure accompanies personal risk mitigation by management (Li & Prabhala, 2007; Heckman, 1979).

Critical Evidence: Regulatory Disruption and Existential Threat

"Our future success depends, in part, on our ability to anticipate and respond effectively to the threat and opportunity presented by digital disruption, 'big data' and data analytics... We may not be successful in anticipating or responding to these developments on a timely and cost-effective basis and our ideas may not be accepted in the marketplace... There has also been increased regulatory scrutiny of the use of 'big data' techniques, machine learning, and artificial intelligence." (Corvel Corporation, 2024)

Theoretical Analysis

This case powerfully shows **Business Model Disruption Theory** (Christensen, 1997). The language reveals management's perception that AI represents both "threat and opportunity." However, the extensive discussion of potential failure ("may not be successful," "ideas may not be accepted") signals genuine uncertainty about the company's ability to adapt to technological change (Teece et al., 1997).

The **Regulatory Uncertainty** dimension adds another layer of risk that **Legitimacy Theory** predicts will create negative stakeholder reactions (Suchman, 1995). Management explicitly acknowledges that "increased regulatory scrutiny" of AI/ML creates compliance risks and potential operational restrictions. This represents mandatory disclosure of factors that could materially impair the business model (Healy & Palepu, 2001).

Self-Selection Bias becomes apparent when considering concurrent insider selling (Roy, 1951; Li & Prabhala, 2007). Management's words emphasize caution and regulatory risk while their actions show personal lack of confidence in company prospects. This behavioral pattern provides rational justification for negative market reactions. Investors correctly interpret the combination as evidence that management belongs to a cohort facing genuine competitive threats rather than strategic opportunities (Akerlof, 1970).

B.3 Additional Sector Cases

B.3.1 Case Study: TTEC Holdings, Inc. (TTEC) - Industrials Sector

- *2023 Form 10-K, Filed: 29 February 2024*
- *AI Score: 31.33 (6th highest)*
- *12-Month Return: -72.99% (Worst performance)*

External Context and Existential Disruption

TTEC represents the most severe case of business model disruption. The company's core human-led customer service offerings face direct replacement by AI automation (Christensen, 1997). The filing accompanied disastrous Q4 2023 earnings with drastically reduced guidance and major client losses.

Critical Evidence: Existential Business Model Threat

"These technology innovations have potential to significantly disrupt our business, reduce business volumes and related revenue unless we are successful in adapting... Deploying AI too rapidly without appropriate controls may result in poor client adoption, liability, and reputational damage." (TTEC Holdings, Inc., 2024)

Theoretical Analysis

This case shows **Existential Disruption Theory** in its purest form (Christensen, 1997; Teece et al., 1997). TTEC faces what we call "mandatory negative signalling." Regulatory requirements force the company to acknowledge that AI "has potential to significantly disrupt our business, reduce business volumes and related revenue" (Healy & Palepu, 2001).

The disclosure reveals management's impossible strategic position. They must simultaneously signal competence (we understand AI threats) while acknowledging potential obsolescence (our business model may become unviable). This creates what signalling theory predicts: when firms attempt to signal contradictory messages, markets interpret the confusion as confirmation of underlying weakness (Spence, 1973; Akerlof, 1970).

Self-Selection Mechanism becomes apparent when considering why TTEC discusses AI so extensively (Roy, 1951; Li & Prabhala, 2007). The company must engage in detailed AI disclosure precisely because AI represents an existential threat requiring regulatory acknowledgment. This creates unavoidable negative signalling where disclosure intensity correlates with business model vulnerability rather than strategic strength.

The market's severely negative reaction (-72.99% return) reflects rational assessment. Companies forced to acknowledge existential AI threats belong to a cohort facing potential obsolescence rather than transformation opportunity. This aligns with **Market for Lemons** theory where adverse selection leads to systematic undervaluation of firms revealing vulnerability (Akerlof, 1970).

B.3.2 Case Study: iRobot Corp. (IRBT) - Consumer Discretionary Sector

- *2022 Form 10-K, Filed: 14 February 2023*
- *AI Score: 31.33 (8th highest)*
- *12-Month Return: -72.99% (Worst performance)*

External Context and Core Competency Loss

iRobot's case shows how AI disclosure can reveal operational vulnerabilities that become critical when external circumstances change. While the failed Amazon acquisition was the direct cause of stock decline, the AI disclosure revealed underlying talent retention problems. These problems explained why the acquisition was necessary rather than opportunistic (Prahalad & Hamel, 1990).

Critical Evidence: Core Competency Erosion

"Competition for hiring these employees is intense, especially with regard to engineers with high levels of experience in... artificial intelligence... we have experienced increased

employee turnover... we expect to continue to experience increased employee turnover in the future." (iRobot Corporation, 2023)

Theoretical Analysis

This disclosure shows **Core Competency Erosion Theory** (Prahalad & Hamel, 1990; Barney, 1991). For a technology company, admitting inability to retain AI talent signals fundamental operational weakness. This undermines the firm's core competitive advantages. The extensive discussion of AI strategy combined with explicit admission of talent retention problems suggests management lacks the human capital necessary to execute their stated strategic vision.

Following **Proprietary Cost Theory** (Verrecchia, 1983), companies typically conceal information about competitive advantages to prevent exploitation by rivals. iRobot's willingness to reveal AI talent retention problems suggests that concealment posed greater legal or regulatory risks than disclosure. This implies that underlying challenges were severe and potentially visible to stakeholders (Admati & Pfleiderer, 2000).

When the Amazon acquisition failed, these disclosed weaknesses became critical determinants of independent viability. Markets possessed clear information suggesting iRobot lacked internal capabilities necessary for success in an AI-driven competitive landscape. This aligns with **Resource-Based View** predictions about the importance of human capital for sustained competitive advantage (Barney, 1991; Wernerfelt, 1984).

Agency Theory implications reflect management's position facing talent exodus (Jensen & Meckling, 1976). The extensive AI disclosure represents impression management designed to show strategic awareness while preparing stakeholders for potential performance impacts from talent losses (Merkl-Davies & Brennan, 2007).

B.4 Summary of Case Study Findings

The analysis of six AI Hyper firms reveals systematic behavioral patterns across all three significant sectors. Each case shows the same fundamental sequence: underlying business pressures create agency incentives for extensive AI disclosure. However, this strategy backfires by revealing vulnerabilities rather than strength.

Health Care Sector: Omnicell and Corvel face regulatory and competitive pressures that force defensive AI disclosure. Omnicell frames AI expansion in terms of increased cybersecurity risks. Corvel explicitly acknowledges potential failure in AI adaptation alongside regulatory scrutiny.

Industrials Sector: 3D Systems and TTEC represent different patterns - failed investment justification versus existential business model threats. 3D Systems explicitly admits their $187.8M AI acquisition was "dilutive." TTEC acknowledges AI may "significantly disrupt our business, reduce business volumes."

Consumer Discretionary Sector: Chegg (analyzed in main text) and iRobot show business model obsolescence and core competency erosion. Chegg admits competitors may develop "superior" technologies. iRobot reveals critical AI talent retention problems.

All cases show self-selection bias where extensive AI disclosure correlates with business vulnerability rather than strategic strength. The consistent negative market reactions validate investor sophistication in distinguishing genuine strategic communication from impression management driven by agency costs.

B.5 Cross-Case Theoretical Synthesis

The comprehensive analysis of six AI Hyper firms across three sectors reveals that extensive AI disclosure consistently occurs under conditions of business pressure rather than strategic strength. Every case shows the same sequence: underlying business challenges create agency pressures for management, leading to impression management strategies that create negative signalling effects through self-selection bias.

B.5.1 Unified Theoretical Framework

The cases validate five interconnected theoretical mechanisms:

Failed Investment Signalling (Spence, 1973; Ross, 1977): Companies use extensive AI discourse to justify value-destroying investments. This creates contradictory signals that markets correctly interpret as evidence of poor capital allocation.

Business Model Disruption (Christensen, 1997; Teece et al., 1997): Firms facing AI-driven obsolescence engage in mandatory negative signalling through regulatory disclosures that reveal existential vulnerabilities.

Core Competency Erosion (Prahalad & Hamel, 1990; Barney, 1991): Companies losing essential capabilities use AI narratives to mask deteriorating advantages while inadvertently revealing vulnerabilities through proprietary cost disclosures.

Legitimacy Theory (Suchman, 1995; Freeman, 1984): AI disclosure signals potential stakeholder concerns about ethics, safety, or regulatory compliance that create legitimacy threats rather than competitive advantages.

Self-Selection Bias (Roy, 1951; Li & Prabhala, 2007; Akerlof, 1970): Companies systematically self-select into extensive AI disclosure when facing business pressures. This creates adverse selection where disclosure intensity correlates with underlying weakness rather than strength.

B.5.2 Market Efficiency and Rational Responses

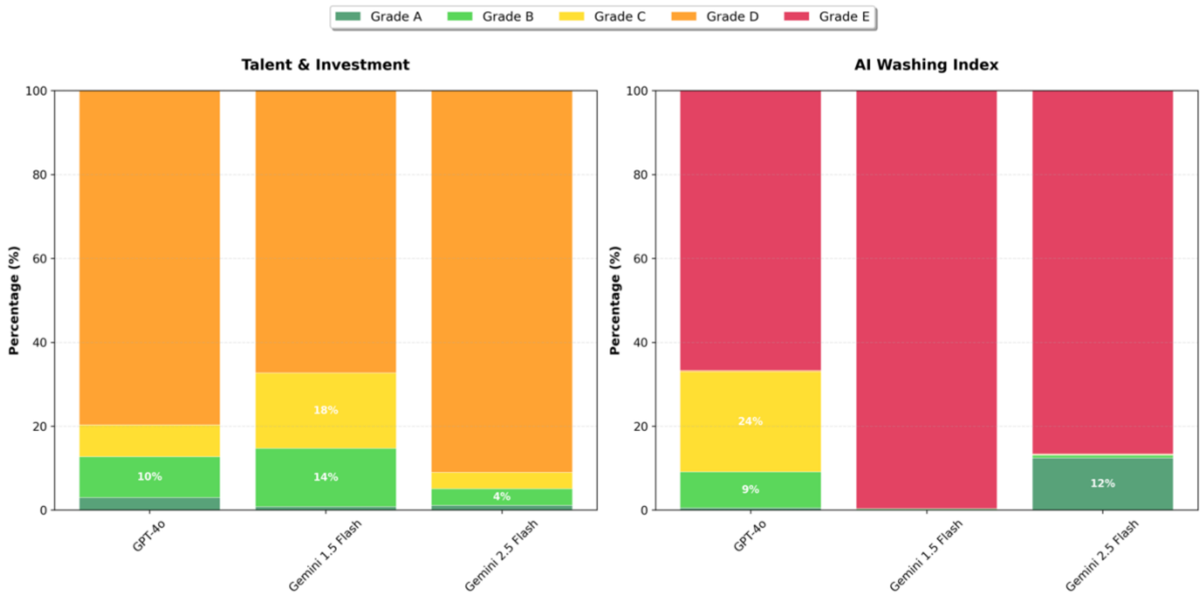
The consistent negative market reactions across all cases suggest sophisticated investor interpretation of behavioral patterns rather than irrational punishment of transparency. Markets appear capable of distinguishing genuine strategic communication from impression management driven by agency costs (Jensen & Meckling, 1976), business model threats (Christensen, 1997), or operational failures (Prahalad & Hamel, 1990).

The subsequent performance of these companies generally validates initial negative market assessments. This confirms that extensive AI disclosure often accompanies rather than contradicts fundamental business weaknesses. This outcome consistency shows market efficiency in processing complex behavioral signals embedded within corporate disclosures (Fama, 1970).

Appendix …: Data Validation and Control

Figure 7:

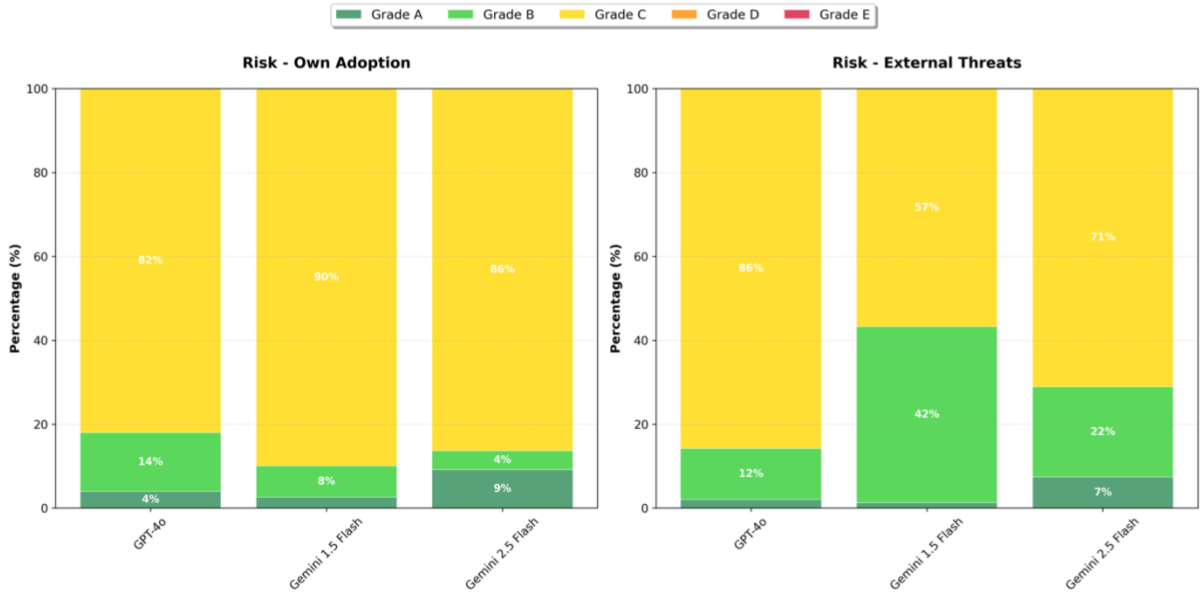
Distribution of AI Factor Grades by LLM Model (Talent & Investment and AI Washing)



Note. Each panel shows frequency of letter grades (A–E) issued by GPT-4o-Mini, Gemini 1.5, and Gemini 2.5.

Figure 8:

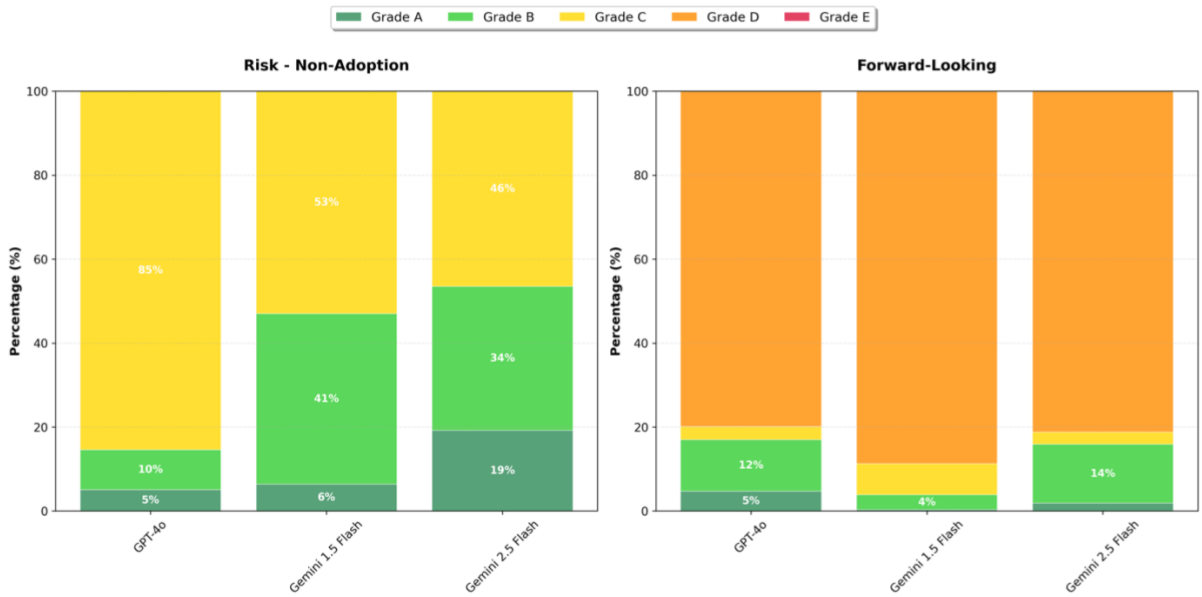
Distribution of AI Factor Grades by LLM Model (Risk: Own Adoption & External Threats)



Note. Each panel shows frequency of letter grades (A–E) issued by GPT-4o-Mini, Gemini 1.5, and Gemini 2.5.

Figure 9:

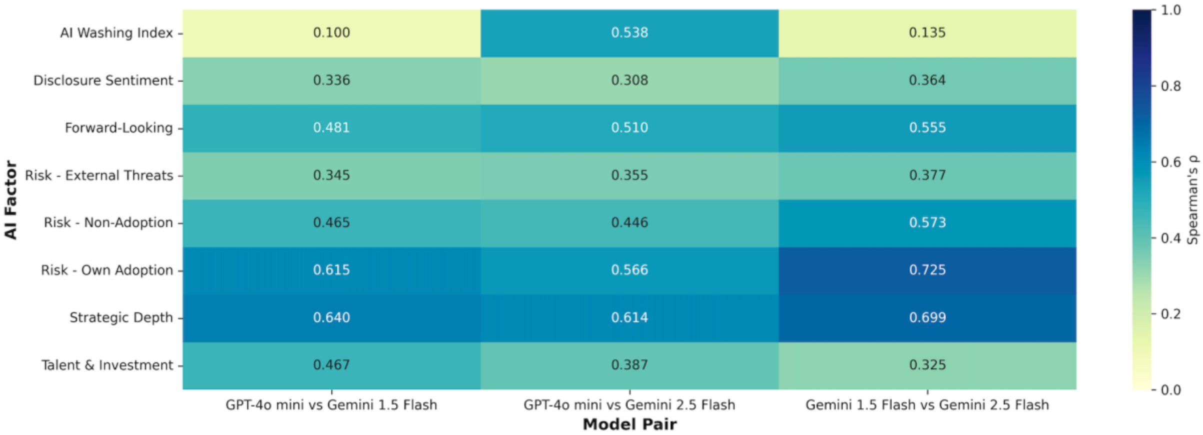
Distribution of AI Factor Grades by LLM Model (Non Adoption Risk & Forward-Looking)



Note. Each panel shows frequency of letter grades (A–E) issued by GPT-4o-Mini, Gemini 1.5, and Gemini 2.5.

Figure 10:

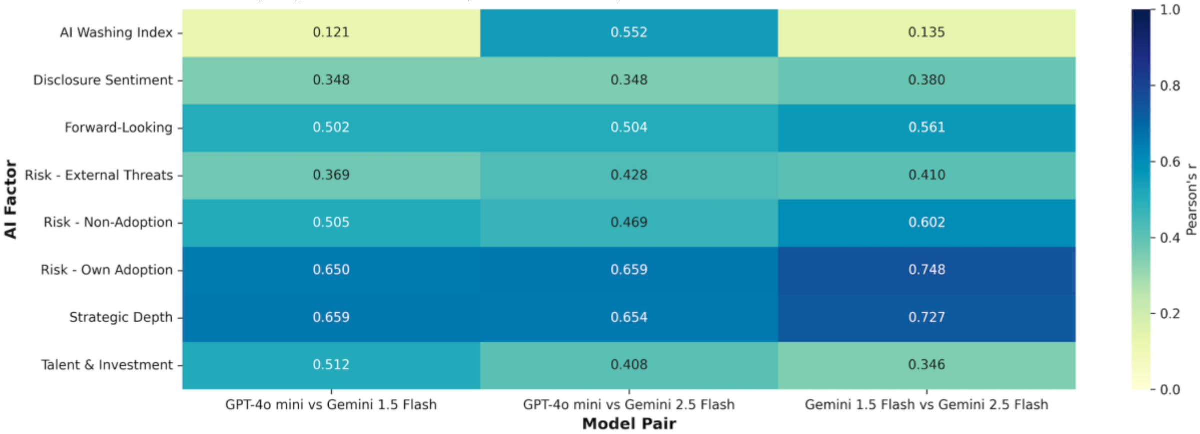
Correlation Heatmap of AI Factors (Spearman's Rho)



Note. Rank-based correlation robust to outliers.

Figure 11:

Correlation Heatmap of AI Factors (Pearson's r)



Note. Pearson's r emphasises linear association—shown for comparison with non-parametric measures.

Table 17:

LLM Agreement: GPT-4o-Mini vs. Gemini Flash 1.5

Factor	KTau	SRho	CKappa	Exact_Match	Within_1	Obs
Strategic Depth	0.59	0.64	0.58	66.60	87.50	3995
Disclosure Sentiment	0.31	0.34	0.31	55.20	69.20	3995
Risk - Own Adoption	0.60	0.61	0.62	85.50	99.50	3995
Risk - External Threats	0.34	0.34	0.31	64.10	99.70	3995
Risk - Non-Adoption	0.45	0.47	0.42	61.60	98.30	3995
Forward-Looking	0.46	0.48	0.39	79.40	88.70	3995
AI Washing Index	0.10	0.10	0.03	66.90	67.30	3995
Talent & Investment	0.44	0.47	0.51	68.50	91.40	3995

Note.

Table 18:
LLM Agreement: Gemini Flash 1.5 vs. 2.5

Factor	KTau	SRho	CKappa	Exact_Match	Within_1	Obs
Strategic Depth	0.66	0.70	0.66	69.60	89.70	3995
Disclosure Sentiment	0.34	0.36	0.33	45.60	56.50	3995
Risk - Own Adoption	0.71	0.72	0.68	88.70	98.00	3995
Risk - External Threats	0.37	0.38	0.40	61.80	99.50	3995
Risk - Non-Adoption	0.54	0.57	0.57	61.30	98.50	3995
Forward-Looking	0.54	0.56	0.46	82.20	91.50	3995
AI Washing Index	0.13	0.14	0.03	86.70	87.00	3995
Talent & Investment	0.31	0.33	0.29	68.30	88.50	3995

Note.

Table 19:
Summary Statistics for Factor Betas (3-, 6-, 9-Month Horizons)

Variable	N	Mean	Std	Min	P25	Median	P75	Max
Factor Beta Controls (3-Month)								
MKTRF	3980	0.9785	0.4607	-0.405	0.741	0.979	1.231	2.146
SMB	3980	0.6751	0.7359	-1.228	0.209	0.654	1.092	2.705
HML	3980	0.1835	0.7712	-1.885	-0.247	0.15	0.569	2.539
RMW	3980	0.1147	0.7979	-2.498	-0.201	0.174	0.523	2.121
CMA	3980	0.0629	0.945	-2.885	-0.367	0.087	0.554	2.619
UMD	3980	-0.0445	0.6134	-1.659	-0.377	-0.052	0.269	1.801
Factor Beta Controls (6-Month)								
MKTRF	3970	0.9921	0.4548	-0.427	0.758	1.001	1.238	2.18
SMB	3970	0.6691	0.7461	-1.271	0.189	0.624	1.092	2.585
HML	3970	0.2028	0.6798	-1.578	-0.195	0.179	0.558	2.012
RMW	3970	0.1298	0.8282	-2.558	-0.213	0.179	0.567	2.271
CMA	3970	0.0103	1.0905	-3.463	-0.458	0.071	0.558	3.015

Variable	N	Mean	Std	Min	P25	Median	P75	Max
UMD	3970	-0.0305	0.5329	-1.458	-0.315	-0.029	0.267	1.404
Factor Beta Controls (9-Month)								
MKTRF	3843	0.9708	0.4706	-0.587	0.75	0.983	1.223	2.124
SMB	3843	0.6748	0.6995	-1.125	0.227	0.649	1.102	2.448
HML	3843	0.1546	0.7864	-2.027	-0.261	0.13	0.554	2.517
RMW	3843	0.1418	0.8669	-2.622	-0.236	0.197	0.599	2.445
CMA	3843	0.0365	1.1002	-3.212	-0.522	0.05	0.619	3.191
UMD	3843	-0.0296	0.615	-1.879	-0.322	-0.014	0.307	1.608

Note. Betas estimated from rolling-window regressions of daily returns on Fama-French factors.

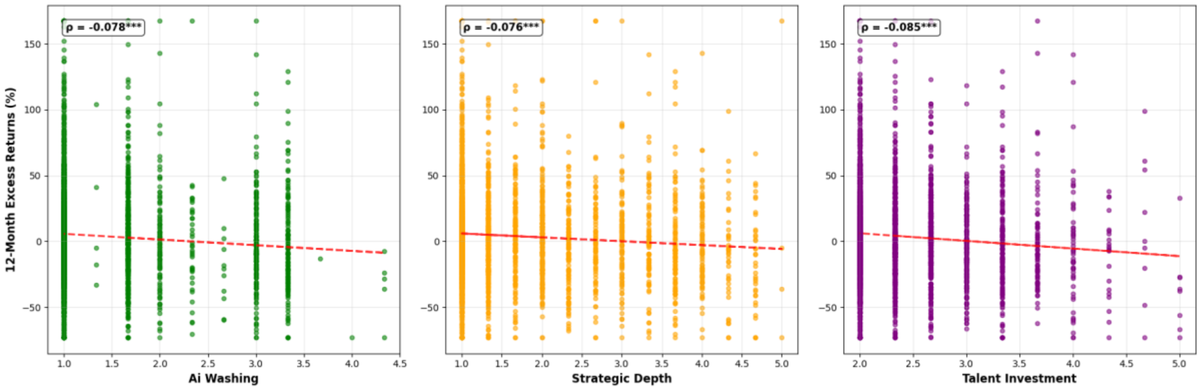
Table 20:
Correlation Matrix: Controls and 12-Month Excess Return

	ER12m	LogMCap	P/B	ROA	βMKTRF	βSMB	βHML	βRMW	βCMA	βUMD
ER12m	1	-0.05	-0.07	0.05	-0.01	0.04	0.05	0.03	-0.02	0.02
LogMCap	-0.05	1	0.43	0.32	-0.03	-0.42	-0.10	-0.01	0.02	0.03
P/B	-0.07	0.43	1	0.26	0.03	-0.19	-0.20	-0.04	-0.06	0.06
ROA	0.05	0.32	0.26	1	0.00	-0.13	0.01	0.13	-0.03	0.19
βMKTRF	-0.01	-0.03	0.03	0.00	1	0.14	0.07	0.00	0.03	0.01
βSMB	0.04	-0.42	-0.19	-0.13	0.14	1	-0.12	0.37	-0.04	0.05
βHML	0.05	-0.10	-0.20	0.01	0.07	-0.12	1	-0.22	-0.37	0.20
βRMW	0.03	-0.01	-0.04	0.13	0.00	0.37	-0.22	1	-0.06	-0.06
βCMA	-0.02	0.02	-0.06	-0.03	0.03	-0.04	-0.37	-0.06	1	-0.11
βUMD	0.02	0.03	0.06	0.19	0.01	0.05	0.20	-0.06	-0.11	1

Note. This table shows the Pearson correlation coefficients between the 12-month excess return and the main control variables.

Appendix C: Univariate Analysis & Additional Regression Tables

Figure 12:
Illustrative Relationships between Individual AI Factors and Returns



Note. Panels plot excess return against selected factor grades.

Table 21:
Regressions of 12-Month Excess Returns on the Talent & Investment Factor with Book Controls

Variable	Base	Model 1	Model 2	Model 3
Talent & Investment	-0.0700***	-0.0657***	-0.0609***	-0.0568***
	(0.0134)	(0.0135)	(0.0136)	(0.0136)
Log Market Cap		-0.0070	-0.0030	-0.0076
		(0.0046)	(0.0050)	(0.0050)
Price-to-Book			-0.0042**	-0.0053***
			(0.0019)	(0.0019)
ROA				1.2031***
				(0.3995)
R^2	0.0446	0.0454	0.047	0.0516
Adj. R^2	0.04	0.0406	0.0418	0.0462
Observations	3,174	3,174	3,174	3,171

Note. Base = industry & year FE; Model 1 adds log market cap; Model 2 adds price-to-book ratio; Model 3 adds ROA and therefore all book controls. Coefficients are percentage-point impacts; heteroskedasticity-robust t-stats in parentheses. *p<0.1, **p<0.05, ***p<0.01.

Table 22:
Regressions of 12-Month Excess Returns on the AI Cumulative Score with Fama-French Factor Controls

Variable	Base	FF3	FF5	FF6
Cum AI Score	-0.0114***	-0.0107***	-0.0106***	-0.0106***
	(0.0020)	(0.0020)	(0.0020)	(0.0020)

Variable	Base	FF3	FF5	FF6
Market Beta		-0.0013 (0.0189)	-0.0049 (0.0193)	-0.0048 (0.0193)
Size Beta		-0.0041 (0.0111)	-0.0016 (0.0113)	-0.0016 (0.0117)
Value Beta		-0.0175 (0.0110)	-0.0141 (0.0122)	-0.0141 (0.0123)
Profitability Beta			-0.0060 (0.0106)	-0.0060 (0.0107)
Investment Beta			0.0076 (0.0084)	0.0076 (0.0084)
Momentum Beta				0.0005 (0.0134)
R^2	0.0459	0.0386	0.0393	0.0393
Adj. R^2	0.0414	0.0328	0.0329	0.0326
Observations	3,174	3,031	3,031	3,031

Note. *p<0.1, **p<0.05, ***p<0.01. Standard errors in parentheses. Base includes Industry and Time Fixed Effects. FF factors use t12 horizon betas.

Table 23:
AI Factors Comparison Book Control Variables vs Fama French 6 Factor (t+9)

AI Factor	Book Control Variables Model	Fama-French 6-Factor Model
AI Washing Index	-0.0323*** (0.0092)	-0.0376*** (0.0089)
Disclosure Sentiment	-0.0254*** (0.0084)	-0.0320*** (0.0081)
Forward-Looking	-0.0377*** (0.0115)	-0.0457*** (0.0113)
Strategic Depth	-0.0202*** (0.0068)	-0.0250*** (0.0066)
Talent & Investment	-0.0480*** (0.0111)	-0.0551*** (0.0107)
Risk External	0.0051 (0.0187)	-0.0025 (0.0185)

AI Factor	Book Control Variables Model	Fama-French 6-Factor Model
Risk Non Adoption	-0.0470*** (0.0151)	-0.0569*** (0.0148)
Risk Own Adoption	-0.0007 (0.0199)	-0.0451** (0.0192)
AI Cumulative Score	-0.0068*** (0.0018)	-0.0082*** (0.0017)
Average R^2	0.0966	0.0811
Average Observations	3,217	3,159

Note. *p<0.1, **p<0.05, ***p<0.01. HC3 robust standard errors in parentheses. Each AI factor is tested in separate regressions to avoid multicollinearity concerns, resulting in 18 distinct regressions (9 AI factors × 2 model specifications). Book Control Variables Model includes log market cap, price-to-book, ROA, plus industry and time fixed effects. Fama-French 6-Factor Model includes market, size, value, profitability, investment, and momentum factor loadings, plus industry and time fixed effects.

Table 24:
AI Factors Comparison Book Control Variables vs Fama French 6 Factor (t+6)

AI Factor	Book Control Variables Model	Fama-French 6-Factor Model
AI Washing Index	-0.0173** (0.0073)	-0.0216*** (0.0071)
Disclosure Sentiment	-0.0139** (0.0063)	-0.0186*** (0.0061)
Forward-Looking	-0.0246*** (0.0089)	-0.0297*** (0.0087)
Strategic Depth	-0.0111** (0.0053)	-0.0148*** (0.0050)
Talent & Investment	-0.0286*** (0.0089)	-0.0328*** (0.0085)
Risk External	-0.0020 (0.0135)	-0.0040 (0.0133)
Risk Non Adoption	-0.0281** (0.0112)	-0.0347*** (0.0112)
Risk Own Adoption	-0.0003612 (0.014)	-0.0307** (0.0137)
AI Cumulative Score	-0.0040*** (0.0014)	-0.0049*** (0.0013)

AI Factor	Book Control Variables Model	Fama-French 6-Factor Model
Average R^2	0.0838	0.0757
Average Observations	3,267	3,251

Note. *p<0.1, **p<0.05, ***p<0.01. HC3 robust standard errors in parentheses. Each AI factor is tested in separate regressions to avoid multicollinearity concerns, resulting in 18 distinct regressions (9 AI factors × 2 model specifications). Book Control Variables Model includes log market cap, price-to-book, ROA, plus industry and time fixed effects. Fama-French 6-Factor Model includes market, size, value, profitability, investment, and momentum factor loadings, plus industry and time fixed effects.

Table 25:
AI Factors Comparison Book Control Variables vs Fama French 6 Factor (t+3)

AI Factor	Book Control Variables Model	Fama-French 6-Factor Model
AI Washing Index	-0.0115**	-0.0137**
	(0.0055)	(0.0054)
Disclosure Sentiment	-0.0112**	-0.0128***
	(0.0047)	(0.0046)
Forward-Looking	-0.0154**	-0.0179***
	(0.0065)	(0.0063)
Strategic Depth	-0.0084**	-0.0100***
	(0.0039)	(0.0038)
Talent & Investment	-0.0209***	-0.0240***
	(0.0069)	(0.0066)
Risk External	-0.0050	-0.0024
	(0.0089)	(0.0090)
Risk Non Adoption	-0.0241***	-0.0261***
	(0.0081)	(0.0082)
Risk Own Adoption	-0.0087	-0.0112
	(0.0100)	(0.0099)
AI Cumulative Score	-0.0029***	-0.0033***
	(0.0011)	(0.0010)
Average R^2	0.1397	0.1304
Average Observations	3,322	3,317

Note. *p<0.1, **p<0.05, ***p<0.01. HC3 robust standard errors in parentheses. Each AI factor is tested in separate regressions to avoid multicollinearity concerns, resulting in 18 distinct regressions (9 AI factors × 2 model specifications). Book Control Variables Model includes log market cap, price-to-book, ROA, plus industry and time fixed effects. Fama-French 6-Factor Model includes market, size, value, profitability, investment, and momentum factor loadings, plus industry and time fixed effects.

Table 26:
Sector-Level Regressions: Strategic Depth Factor

	(Controls Model)			(FF6 Model)			
Sector	Coefficient	P-Value	R	Coefficient	P-Value	R	N
Communication	-0.15% (5.47)	0.978	0.125	-2.90% (5.13)	0.572	0.256	102
Consumer Discretionary	-6.61%*** (1.90)	4.94E-04	0.076	-7.27%*** (1.87)	1.05E-04	0.063	489
Consumer Staples	-1.15% (2.65)	0.665	0.095	-1.99% (2.61)	0.446	0.044	187
Energy	8.98% (7.67)	0.241	0.105	10.66% (6.95)	0.125	0.184	131
Financials	1.36% (2.82)	0.628	0.121	2.15% (2.48)	0.385	0.125	403
Health Care	-4.91%** (2.01)	0.014	0.032	-2.59% (1.94)	0.183	0.042	276
Industrials	-6.57%*** (1.45)	6.14E-06	0.06	-6.13%*** (1.46)	2.78E-05	0.055	658
Information Technology	0.11% (1.85)	0.954	0.035	-0.84% (1.85)	0.648	0.03	432
Materials	-0.73% (4.09)	0.858	0.118	-5.39% (3.61)	0.136	0.084	184
Real Estate	-1.45% (3.68)	0.693	0.26	-0.17% (3.47)	0.96	0.304	244
Utilities	-1.21% (3.22)	0.707	0.264	-3.50% (2.63)	0.184	0.237	68

Note. See Table 17 note for statistical conventions.

Table 27:
Sector-Level Regressions: Disclosure Sentiment Factor

	(Controls Model)			(FF6 Model)			
Sector	Coefficient	P-Value	R	Coefficient	P-Value	R	N

	(Controls Model)			(FF6 Model)			
Communication	-1.04% (6.37)	0.87	0.125	-2.93% (5.61)	0.601	0.255	102
Consumer Discretionary	-7.85%*** (2.38)	9.92E-04	0.074	-8.68%*** (2.37)	2.52E-04	0.061	489
Consumer Staples	-3.06% (3.14)	0.33	0.099	-3.92% (3.09)	0.204	0.05	187
Energy	6.69% (6.84)	0.328	0.098	7.80% (6.58)	0.236	0.173	131
Financials	-0.99% (3.44)	0.774	0.121	-0.71% (3.22)	0.826	0.123	403
Health Care	-7.36%** (2.98)	0.013	0.035	-3.97% (2.97)	0.181	0.043	276
Industrials	-6.55%*** (1.90)	5.73E-04	0.049	-6.28%*** (1.93)	0.001	0.045	658
Information Technology	0.93% (2.62)	0.722	0.036	-0.38% (2.67)	0.887	0.03	432
Materials	-0.76% (4.25)	0.858	0.118	-3.52% (3.87)	0.363	0.081	184
Real Estate	-1.66% (2.19)	0.449	0.261	0.13% (1.96)	0.948	0.304	244
Utilities	-1.67% (3.27)	0.61	0.264	-2.62% (3.05)	0.39	0.228	68

Table 28:
Sector-Level Regressions: Talent & Investment Factor

	(Controls Model)			(FF6 Model)			
Sector	Coefficient	P-Value	R	Coefficient	P-Value	R	N
Communication	3.55% (6.14)	0.563	0.127	-5.91% (7.26)	0.415	0.261	102
Consumer Discretionary	-9.46%*** (3.42)	0.006	0.072	-10.25%*** (3.51)	0.004	0.057	489

	(Controls Model)			(FF6 Model)			
Consumer Staples	-7.37% (4.79)	0.124	0.102	-8.35% (5.14)	0.104	0.052	187
Energy	-11.22% (18.50)	0.544	0.091	-3.61% (16.96)	0.831	0.16	131
Financials	1.39% (4.09)	0.734	0.121	2.16% (3.80)	0.571	0.123	403
Health Care	-11.48%** (4.52)	0.011	0.034	-6.73% (4.58)	0.142	0.044	276
Industrials	-9.19%*** (2.64)	4.85E-04	0.052	-9.57%*** (2.62)	2.64E-04	0.05	658
Information Technology	-0.78% (3.12)	0.803	0.035	-2.99% (3.09)	0.334	0.032	432
Materials	-11.66% (8.17)	0.154	0.128	-14.35% (9.07)	0.114	0.095	184
Real Estate	-1.93% (6.88)	0.779	0.26	0.31% (6.72)	0.963	0.304	244
Utilities	2.80% (6.55)	0.669	0.264	-5.44% (3.38)	0.108	0.234	68

Table 29:
Sector-Level Regressions: AI Washing Factor

	(Controls Model)			(FF6 Model)			
Sector	Coefficient	P-Value	R	Coefficient	P-Value	R	N
Communication	-2.90% (6.31)	0.645	0.126	-7.08% (5.62)	0.208	0.264	102
Consumer Discretionary	-6.88%** (2.73)	0.012	0.066	-8.13%*** (2.76)	0.003	0.053	489
Consumer Staples	-2.09% (3.86)	0.588	0.096	-3.09% (3.90)	0.428	0.045	187
Energy	3.96% (8.31)	0.633	0.09	7.09% (8.23)	0.389	0.166	131
Financials	4.00% (6.63)	0.546	0.123	6.78% (6.16)	0.271	0.129	403

	(Controls Model)			(FF6 Model)			
Health Care	-6.76%** (2.81)	0.016	0.032	-0.126828	0.084	0.045	276
Industrials	-8.02%*** (1.87)	1.81E-05	0.055	-7.83%*** (1.86)	2.58E-05	0.052	658
Information Technology	-1.79% (2.44)	0.465	0.037	-3.12% (2.37)	0.188	0.034	432
Materials	-0.146118	0.091	0.123	-7.67%*** (2.97)	0.01	0.089	184
Real Estate	-1.15% (3.76)	0.76	0.26	0.36% (3.48)	0.919	0.304	244
Utilities	-1.87% (3.34)	0.576	0.265	-5.16%** (2.07)	0.012	0.243	68

Table 30:
Sector-Level Regressions: Forward-Looking Factor

	(Controls Model)			(FF6 Model)			
Sector	Coefficient	P-Value	R	Coefficient	P-Value	R	N
Communication	-1.02% (6.66)	0.878	0.125	-3.56% (6.21)	0.567	0.255	102
Consumer Discretionary	-8.76%*** (3.22)	0.007	0.068	-10.06%*** (3.17)	0.002	0.054	489
Consumer Staples	-2.17% (4.30)	0.614	0.096	-4.48% (4.25)	0.292	0.046	187
Energy	-7.87% (12.38)	0.525	0.092	-2.01% (11.20)	0.858	0.159	131
Financials	3.61% (4.94)	0.465	0.122	2.66% (4.18)	0.524	0.124	403
Health Care	-8.43%** (3.30)	0.011	0.032	-0.193848	0.072	0.046	276
Industrials	-10.85%*** (2.71)	6.07E-05	0.057	-10.38%*** (2.69)	1.15E-04	0.052	658
Information Technology	-1.29% (3.02)	0.669	0.036	-3.41% (3.06)	0.264	0.033	432

	(Controls Model)			(FF6 Model)			
Materials	-6.93% (8.43)	0.411	0.121	-10.40% (8.34)	0.212	0.085	184
Real Estate	-4.47% (6.00)	0.456	0.261	-1.61% (5.86)	0.784	0.304	244
Utilities	-3.40% (4.33)	0.433	0.267	-7.05%** (2.94)	0.016	0.244	68

Table 31:
Sector-Level Regressions: Risk External Factor

	(Controls Model)			(FF6 Model)			
Sector	Coefficient	P-Value	R	Coefficient	P-Value	R	N
Communication	-10.29% (10.95)	0.348	0.131	-15.06% (10.66)	0.158	0.269	102
Consumer Discretionary	0.08% (7.92)	0.992	0.053	-0.39% (7.92)	0.96	0.033	489
Consumer Staples	5.17% (10.21)	0.613	0.097	1.27% (9.18)	0.89	0.042	187
Energy	19.72% (13.61)	0.147	0.1	9.63% (14.26)	0.499	0.162	131
Financials	1.14% (4.19)	0.786	0.121	0.06% (4.63)	0.989	0.123	403
Health Care	-11.97% (7.28)	0.1	0.024	-3.53% (7.25)	0.626	0.038	276
Industrials	1.45% (4.50)	0.747	0.035	0.59% (4.45)	0.895	0.032	658
Information Technology	-3.56% (4.84)	0.463	0.036	-1.32% (5.17)	0.799	0.03	432
Materials	-4.98% (13.35)	0.709	0.12	-11.96% (13.50)	0.376	0.085	184
Real Estate	-4.98% (5.75)	0.387	0.263	-3.01% (5.35)	0.574	0.305	244
Utilities	-6.71% (9.97)	0.501	0.273	-4.37% (8.85)	0.621	0.227	68

Table 32:
Sector-Level Regressions: Disclosure Sentiment Factor Risk Non Adoption

	(Controls Model)			(FF6 Model)			
Sector	Coefficient	P-Value	R	Coefficient	P-Value	R	N
Communication	-5.78% (11.63)	0.619	0.127	-13.76% (10.47)	0.189	0.268	102
Consumer Discretionary	-13.70%*** (3.93)	4.89E-04	0.074	-13.91%*** (3.84)	2.89E-04	0.057	489
Consumer Staples	10.07% (10.37)	0.332	0.104	8.47% (10.08)	0.401	0.05	187
Energy	-1.59% (13.77)	0.908	0.088	-0.93% (13.19)	0.944	0.159	131
Financials	-1.01% (3.39)	0.766	0.121	-0.80% (3.60)	0.823	0.123	403
Health Care	-0.571938	0.06	0.028	-4.08% (5.66)	0.47	0.039	276
Industrials	-9.35%*** (3.17)	0.003	0.048	-9.11%*** (3.30)	0.006	0.044	658
Information Technology	2.34% (4.51)	0.603	0.036	0.05% (4.81)	0.991	0.03	432
Materials	3.82% (8.54)	0.655	0.119	-1.82% (9.13)	0.842	0.078	184
Real Estate	-4.96% (4.47)	0.267	0.263	-2.85% (4.76)	0.55	0.305	244
Utilities	-4.62% (10.16)	0.649	0.266	-7.00% (9.30)	0.452	0.231	68

Table 33:
Sector-Level Regressions: Risk Own Adoption Factor

	(Controls Model)			(FF6 Model)			
Sector	Coefficient	P-Value	R	Coefficient	P-Value	R	N
Communication	11.79% (14.11)	0.404	0.135	2.09% (12.22)	0.865	0.253	102
Consumer Discretionary	-15.98%*** (6.14)	0.009	0.066	-18.31%*** (6.25)	0.003	0.052	489

	(Controls Model)			(FF6 Model)			
Consumer Staples	5.98% (13.56)	0.659	0.095	-3.19% (15.09)	0.833	0.042	187
Energy	18.90% (20.17)	0.349	0.092	20.59% (15.88)	0.195	0.165	131
Financials	3.29% (4.07)	0.419	0.121	5.02% (3.73)	0.178	0.125	403
Health Care	-0.395601	0.098	0.021	-3.36% (4.78)	0.483	0.038	276
Industrials	-11.37%*** (4.40)	0.01	0.044	-12.19%*** (4.32)	0.005	0.043	658
Information Technology	-2.89% (4.56)	0.525	0.036	-6.27% (4.78)	0.189	0.035	432
Materials	4.91% (18.44)	0.79	0.119	-5.70% (55.37)	0.918	0.078	184
Real Estate	-6.48% (8.03)	0.42	0.262	-3.92% (8.75)	0.655	0.305	244
Utilities	9.27% (15.06)	0.538	0.268	0.80% (18.71)	0.966	0.223	68

Appendix D: Robustness Test Results

Table 34:
Summary Statistics Before Winsorization

Variable	N	Mean	Std	Min	Med.	Max	Skew	Kurt
3m ExRet	3325	0.29%	21.06%	-91.98%	-0.84%	255.29%	1.96	17.95
6m ExRet	3270	2.76%	28.99%	-92.70%	0.29%	398.43%	2.65	25.13
9m ExRet	3220	5.18%	38.88%	-93.48%	1.60%	452.26%	2.67	19.63
12m ExRet	3174	4.89%	42.97%	-94.35%	0.13%	490.26%	2.87	19.59
Log M Cap	3989	22.14	1.68	16.22	22.02	28.85	0.37	0.11
P/B	3985	3.29	83.50	-4,471.06	2.35	1,351.32	-40.92	2,168.53
ROA	3984	1.22%	3.29%	-40.59%	1.06%	82.24%	2.57	113.71
MKTRF Beta	3980	0.98	0.72	-7.54	0.98	13.44	0.83	49.49

Variable	N	Mean	Std	Min	Med.	Max	Skew	Kurt
SMB Beta	3980	0.67	1.13	-10.25	0.65	15.26	0.07	29.64
HML Beta	3980	0.18	1.23	-15.70	0.15	17.10	0.73	45.26
RMW Beta	3980	0.10	1.21	-13.50	0.17	9.36	-1.63	27.05
CMA Beta	3980	0.05	1.67	-19.13	0.09	19.94	-0.31	47.40
UMD Beta	3980	-0.05	0.88	-10.97	-0.05	10.12	-0.37	33.22

Table 35:
Summary Statistics After Winsorization

Variable	N	Mean	Std	Min	Med.	Max	Skew	Kurt
3m ExRet	3325	0.03%	18.50%	-46.47%	-0.84%	62.45%	0.43	1.08
6m ExRet	3270	2.35%	25.41%	-57.54%	0.29%	97.54%	0.74	1.94
9m ExRet	3220	4.51%	33.99%	-66.74%	1.60%	138.90%	0.98	2.32
12m ExRet	3174	4.12%	37.60%	-72.99%	0.13%	167.17%	1.26	3.57
Log M Cap	3989	22.14	1.64	18.91	22.02	26.41	0.31	-0.40
P/B	3985	4.30	9.68	-27.94	2.35	63.95	3.19	19.28
ROA	3984	1.22%	2.36%	-8.00%	1.06%	8.66%	-0.36	3.73
MKTRF Beta	3980	0.97	0.53	-1.39	0.98	2.66	-0.80	4.88
SMB Beta	3980	0.66	0.86	-2.97	0.65	3.61	-0.29	4.24
HML Beta	3980	0.18	0.87	-3.15	0.15	3.54	0.16	4.15
RMW Beta	3980	0.11	0.95	-4.09	0.17	3.54	-0.89	6.09
CMA Beta	3980	0.06	1.11	-4.83	0.09	4.05	-0.63	5.64
UMD Beta	3980	-0.04	0.69	-2.37	-0.05	2.79	0.46	4.04

Table 36:
Robustness-Check Summary Table

Test	P value	Result
Breusch-Pagan	6.10E-15	Heteroskedasticity
White Test	6.17E-19	Heteroskedasticity
Durbin-Watson	1.91474604	No autocorrelation

Table 37:
Variance-Inflation Factors for Main Regression Models

AI Factor	Factor Clean	Factor Type	VIF
AI_Washing	AI Washing	AI Factor	3.30041114
AI_Washing	Beta Mktrf	FF Factor	4.09168705
AI_Washing	Beta Smb	FF Factor	2.09960645
AI_Washing	Beta Hml	FF Factor	1.30145539
AI_Washing	Beta Rmw	FF Factor	1.24385144
AI_Washing	Beta Cma	FF Factor	1.25733613
AI_Washing	Beta Umd	FF Factor	1.06685207
Disclosure_Sentiment	Disclosure Sentiment	AI Factor	4.19660295
Disclosure_Sentiment	Beta Mktrf	FF Factor	4.84900674
Disclosure_Sentiment	Beta Smb	FF Factor	2.10739769
Disclosure_Sentiment	Beta Hml	FF Factor	1.28850189
Disclosure_Sentiment	Beta Rmw	FF Factor	1.24730498
Disclosure_Sentiment	Beta Cma	FF Factor	1.25032622
Disclosure_Sentiment	Beta Umd	FF Factor	1.06688475
Forward_Looking	Forward-Looking	AI Factor	4.90696466
Forward_Looking	Beta Mktrf	FF Factor	5.23418426
Forward_Looking	Beta Smb	FF Factor	2.14951961
Forward_Looking	Beta Hml	FF Factor	1.28442932
Forward_Looking	Beta Rmw	FF Factor	1.24723702
Forward_Looking	Beta Cma	FF Factor	1.25058684

AI Factor	Factor Clean	Factor Type	VIF
Forward_Looking	Beta Umd	FF Factor	1.06847623
Strategic_Depth	Strategic Depth	AI Factor	2.85300499
Strategic_Depth	Beta Mktrf	FF Factor	3.80262312
Strategic_Depth	Beta Smb	FF Factor	2.08600494
Strategic_Depth	Beta Hml	FF Factor	1.30757816
Strategic_Depth	Beta Rmw	FF Factor	1.24364697
Strategic_Depth	Beta Cma	FF Factor	1.25765947
Strategic_Depth	Beta Umd	FF Factor	1.06685345
Talent_Investment	Talent & Investment	AI Factor	4.95242672
Talent_Investment	Beta Mktrf	FF Factor	5.3857436
Talent_Investment	Beta Smb	FF Factor	2.13364521
Talent_Investment	Beta Hml	FF Factor	1.29163016
Talent_Investment	Beta Rmw	FF Factor	1.24800857
Talent_Investment	Beta Cma	FF Factor	1.25203181
Talent_Investment	Beta Umd	FF Factor	1.06702905
Risk_External	Risk External	AI Factor	5.42273201
Risk_External	Beta Mktrf	FF Factor	5.48647869
Risk_External	Beta Smb	FF Factor	2.17950028
Risk_External	Beta Hml	FF Factor	1.27246113
Risk_External	Beta Rmw	FF Factor	1.25136427
Risk_External	Beta Cma	FF Factor	1.24729342
Risk_External	Beta Umd	FF Factor	1.06891629
Risk_Non_Adoption	Risk Non Adoption	AI Factor	5.44791123
Risk_Non_Adoption	Beta Mktrf	FF Factor	5.57401007
Risk_Non_Adoption	Beta Smb	FF Factor	2.17194551
Risk_Non_Adoption	Beta Hml	FF Factor	1.27597782

AI Factor	Factor Clean	Factor Type	VIF
Risk_Non_Adoption	Beta Rmw	FF Factor	1.24976934
Risk_Non_Adoption	Beta Cma	FF Factor	1.24804522
Risk_Non_Adoption	Beta Umd	FF Factor	1.06887092
Risk_Own_Adoption	Risk Own Adoption	AI Factor	5.5209148
Risk_Own_Adoption	Beta Mktrf	FF Factor	5.60099503
Risk_Own_Adoption	Beta Smb	FF Factor	2.18019879
Risk_Own_Adoption	Beta Hml	FF Factor	1.27607625
Risk_Own_Adoption	Beta Rmw	FF Factor	1.24988556
Risk_Own_Adoption	Beta Cma	FF Factor	1.24759023
Risk_Own_Adoption	Beta Umd	FF Factor	1.06907371

Note. VIF > 10 (shaded) signals potential multicollinearity.